

ENGINEERING MASTER CURRICULUM

Polytechnic / Diploma → B.Tech / B.E. → M.Tech / M.E.

Structure: Course → Semester → Subject → Unit/Chapter → Topic

A clean, standardized India-oriented master curriculum map. It is designed as a broad reference across major engineering branches. Individual universities, state boards and regulation years can differ.

Reference basis: AICTE model curriculum documents and current engineering curriculum frameworks.
AICTE notes that institutions may adapt model curricula to their applicable programs and needs.

How to Read This PDF

- Every course is separated from every other course.
- Every course is divided semester-by-semester.
- Every semester contains subjects.
- Every subject is broken into five Units/Chapters with topic-level coverage.
- Project, laboratory, internship, seminar and thesis entries use practical competency maps.
- This is a standardized master map, not a replacement for a particular university's official syllabus.

Source notes

AICTE's first-year UG model curriculum provides common curriculum contents such as Chemistry, Physics, Mathematics, Programming for Problem Solving, English, Engineering Graphics, Workshop/Manufacturing Practices and Basic Electrical Engineering. ■cite■turn0search33■

AICTE's PG model curriculum contains M.Tech specializations with semester-wise cores, electives, research methodology, laboratories, mini projects and dissertation/research work. Examples include CSE/IoT and Mechanical Design Engineering. ■cite■turn0search34■turn0search36■

AICTE publishes branch-specific UG model curricula; for example, the 2024 Civil Engineering model curriculum is a four-year, 160-credit structure. ■cite■turn0search35■

PART I — POLYTECHNIC / DIPLOMA

Major diploma engineering courses are organized here into a six-semester model. Actual state-board structures may use different semester counts, subject names or elective patterns.

Polytechnic – Computer Science & Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Programming Fundamentals
6	Workshop Practice

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

6. Workshop Practice

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Digital Electronics
3	Programming Fundamentals
4	Computer Organization
5	Communication Skills
6	Computer Laboratory

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates

Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders

Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers

Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic

Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

3. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

4. Computer Organization

Unit 1 – Computer Basics: Instruction execution; CPU organization; Performance; Data representation

Unit 2 – Digital Arithmetic: Integer arithmetic; Floating point; ALU

Unit 3 – Processor: Instruction formats; Addressing modes; Control unit; Pipelining

Unit 4 – Memory: Cache; Main memory; Virtual memory; Memory hierarchy

Unit 5 – I/O: I/O interfaces; Interrupts; DMA; Buses

5. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Computer Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 3

No.	Subject
1	Data Structures
2	Object-Oriented Programming
3	Digital Electronics
4	Database Management Systems
5	Computer Networks
6	Data Structures Laboratory

1. Data Structures

Unit 1 – Linear Structures: Arrays; Linked lists; Stacks; Queues; Circular queues
Unit 2 – Trees: Binary trees; BST; AVL trees; Heaps; Tree traversal
Unit 3 – Graphs: Graph representations; BFS; DFS; Shortest paths; Spanning trees
Unit 4 – Hashing: Hash functions; Collision handling; Hash tables
Unit 5 – Applications: Searching; Sorting; Priority queues; Complexity analysis

2. Object-Oriented Programming

Unit 1 – OOP Foundations: Classes and objects; Constructors; Methods; Encapsulation
Unit 2 – Inheritance: Inheritance models; Method overriding; Interfaces
Unit 3 – Polymorphism: Overloading; Overriding; Dynamic binding
Unit 4 – Exceptions and Files: Exception handling; Streams; File I/O
Unit 5 – Modern OOP: Generics; Collections; Packages; Basic design principles

3. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates
Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders
Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers
Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic
Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

4. Database Management Systems

Unit 1 – Database Foundations: DBMS architecture; Data models; Schema and instances; Data independence
Unit 2 – Relational Model: Relations; Keys; Constraints; Relational algebra
Unit 3 – SQL: DDL; DML; Queries; Joins; Subqueries; Views
Unit 4 – Design: ER modeling; Functional dependencies; Normalization; Decomposition
Unit 5 – Transactions: ACID; Concurrency; Locking; Deadlocks; Recovery

5. Computer Networks

Unit 1 – Networking Basics: Network types; Topologies; OSI model; TCP/IP model
Unit 2 – Data Link: Framing; Error control; MAC; Ethernet
Unit 3 – Network Layer: IP addressing; Subnetting; Routing; IPv4/IPv6
Unit 4 – Transport: TCP; UDP; Flow control; Congestion control
Unit 5 – Applications: DNS; HTTP/HTTPS; Email; DHCP; Network security basics

6. Data Structures Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice
Unit 2 – Fundamentals: Core experiments; Observation; Data collection
Unit 3 – Analysis: Calculations; Graphs; Error analysis
Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation
Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Operating Systems
2	Computer Organization
3	Web Technologies
4	Software Engineering
5	Microprocessors
6	Programming Laboratory

1. Operating Systems

Unit 1 – OS Fundamentals: OS services; System calls; Processes; Threads

Unit 2 – CPU Management: Scheduling; Synchronization; Critical sections; Deadlocks

Unit 3 – Memory: Allocation; Paging; Segmentation; Virtual memory

Unit 4 – Storage: File systems; Disk scheduling; I/O systems

Unit 5 – Security: Protection; Access control; Authentication; OS security

2. Computer Organization

Unit 1 – Computer Basics: Instruction execution; CPU organization; Performance; Data representation

Unit 2 – Digital Arithmetic: Integer arithmetic; Floating point; ALU

Unit 3 – Processor: Instruction formats; Addressing modes; Control unit; Pipelining

Unit 4 – Memory: Cache; Main memory; Virtual memory; Memory hierarchy

Unit 5 – I/O: I/O interfaces; Interrupts; DMA; Buses

3. Web Technologies

Unit 1 – Web Basics: HTTP; Browsers; HTML; Semantic markup

Unit 2 – Styling: CSS; Layouts; Responsive design; Accessibility

Unit 3 – Client Programming: JavaScript; DOM; Events; Fetch/API

Unit 4 – Server Programming: REST; Authentication; Sessions; Database integration

Unit 5 – Modern Web: Framework concepts; Security; Testing; Deployment

4. Software Engineering

Unit 1 – Software Process: Software characteristics; Process models; Agile; DevOps overview

Unit 2 – Requirements: Elicitation; SRS; Use cases; Validation

Unit 3 – Design: Architecture; Modularity; UML; Design principles

Unit 4 – Testing: Test levels; Black-box testing; White-box testing; Automation concepts

Unit 5 – Management: Estimation; Scheduling; Risk; Configuration management; Quality

5. Microprocessors

Unit 1: Architecture; Instruction set

Unit 2: Assembly programming

Unit 3: Memory and I/O

Unit 4: Interrupts; Timers

Unit 5: Interfacing; Applications

6. Programming Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Computer Networks
2	Cloud Computing
3	Cyber Security
4	Computer Graphics
5	Internet of Things
6	Project/Industrial Training

1. Computer Networks

Unit 1 – Networking Basics: Network types; Topologies; OSI model; TCP/IP model

Unit 2 – Data Link: Framing; Error control; MAC; Ethernet

Unit 3 – Network Layer: IP addressing; Subnetting; Routing; IPv4/IPv6

Unit 4 – Transport: TCP; UDP; Flow control; Congestion control

Unit 5 – Applications: DNS; HTTP/HTTPS; Email; DHCP; Network security basics

2. Cloud Computing

Unit 1 – Cloud Foundations: Service models; Deployment models; Virtualization; Elasticity

Unit 2 – Cloud Architecture: Compute; Storage; Networking; Identity

Unit 3 – Cloud Platforms: IaaS; PaaS; SaaS; Containers

Unit 4 – Cloud Security: IAM; Encryption; Security monitoring; Compliance

Unit 5 – Cloud Applications: Serverless; Autoscaling; DevOps; Cost management

3. Cyber Security

Unit 1 – Security Foundations: CIA triad; Threats; Vulnerabilities; Risk

Unit 2 – Cryptography: Symmetric encryption; Asymmetric encryption; Hashing; Digital signatures

Unit 3 – Network Security: Firewalls; IDS/IPS; VPN; Secure protocols

Unit 4 – Application Security: Authentication; Authorization; Web vulnerabilities; Secure coding

Unit 5 – Governance: Incident response; Forensics basics; Policies; Security standards

4. Computer Graphics

Unit 1 – Graphics Basics: Display systems; Scan conversion; Line and circle algorithms

Unit 2 – 2D Graphics: Transformations; Clipping; Windowing; Viewing

Unit 3 – 3D Graphics: 3D transformations; Projections; Viewing pipeline

Unit 4 – Modeling: Curves; Surfaces; Meshes; Lighting

Unit 5 – Rendering: Shading; Textures; Rasterization; Animation basics

5. Internet of Things

Unit 1 – IoT Foundations: IoT architecture; Sensors; Actuators; Edge devices

Unit 2 – Connectivity: Wi-Fi; Bluetooth; Zigbee; LPWAN; MQTT

Unit 3 – IoT Platforms: Gateways; Cloud integration; Data pipelines; Device management

Unit 4 – Analytics: Stream processing; Time-series data; Edge analytics; AI at edge

Unit 5 – Security: Device identity; Secure boot; Encryption; OTA updates

6. Project/Industrial Training

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

Semester 6

No.	Subject
1	Advanced Programming
2	Big Data Analytics
3	AI/ML Fundamentals
4	Web Technologies
5	Project
6	Industrial Training/Seminar

1. Advanced Programming

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Big Data Analytics

Unit 1 – Big Data: Characteristics; Data sources; Distributed storage; Batch vs stream

Unit 2 – Distributed Processing: MapReduce; Spark concepts; Data partitioning

Unit 3 – NoSQL: Key-value; Document; Column; Graph databases

Unit 4 – Analytics: Data preparation; Descriptive analytics; Predictive analytics; Visualization

Unit 5 – Platforms: Data lakes; Pipelines; Governance; Security

3. AI/ML Fundamentals

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Web Technologies

Unit 1 – Web Basics: HTTP; Browsers; HTML; Semantic markup

Unit 2 – Styling: CSS; Layouts; Responsive design; Accessibility

Unit 3 – Client Programming: JavaScript; DOM; Events; Fetch/API

Unit 4 – Server Programming: REST; Authentication; Sessions; Database integration

Unit 5 – Modern Web: Framework concepts; Security; Testing; Deployment

5. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

6. Industrial Training/Seminar

Unit 1 – Orientation: Organization; Safety; Tools; Professional conduct

Unit 2 – Technical Exposure: Workflow; Systems; Standards; Documentation

Unit 3 – Practical Work: Assigned tasks; Implementation; Testing

Unit 4 – Industry Practices: Quality; Version control; Teamwork; Communication

Unit 5 – Report: Learning outcomes; Case study; Summary; Presentation

Polytechnic – Information Technology

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Programming Fundamentals
6	Workshop Practice

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

6. Workshop Practice

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Programming Fundamentals
3	Digital Electronics
4	Computer Organization
5	Communication Skills
6	Computer Laboratory

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

3. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates

Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders

Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers

Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic

Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

4. Computer Organization

Unit 1 – Computer Basics: Instruction execution; CPU organization; Performance; Data representation

Unit 2 – Digital Arithmetic: Integer arithmetic; Floating point; ALU

Unit 3 – Processor: Instruction formats; Addressing modes; Control unit; Pipelining

Unit 4 – Memory: Cache; Main memory; Virtual memory; Memory hierarchy

Unit 5 – I/O: I/O interfaces; Interrupts; DMA; Buses

5. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Computer Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 3

No.	Subject
1	Data Structures
2	Object-Oriented Programming
3	Database Management Systems
4	Computer Networks
5	Web Technologies
6	Laboratory

1. Data Structures

Unit 1 – Linear Structures: Arrays; Linked lists; Stacks; Queues; Circular queues
Unit 2 – Trees: Binary trees; BST; AVL trees; Heaps; Tree traversal
Unit 3 – Graphs: Graph representations; BFS; DFS; Shortest paths; Spanning trees
Unit 4 – Hashing: Hash functions; Collision handling; Hash tables
Unit 5 – Applications: Searching; Sorting; Priority queues; Complexity analysis

2. Object-Oriented Programming

Unit 1 – OOP Foundations: Classes and objects; Constructors; Methods; Encapsulation
Unit 2 – Inheritance: Inheritance models; Method overriding; Interfaces
Unit 3 – Polymorphism: Overloading; Overriding; Dynamic binding
Unit 4 – Exceptions and Files: Exception handling; Streams; File I/O
Unit 5 – Modern OOP: Generics; Collections; Packages; Basic design principles

3. Database Management Systems

Unit 1 – Database Foundations: DBMS architecture; Data models; Schema and instances; Data independence
Unit 2 – Relational Model: Relations; Keys; Constraints; Relational algebra
Unit 3 – SQL: DDL; DML; Queries; Joins; Subqueries; Views
Unit 4 – Design: ER modeling; Functional dependencies; Normalization; Decomposition
Unit 5 – Transactions: ACID; Concurrency; Locking; Deadlocks; Recovery

4. Computer Networks

Unit 1 – Networking Basics: Network types; Topologies; OSI model; TCP/IP model
Unit 2 – Data Link: Framing; Error control; MAC; Ethernet
Unit 3 – Network Layer: IP addressing; Subnetting; Routing; IPv4/IPv6
Unit 4 – Transport: TCP; UDP; Flow control; Congestion control
Unit 5 – Applications: DNS; HTTP/HTTPS; Email; DHCP; Network security basics

5. Web Technologies

Unit 1 – Web Basics: HTTP; Browsers; HTML; Semantic markup
Unit 2 – Styling: CSS; Layouts; Responsive design; Accessibility
Unit 3 – Client Programming: JavaScript; DOM; Events; Fetch/API
Unit 4 – Server Programming: REST; Authentication; Sessions; Database integration
Unit 5 – Modern Web: Framework concepts; Security; Testing; Deployment

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice
Unit 2 – Fundamentals: Core experiments; Observation; Data collection
Unit 3 – Analysis: Calculations; Graphs; Error analysis
Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation
Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Operating Systems
2	Software Engineering
3	Web Technologies
4	Computer Networks
5	DBMS
6	Laboratory

1. Operating Systems

Unit 1 – OS Fundamentals: OS services; System calls; Processes; Threads

Unit 2 – CPU Management: Scheduling; Synchronization; Critical sections; Deadlocks

Unit 3 – Memory: Allocation; Paging; Segmentation; Virtual memory

Unit 4 – Storage: File systems; Disk scheduling; I/O systems

Unit 5 – Security: Protection; Access control; Authentication; OS security

2. Software Engineering

Unit 1 – Software Process: Software characteristics; Process models; Agile; DevOps overview

Unit 2 – Requirements: Elicitation; SRS; Use cases; Validation

Unit 3 – Design: Architecture; Modularity; UML; Design principles

Unit 4 – Testing: Test levels; Black-box testing; White-box testing; Automation concepts

Unit 5 – Management: Estimation; Scheduling; Risk; Configuration management; Quality

3. Web Technologies

Unit 1 – Web Basics: HTTP; Browsers; HTML; Semantic markup

Unit 2 – Styling: CSS; Layouts; Responsive design; Accessibility

Unit 3 – Client Programming: JavaScript; DOM; Events; Fetch/API

Unit 4 – Server Programming: REST; Authentication; Sessions; Database integration

Unit 5 – Modern Web: Framework concepts; Security; Testing; Deployment

4. Computer Networks

Unit 1 – Networking Basics: Network types; Topologies; OSI model; TCP/IP model

Unit 2 – Data Link: Framing; Error control; MAC; Ethernet

Unit 3 – Network Layer: IP addressing; Subnetting; Routing; IPv4/IPv6

Unit 4 – Transport: TCP; UDP; Flow control; Congestion control

Unit 5 – Applications: DNS; HTTP/HTTPS; Email; DHCP; Network security basics

5. DBMS

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Cloud Computing
2	Cyber Security
3	Big Data Analytics
4	Internet of Things
5	Mobile Computing
6	Project

1. Cloud Computing

Unit 1 – Cloud Foundations: Service models; Deployment models; Virtualization; Elasticity

Unit 2 – Cloud Architecture: Compute; Storage; Networking; Identity

Unit 3 – Cloud Platforms: IaaS; PaaS; SaaS; Containers

Unit 4 – Cloud Security: IAM; Encryption; Security monitoring; Compliance

Unit 5 – Cloud Applications: Serverless; Autoscaling; DevOps; Cost management

2. Cyber Security

Unit 1 – Security Foundations: CIA triad; Threats; Vulnerabilities; Risk

Unit 2 – Cryptography: Symmetric encryption; Asymmetric encryption; Hashing; Digital signatures

Unit 3 – Network Security: Firewalls; IDS/IPS; VPN; Secure protocols

Unit 4 – Application Security: Authentication; Authorization; Web vulnerabilities; Secure coding

Unit 5 – Governance: Incident response; Forensics basics; Policies; Security standards

3. Big Data Analytics

Unit 1 – Big Data: Characteristics; Data sources; Distributed storage; Batch vs stream

Unit 2 – Distributed Processing: MapReduce; Spark concepts; Data partitioning

Unit 3 – NoSQL: Key-value; Document; Column; Graph databases

Unit 4 – Analytics: Data preparation; Descriptive analytics; Predictive analytics; Visualization

Unit 5 – Platforms: Data lakes; Pipelines; Governance; Security

4. Internet of Things

Unit 1 – IoT Foundations: IoT architecture; Sensors; Actuators; Edge devices

Unit 2 – Connectivity: Wi-Fi; Bluetooth; Zigbee; LPWAN; MQTT

Unit 3 – IoT Platforms: Gateways; Cloud integration; Data pipelines; Device management

Unit 4 – Analytics: Stream processing; Time-series data; Edge analytics; AI at edge

Unit 5 – Security: Device identity; Secure boot; Encryption; OTA updates

5. Mobile Computing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

Semester 6

No.	Subject
1	Advanced Web Development
2	Data Analytics
3	AI/ML Fundamentals
4	Cloud Computing
5	Project
6	Industrial Training/Seminar

1. Advanced Web Development

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Data Analytics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. AI/ML Fundamentals

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Cloud Computing

Unit 1 – Cloud Foundations: Service models; Deployment models; Virtualization; Elasticity

Unit 2 – Cloud Architecture: Compute; Storage; Networking; Identity

Unit 3 – Cloud Platforms: IaaS; PaaS; SaaS; Containers

Unit 4 – Cloud Security: IAM; Encryption; Security monitoring; Compliance

Unit 5 – Cloud Applications: Serverless; Autoscaling; DevOps; Cost management

5. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

6. Industrial Training/Seminar

Unit 1 – Orientation: Organization; Safety; Tools; Professional conduct

Unit 2 – Technical Exposure: Workflow; Systems; Standards; Documentation

Unit 3 – Practical Work: Assigned tasks; Implementation; Testing

Unit 4 – Industry Practices: Quality; Version control; Teamwork; Communication

Unit 5 – Report: Learning outcomes; Case study; Summary; Presentation

Polytechnic – Electronics & Communication Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Basic Electrical Engineering
6	Workshop Practice

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Basic Electrical Engineering

Unit 1 – DC Circuits: Voltage and current; Ohm's law; KCL/KVL; Network reduction

Unit 2 – AC Circuits: Sinusoids; Phasors; Impedance; Power factor; Three-phase systems

Unit 3 – Machines: Transformers; DC machines; Induction motors; Synchronous machines

Unit 4 – Measurements: Meters; Bridges; Errors; Power and energy measurement

Unit 5 – Safety and Power: Earthing; Protection; Energy systems; Electrical safety

6. Workshop Practice

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Circuit Theory
3	Digital Electronics
4	Electronic Devices
5	Communication Skills
6	Electronics Laboratory

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Circuit Theory

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates

Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders

Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers

Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic

Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

4. Electronic Devices

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Electronics Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 3

No.	Subject
1	Analog Electronics
2	Digital Electronics
3	Signals and Systems
4	Microprocessors
5	Network Theory
6	Electronics Laboratory

1. Analog Electronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates

Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders

Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers

Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic

Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

3. Signals and Systems

Unit 1 – Signals: Continuous/discrete signals; Energy and power; Basic operations

Unit 2 – Systems: LTI systems; Convolution; Causality; Stability

Unit 3 – Fourier Analysis: Fourier series; Fourier transform; Properties

Unit 4 – Laplace/Z Transform: Laplace transform; Z transform; ROC; Applications

Unit 5 – Sampling: Sampling theorem; Reconstruction; Aliasing; Practical systems

4. Microprocessors

Unit 1: Architecture; Instruction set

Unit 2: Assembly programming

Unit 3: Memory and I/O

Unit 4: Interrupts; Timers

Unit 5: Interfacing; Applications

5. Network Theory

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Electronics Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Communication Systems
2	Microcontrollers
3	Electromagnetic Theory
4	Control Systems
5	Digital Signal Processing
6	Laboratory

1. Communication Systems

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Microcontrollers

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Electromagnetic Theory

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Control Systems

Unit 1: Modeling; Transfer functions; Block diagrams

Unit 2: Time response; Stability

Unit 3: Root locus; Frequency response

Unit 4: State-space methods

Unit 5: Controllers; Industrial applications

5. Digital Signal Processing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Embedded Systems
2	VLSI Basics
3	Wireless Communication
4	IoT
5	Antenna Basics
6	Project

1. Embedded Systems

Unit 1: Embedded architecture; Microcontrollers

Unit 2: Real-time concepts; RTOS

Unit 3: Interfaces; Peripherals

Unit 4: Embedded networking

Unit 5: Testing; Deployment

2. VLSI Basics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Wireless Communication

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. IoT

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Antenna Basics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

Semester 6

No.	Subject
1	Advanced Communication
2	Embedded Applications
3	Industrial Electronics
4	Project
5	Industrial Training
6	Seminar

1. Advanced Communication

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Embedded Applications

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Industrial Electronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

5. Industrial Training

Unit 1 – Orientation: Organization; Safety; Tools; Professional conduct

Unit 2 – Technical Exposure: Workflow; Systems; Standards; Documentation

Unit 3 – Practical Work: Assigned tasks; Implementation; Testing

Unit 4 – Industry Practices: Quality; Version control; Teamwork; Communication

Unit 5 – Report: Learning outcomes; Case study; Summary; Presentation

6. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

Polytechnic – Electrical & Electronics Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Basic Electrical Engineering
6	Workshop Practice

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Basic Electrical Engineering

Unit 1 – DC Circuits: Voltage and current; Ohm's law; KCL/KVL; Network reduction

Unit 2 – AC Circuits: Sinusoids; Phasors; Impedance; Power factor; Three-phase systems

Unit 3 – Machines: Transformers; DC machines; Induction motors; Synchronous machines

Unit 4 – Measurements: Meters; Bridges; Errors; Power and energy measurement

Unit 5 – Safety and Power: Earthing; Protection; Energy systems; Electrical safety

6. Workshop Practice

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Circuit Theory
3	Digital Electronics
4	Electrical Measurements
5	Communication Skills
6	Electrical Laboratory

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Circuit Theory

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates

Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders

Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers

Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic

Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

4. Electrical Measurements

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Electrical Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 3

No.	Subject
1	Electrical Machines I
2	Power Systems I
3	Analog Electronics
4	Electrical Measurements
5	Microprocessors
6	Laboratory

1. Electrical Machines I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Power Systems I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Analog Electronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Electrical Measurements

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Microprocessors

Unit 1: Architecture; Instruction set

Unit 2: Assembly programming

Unit 3: Memory and I/O

Unit 4: Interrupts; Timers

Unit 5: Interfacing; Applications

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Electrical Machines II
2	Power Systems II
3	Control Systems
4	Power Electronics
5	Microcontrollers
6	Laboratory

1. Electrical Machines II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Power Systems II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Control Systems

Unit 1: Modeling; Transfer functions; Block diagrams

Unit 2: Time response; Stability

Unit 3: Root locus; Frequency response

Unit 4: State-space methods

Unit 5: Controllers; Industrial applications

4. Power Electronics

Unit 1: Power semiconductor devices

Unit 2: Controlled rectifiers

Unit 3: DC-DC converters

Unit 4: Inverters

Unit 5: Drives; Applications

5. Microcontrollers

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Renewable Energy
2	Industrial Drives
3	Protection
4	Embedded/IoT Applications
5	Electrical Design
6	Project

1. Renewable Energy

Unit 1: Solar energy

Unit 2: Wind energy

Unit 3: Hydro and biomass

Unit 4: Fuel cells; Geothermal

Unit 5: Grid integration; Storage

2. Industrial Drives

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Protection

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Embedded/IoT Applications

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Electrical Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

Semester 6

No.	Subject
1	Advanced Power Systems
2	Electric Vehicles
3	Smart Grid Basics
4	Project
5	Industrial Training
6	Seminar

1. Advanced Power Systems

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Electric Vehicles

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Smart Grid Basics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

5. Industrial Training

Unit 1 – Orientation: Organization; Safety; Tools; Professional conduct

Unit 2 – Technical Exposure: Workflow; Systems; Standards; Documentation

Unit 3 – Practical Work: Assigned tasks; Implementation; Testing

Unit 4 – Industry Practices: Quality; Version control; Teamwork; Communication

Unit 5 – Report: Learning outcomes; Case study; Summary; Presentation

6. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

Polytechnic – Mechanical Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Engineering Mechanics
6	Workshop Practice

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Engineering Mechanics

Unit 1 – Statics: Forces and moments; Free-body diagrams; Equilibrium; Friction

Unit 2 – Structures: Trusses; Frames; Centroids; Moments of inertia

Unit 3 – Kinematics: Particle motion; Rigid body motion; Relative motion

Unit 4 – Dynamics: Newton's laws; Work-energy; Impulse-momentum

Unit 5 – Applications: Virtual work; Simple machines; Engineering applications

6. Workshop Practice

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Strength of Materials
3	Thermodynamics
4	Manufacturing Processes
5	Materials Science
6	Mechanical Laboratory

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Strength of Materials

Unit 1 – Stress and Strain: Normal stress; Shear stress; Strain; Hooke's law; Elastic constants

Unit 2 – Bending: Shear force; Bending moment; Bending stress; Beam deflection

Unit 3 – Torsion: Torsion equation; Shafts; Power transmission

Unit 4 – Columns: Buckling; Euler theory; Rankine theory

Unit 5 – Failure: Principal stresses; Mohr's circle; Failure theories

3. Thermodynamics

Unit 1 – Fundamentals: Systems and properties; Processes; Zeroth law; First law

Unit 2 – Second Law: Entropy; Reversibility; Carnot cycle; Clausius inequality

Unit 3 – Properties: Pure substances; Steam tables; Ideal gases; Real gases

Unit 4 – Cycles: Otto; Diesel; Dual; Rankine cycles

Unit 5 – Applications: Nozzles; Compressors; Turbines; Heat engines

4. Manufacturing Processes

Unit 1: Casting

Unit 2: Forming

Unit 3: Machining

Unit 4: Joining

Unit 5: Advanced manufacturing

5. Materials Science

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Mechanical Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 3

No.	Subject
1	Fluid Mechanics
2	Theory of Machines
3	Machine Drawing
4	Manufacturing Technology
5	Thermal Engineering
6	Laboratory

1. Fluid Mechanics

Unit 1: Fluid properties; Pressure

Unit 2: Fluid kinematics

Unit 3: Bernoulli equation

Unit 4: Pipe flow

Unit 5: Boundary layers; Turbomachinery basics

2. Theory of Machines

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Machine Drawing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Manufacturing Technology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Thermal Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Machine Design
2	Heat Transfer
3	CAD/CAM
4	Metrology
5	Production Technology
6	Laboratory

1. Machine Design

Unit 1: Design principles; Failure theories

Unit 2: Shafts; Keys; Couplings

Unit 3: Gears

Unit 4: Bearings; Springs

Unit 5: Design optimization; CAD

2. Heat Transfer

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. CAD/CAM

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Metrology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Production Technology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	CNC Technology
2	Industrial Engineering
3	Refrigeration & Air Conditioning
4	Automobile Basics
5	Robotics/Mechatronics
6	Project

1. CNC Technology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Industrial Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Refrigeration & Air Conditioning

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Automobile Basics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Robotics/Mechatronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

Semester 6

No.	Subject
1	Advanced Manufacturing
2	Maintenance
3	Quality Control
4	Project
5	Industrial Training
6	Seminar

1. Advanced Manufacturing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Maintenance

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Quality Control

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

5. Industrial Training

Unit 1 – Orientation: Organization; Safety; Tools; Professional conduct

Unit 2 – Technical Exposure: Workflow; Systems; Standards; Documentation

Unit 3 – Practical Work: Assigned tasks; Implementation; Testing

Unit 4 – Industry Practices: Quality; Version control; Teamwork; Communication

Unit 5 – Report: Learning outcomes; Case study; Summary; Presentation

6. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

Polytechnic – Civil Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Engineering Mechanics
6	Workshop Practice

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Engineering Mechanics

Unit 1 – Statics: Forces and moments; Free-body diagrams; Equilibrium; Friction

Unit 2 – Structures: Trusses; Frames; Centroids; Moments of inertia

Unit 3 – Kinematics: Particle motion; Rigid body motion; Relative motion

Unit 4 – Dynamics: Newton's laws; Work-energy; Impulse-momentum

Unit 5 – Applications: Virtual work; Simple machines; Engineering applications

6. Workshop Practice

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Surveying I
3	Building Materials
4	Strength of Materials
5	Communication Skills
6	Civil Laboratory

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Surveying I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Building Materials

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Strength of Materials

Unit 1 – Stress and Strain: Normal stress; Shear stress; Strain; Hooke's law; Elastic constants

Unit 2 – Bending: Shear force; Bending moment; Bending stress; Beam deflection

Unit 3 – Torsion: Torsion equation; Shafts; Power transmission

Unit 4 – Columns: Buckling; Euler theory; Rankine theory

Unit 5 – Failure: Principal stresses; Mohr's circle; Failure theories

5. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Civil Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 3

No.	Subject
1	Surveying II
2	Fluid Mechanics
3	Concrete Technology
4	Structural Analysis
5	Geotechnical Engineering I
6	Laboratory

1. Surveying II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Fluid Mechanics

Unit 1: Fluid properties; Pressure

Unit 2: Fluid kinematics

Unit 3: Bernoulli equation

Unit 4: Pipe flow

Unit 5: Boundary layers; Turbomachinery basics

3. Concrete Technology

Unit 1: Cement; Aggregates

Unit 2: Fresh concrete

Unit 3: Mix design

Unit 4: Strength and durability

Unit 5: Special concretes

4. Structural Analysis

Unit 1: Determinate structures

Unit 2: Deflection

Unit 3: Indeterminate structures

Unit 4: Matrix methods

Unit 5: Influence lines; Applications

5. Geotechnical Engineering I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Geotechnical Engineering II
2	RCC Design
3	Transportation Engineering
4	Hydraulics
5	Environmental Engineering
6	Laboratory

1. Geotechnical Engineering II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. RCC Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Transportation Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Hydraulics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Environmental Engineering

Unit 1: Water quality

Unit 2: Water treatment

Unit 3: Wastewater treatment

Unit 4: Air pollution

Unit 5: Solid waste management

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Steel Structures
2	Water Resources
3	Construction Management
4	Highway Engineering
5	Estimation and Costing
6	Project

1. Steel Structures

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Water Resources

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Construction Management

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Highway Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Estimation and Costing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

Semester 6

No.	Subject
1	Advanced Structural Topics
2	Urban Infrastructure
3	Project
4	Industrial Training
5	Seminar
6	Professional Practice

1. Advanced Structural Topics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Urban Infrastructure

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

4. Industrial Training

Unit 1 – Orientation: Organization; Safety; Tools; Professional conduct

Unit 2 – Technical Exposure: Workflow; Systems; Standards; Documentation

Unit 3 – Practical Work: Assigned tasks; Implementation; Testing

Unit 4 – Industry Practices: Quality; Version control; Teamwork; Communication

Unit 5 – Report: Learning outcomes; Case study; Summary; Presentation

5. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

6. Professional Practice

Unit 1 – Professional Skills: Communication; Teamwork; Time management

Unit 2 – Ethics: Professional ethics; Responsible engineering; Safety

Unit 3 – Industry: Standards; Quality; Documentation

Unit 4 – Career Skills: Resume; Interview; Portfolio; Continuous learning

Unit 5 – Practice: Case studies; Problem solving; Professional conduct

Polytechnic – Automobile Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Engineering Mechanics
6	Workshop Practice

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Engineering Mechanics

Unit 1 – Statics: Forces and moments; Free-body diagrams; Equilibrium; Friction

Unit 2 – Structures: Trusses; Frames; Centroids; Moments of inertia

Unit 3 – Kinematics: Particle motion; Rigid body motion; Relative motion

Unit 4 – Dynamics: Newton's laws; Work-energy; Impulse-momentum

Unit 5 – Applications: Virtual work; Simple machines; Engineering applications

6. Workshop Practice

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Thermodynamics
3	Automotive Materials
4	Automotive Electrical Basics
5	Manufacturing Processes
6	Laboratory

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Thermodynamics

Unit 1 – Fundamentals: Systems and properties; Processes; Zeroth law; First law

Unit 2 – Second Law: Entropy; Reversibility; Carnot cycle; Clausius inequality

Unit 3 – Properties: Pure substances; Steam tables; Ideal gases; Real gases

Unit 4 – Cycles: Otto; Diesel; Dual; Rankine cycles

Unit 5 – Applications: Nozzles; Compressors; Turbines; Heat engines

3. Automotive Materials

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Automotive Electrical Basics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Manufacturing Processes

Unit 1: Casting

Unit 2: Forming

Unit 3: Machining

Unit 4: Joining

Unit 5: Advanced manufacturing

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 3

No.	Subject
1	Automotive Engines
2	Automotive Chassis
3	Automotive Transmission
4	Fluid Mechanics
5	Machine Design
6	Laboratory

1. Automotive Engines

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Automotive Chassis

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Automotive Transmission

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Fluid Mechanics

Unit 1: Fluid properties; Pressure

Unit 2: Fluid kinematics

Unit 3: Bernoulli equation

Unit 4: Pipe flow

Unit 5: Boundary layers; Turbomachinery basics

5. Machine Design

Unit 1: Design principles; Failure theories

Unit 2: Shafts; Keys; Couplings

Unit 3: Gears

Unit 4: Bearings; Springs

Unit 5: Design optimization; CAD

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Vehicle Electrical Systems
2	Automotive Electronics
3	Vehicle Dynamics
4	Automotive Manufacturing
5	Maintenance Engineering
6	Laboratory

1. Vehicle Electrical Systems

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Automotive Electronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Vehicle Dynamics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Automotive Manufacturing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Maintenance Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Emission Control
2	Electric Vehicles
3	Hybrid Vehicles
4	Automotive Diagnostics
5	CAD/CAM
6	Project

1. Emission Control

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Electric Vehicles

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Hybrid Vehicles

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Automotive Diagnostics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. CAD/CAM

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

Semester 6

No.	Subject
1	Advanced Vehicle Systems
2	EV Diagnostics
3	Project
4	Industrial Training
5	Seminar
6	Service Management

1. Advanced Vehicle Systems

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. EV Diagnostics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

4. Industrial Training

Unit 1 – Orientation: Organization; Safety; Tools; Professional conduct

Unit 2 – Technical Exposure: Workflow; Systems; Standards; Documentation

Unit 3 – Practical Work: Assigned tasks; Implementation; Testing

Unit 4 – Industry Practices: Quality; Version control; Teamwork; Communication

Unit 5 – Report: Learning outcomes; Case study; Summary; Presentation

5. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

6. Service Management

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Polytechnic – Mechatronics

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Engineering Mechanics
6	Workshop Practice

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Engineering Mechanics

Unit 1 – Statics: Forces and moments; Free-body diagrams; Equilibrium; Friction

Unit 2 – Structures: Trusses; Frames; Centroids; Moments of inertia

Unit 3 – Kinematics: Particle motion; Rigid body motion; Relative motion

Unit 4 – Dynamics: Newton's laws; Work-energy; Impulse-momentum

Unit 5 – Applications: Virtual work; Simple machines; Engineering applications

6. Workshop Practice

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Circuit Theory
3	Digital Electronics
4	Manufacturing Processes
5	Programming Fundamentals
6	Laboratory

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Circuit Theory

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates

Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders

Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers

Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic

Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

4. Manufacturing Processes

Unit 1: Casting

Unit 2: Forming

Unit 3: Machining

Unit 4: Joining

Unit 5: Advanced manufacturing

5. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 3

No.	Subject
1	Sensors and Transducers
2	Microcontrollers
3	Pneumatics and Hydraulics
4	Control Systems
5	Mechanical Design
6	Laboratory

1. Sensors and Transducers

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Microcontrollers

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Pneumatics and Hydraulics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Control Systems

Unit 1: Modeling; Transfer functions; Block diagrams

Unit 2: Time response; Stability

Unit 3: Root locus; Frequency response

Unit 4: State-space methods

Unit 5: Controllers; Industrial applications

5. Mechanical Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	PLC and Automation
2	Robotics
3	Embedded Systems
4	Industrial Instrumentation
5	CAD/CAM
6	Laboratory

1. PLC and Automation

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Robotics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Embedded Systems

Unit 1: Embedded architecture; Microcontrollers

Unit 2: Real-time concepts; RTOS

Unit 3: Interfaces; Peripherals

Unit 4: Embedded networking

Unit 5: Testing; Deployment

4. Industrial Instrumentation

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. CAD/CAM

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Industrial Automation
2	IoT
3	Machine Vision Basics
4	Motion Control
5	Mechatronic System Design
6	Project

1. Industrial Automation

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. IoT

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Machine Vision Basics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Motion Control

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Mechatronic System Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

Semester 6

No.	Subject
1	Advanced Robotics
2	Smart Manufacturing
3	Project
4	Industrial Training
5	Seminar
6	Maintenance

1. Advanced Robotics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Smart Manufacturing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

4. Industrial Training

Unit 1 – Orientation: Organization; Safety; Tools; Professional conduct

Unit 2 – Technical Exposure: Workflow; Systems; Standards; Documentation

Unit 3 – Practical Work: Assigned tasks; Implementation; Testing

Unit 4 – Industry Practices: Quality; Version control; Teamwork; Communication

Unit 5 – Report: Learning outcomes; Case study; Summary; Presentation

5. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

6. Maintenance

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Polytechnic – Artificial Intelligence & Machine Learning

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Programming Fundamentals
6	Workshop Practice

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

6. Workshop Practice

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Programming Fundamentals
3	Basic Electronics
4	Communication Skills
5	Engineering Mechanics
6	Laboratory

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

3. Basic Electronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Engineering Mechanics

Unit 1 – Statics: Forces and moments; Free-body diagrams; Equilibrium; Friction

Unit 2 – Structures: Trusses; Frames; Centroids; Moments of inertia

Unit 3 – Kinematics: Particle motion; Rigid body motion; Relative motion

Unit 4 – Dynamics: Newton's laws; Work-energy; Impulse-momentum

Unit 5 – Applications: Virtual work; Simple machines; Engineering applications

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 3

No.	Subject
1	Data Structures
2	Digital Electronics
3	Database Management Systems
4	Computer Networks
5	Core Technical Subject I
6	Laboratory

1. Data Structures

Unit 1 – Linear Structures: Arrays; Linked lists; Stacks; Queues; Circular queues
Unit 2 – Trees: Binary trees; BST; AVL trees; Heaps; Tree traversal
Unit 3 – Graphs: Graph representations; BFS; DFS; Shortest paths; Spanning trees
Unit 4 – Hashing: Hash functions; Collision handling; Hash tables
Unit 5 – Applications: Searching; Sorting; Priority queues; Complexity analysis

2. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates
Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders
Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers
Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic
Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

3. Database Management Systems

Unit 1 – Database Foundations: DBMS architecture; Data models; Schema and instances; Data independence
Unit 2 – Relational Model: Relations; Keys; Constraints; Relational algebra
Unit 3 – SQL: DDL; DML; Queries; Joins; Subqueries; Views
Unit 4 – Design: ER modeling; Functional dependencies; Normalization; Decomposition
Unit 5 – Transactions: ACID; Concurrency; Locking; Deadlocks; Recovery

4. Computer Networks

Unit 1 – Networking Basics: Network types; Topologies; OSI model; TCP/IP model
Unit 2 – Data Link: Framing; Error control; MAC; Ethernet
Unit 3 – Network Layer: IP addressing; Subnetting; Routing; IPv4/IPv6
Unit 4 – Transport: TCP; UDP; Flow control; Congestion control
Unit 5 – Applications: DNS; HTTP/HTTPS; Email; DHCP; Network security basics

5. Core Technical Subject I

Unit 1 – Foundations: Core concepts; Terminology; Architecture
Unit 2 – Methods: Algorithms; Design methods; Implementation
Unit 3 – Applications: Use cases; Tools; Case studies
Unit 4 – Evaluation: Performance; Testing; Security/quality
Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice
Unit 2 – Fundamentals: Core experiments; Observation; Data collection
Unit 3 – Analysis: Calculations; Graphs; Error analysis
Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation
Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Operating Systems
2	Software Engineering
3	Core Technical Subject II
4	Core Technical Subject III
5	Core Technical Subject IV
6	Laboratory

1. Operating Systems

Unit 1 – OS Fundamentals: OS services; System calls; Processes; Threads

Unit 2 – CPU Management: Scheduling; Synchronization; Critical sections; Deadlocks

Unit 3 – Memory: Allocation; Paging; Segmentation; Virtual memory

Unit 4 – Storage: File systems; Disk scheduling; I/O systems

Unit 5 – Security: Protection; Access control; Authentication; OS security

2. Software Engineering

Unit 1 – Software Process: Software characteristics; Process models; Agile; DevOps overview

Unit 2 – Requirements: Elicitation; SRS; Use cases; Validation

Unit 3 – Design: Architecture; Modularity; UML; Design principles

Unit 4 – Testing: Test levels; Black-box testing; White-box testing; Automation concepts

Unit 5 – Management: Estimation; Scheduling; Risk; Configuration management; Quality

3. Core Technical Subject II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Core Technical Subject III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Core Technical Subject IV

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Advanced Core I
2	Advanced Core II
3	Elective I
4	Elective II
5	Industry Skills
6	Project

1. Advanced Core I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Core II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Industry Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

Semester 6

No.	Subject
1	Advanced Core III
2	Elective III
3	Project
4	Industrial Training
5	Seminar
6	Professional Practice

1. Advanced Core III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Elective III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

4. Industrial Training

Unit 1 – Orientation: Organization; Safety; Tools; Professional conduct

Unit 2 – Technical Exposure: Workflow; Systems; Standards; Documentation

Unit 3 – Practical Work: Assigned tasks; Implementation; Testing

Unit 4 – Industry Practices: Quality; Version control; Teamwork; Communication

Unit 5 – Report: Learning outcomes; Case study; Summary; Presentation

5. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

6. Professional Practice

Unit 1 – Professional Skills: Communication; Teamwork; Time management

Unit 2 – Ethics: Professional ethics; Responsible engineering; Safety

Unit 3 – Industry: Standards; Quality; Documentation

Unit 4 – Career Skills: Resume; Interview; Portfolio; Continuous learning

Unit 5 – Practice: Case studies; Problem solving; Professional conduct

Polytechnic – Data Science

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Programming Fundamentals
6	Workshop Practice

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

6. Workshop Practice

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Programming Fundamentals
3	Basic Electronics
4	Communication Skills
5	Engineering Mechanics
6	Laboratory

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

3. Basic Electronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Engineering Mechanics

Unit 1 – Statics: Forces and moments; Free-body diagrams; Equilibrium; Friction

Unit 2 – Structures: Trusses; Frames; Centroids; Moments of inertia

Unit 3 – Kinematics: Particle motion; Rigid body motion; Relative motion

Unit 4 – Dynamics: Newton's laws; Work-energy; Impulse-momentum

Unit 5 – Applications: Virtual work; Simple machines; Engineering applications

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 3

No.	Subject
1	Data Structures
2	Digital Electronics
3	Database Management Systems
4	Computer Networks
5	Core Technical Subject I
6	Laboratory

1. Data Structures

Unit 1 – Linear Structures: Arrays; Linked lists; Stacks; Queues; Circular queues
Unit 2 – Trees: Binary trees; BST; AVL trees; Heaps; Tree traversal
Unit 3 – Graphs: Graph representations; BFS; DFS; Shortest paths; Spanning trees
Unit 4 – Hashing: Hash functions; Collision handling; Hash tables
Unit 5 – Applications: Searching; Sorting; Priority queues; Complexity analysis

2. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates
Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders
Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers
Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic
Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

3. Database Management Systems

Unit 1 – Database Foundations: DBMS architecture; Data models; Schema and instances; Data independence
Unit 2 – Relational Model: Relations; Keys; Constraints; Relational algebra
Unit 3 – SQL: DDL; DML; Queries; Joins; Subqueries; Views
Unit 4 – Design: ER modeling; Functional dependencies; Normalization; Decomposition
Unit 5 – Transactions: ACID; Concurrency; Locking; Deadlocks; Recovery

4. Computer Networks

Unit 1 – Networking Basics: Network types; Topologies; OSI model; TCP/IP model
Unit 2 – Data Link: Framing; Error control; MAC; Ethernet
Unit 3 – Network Layer: IP addressing; Subnetting; Routing; IPv4/IPv6
Unit 4 – Transport: TCP; UDP; Flow control; Congestion control
Unit 5 – Applications: DNS; HTTP/HTTPS; Email; DHCP; Network security basics

5. Core Technical Subject I

Unit 1 – Foundations: Core concepts; Terminology; Architecture
Unit 2 – Methods: Algorithms; Design methods; Implementation
Unit 3 – Applications: Use cases; Tools; Case studies
Unit 4 – Evaluation: Performance; Testing; Security/quality
Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice
Unit 2 – Fundamentals: Core experiments; Observation; Data collection
Unit 3 – Analysis: Calculations; Graphs; Error analysis
Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation
Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Operating Systems
2	Software Engineering
3	Core Technical Subject II
4	Core Technical Subject III
5	Core Technical Subject IV
6	Laboratory

1. Operating Systems

Unit 1 – OS Fundamentals: OS services; System calls; Processes; Threads

Unit 2 – CPU Management: Scheduling; Synchronization; Critical sections; Deadlocks

Unit 3 – Memory: Allocation; Paging; Segmentation; Virtual memory

Unit 4 – Storage: File systems; Disk scheduling; I/O systems

Unit 5 – Security: Protection; Access control; Authentication; OS security

2. Software Engineering

Unit 1 – Software Process: Software characteristics; Process models; Agile; DevOps overview

Unit 2 – Requirements: Elicitation; SRS; Use cases; Validation

Unit 3 – Design: Architecture; Modularity; UML; Design principles

Unit 4 – Testing: Test levels; Black-box testing; White-box testing; Automation concepts

Unit 5 – Management: Estimation; Scheduling; Risk; Configuration management; Quality

3. Core Technical Subject II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Core Technical Subject III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Core Technical Subject IV

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Advanced Core I
2	Advanced Core II
3	Elective I
4	Elective II
5	Industry Skills
6	Project

1. Advanced Core I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Core II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Industry Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

Semester 6

No.	Subject
1	Advanced Core III
2	Elective III
3	Project
4	Industrial Training
5	Seminar
6	Professional Practice

1. Advanced Core III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Elective III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

4. Industrial Training

Unit 1 – Orientation: Organization; Safety; Tools; Professional conduct

Unit 2 – Technical Exposure: Workflow; Systems; Standards; Documentation

Unit 3 – Practical Work: Assigned tasks; Implementation; Testing

Unit 4 – Industry Practices: Quality; Version control; Teamwork; Communication

Unit 5 – Report: Learning outcomes; Case study; Summary; Presentation

5. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

6. Professional Practice

Unit 1 – Professional Skills: Communication; Teamwork; Time management

Unit 2 – Ethics: Professional ethics; Responsible engineering; Safety

Unit 3 – Industry: Standards; Quality; Documentation

Unit 4 – Career Skills: Resume; Interview; Portfolio; Continuous learning

Unit 5 – Practice: Case studies; Problem solving; Professional conduct

Polytechnic – Biomedical Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Programming Fundamentals
6	Workshop Practice

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

6. Workshop Practice

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Programming Fundamentals
3	Basic Electronics
4	Communication Skills
5	Engineering Mechanics
6	Laboratory

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

3. Basic Electronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Engineering Mechanics

Unit 1 – Statics: Forces and moments; Free-body diagrams; Equilibrium; Friction

Unit 2 – Structures: Trusses; Frames; Centroids; Moments of inertia

Unit 3 – Kinematics: Particle motion; Rigid body motion; Relative motion

Unit 4 – Dynamics: Newton's laws; Work-energy; Impulse-momentum

Unit 5 – Applications: Virtual work; Simple machines; Engineering applications

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 3

No.	Subject
1	Data Structures
2	Digital Electronics
3	Database Management Systems
4	Computer Networks
5	Core Technical Subject I
6	Laboratory

1. Data Structures

Unit 1 – Linear Structures: Arrays; Linked lists; Stacks; Queues; Circular queues
Unit 2 – Trees: Binary trees; BST; AVL trees; Heaps; Tree traversal
Unit 3 – Graphs: Graph representations; BFS; DFS; Shortest paths; Spanning trees
Unit 4 – Hashing: Hash functions; Collision handling; Hash tables
Unit 5 – Applications: Searching; Sorting; Priority queues; Complexity analysis

2. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates
Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders
Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers
Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic
Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

3. Database Management Systems

Unit 1 – Database Foundations: DBMS architecture; Data models; Schema and instances; Data independence
Unit 2 – Relational Model: Relations; Keys; Constraints; Relational algebra
Unit 3 – SQL: DDL; DML; Queries; Joins; Subqueries; Views
Unit 4 – Design: ER modeling; Functional dependencies; Normalization; Decomposition
Unit 5 – Transactions: ACID; Concurrency; Locking; Deadlocks; Recovery

4. Computer Networks

Unit 1 – Networking Basics: Network types; Topologies; OSI model; TCP/IP model
Unit 2 – Data Link: Framing; Error control; MAC; Ethernet
Unit 3 – Network Layer: IP addressing; Subnetting; Routing; IPv4/IPv6
Unit 4 – Transport: TCP; UDP; Flow control; Congestion control
Unit 5 – Applications: DNS; HTTP/HTTPS; Email; DHCP; Network security basics

5. Core Technical Subject I

Unit 1 – Foundations: Core concepts; Terminology; Architecture
Unit 2 – Methods: Algorithms; Design methods; Implementation
Unit 3 – Applications: Use cases; Tools; Case studies
Unit 4 – Evaluation: Performance; Testing; Security/quality
Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice
Unit 2 – Fundamentals: Core experiments; Observation; Data collection
Unit 3 – Analysis: Calculations; Graphs; Error analysis
Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation
Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Operating Systems
2	Software Engineering
3	Core Technical Subject II
4	Core Technical Subject III
5	Core Technical Subject IV
6	Laboratory

1. Operating Systems

Unit 1 – OS Fundamentals: OS services; System calls; Processes; Threads

Unit 2 – CPU Management: Scheduling; Synchronization; Critical sections; Deadlocks

Unit 3 – Memory: Allocation; Paging; Segmentation; Virtual memory

Unit 4 – Storage: File systems; Disk scheduling; I/O systems

Unit 5 – Security: Protection; Access control; Authentication; OS security

2. Software Engineering

Unit 1 – Software Process: Software characteristics; Process models; Agile; DevOps overview

Unit 2 – Requirements: Elicitation; SRS; Use cases; Validation

Unit 3 – Design: Architecture; Modularity; UML; Design principles

Unit 4 – Testing: Test levels; Black-box testing; White-box testing; Automation concepts

Unit 5 – Management: Estimation; Scheduling; Risk; Configuration management; Quality

3. Core Technical Subject II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Core Technical Subject III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Core Technical Subject IV

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Advanced Core I
2	Advanced Core II
3	Elective I
4	Elective II
5	Industry Skills
6	Project

1. Advanced Core I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Core II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Industry Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

Semester 6

No.	Subject
1	Advanced Core III
2	Elective III
3	Project
4	Industrial Training
5	Seminar
6	Professional Practice

1. Advanced Core III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Elective III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

4. Industrial Training

Unit 1 – Orientation: Organization; Safety; Tools; Professional conduct

Unit 2 – Technical Exposure: Workflow; Systems; Standards; Documentation

Unit 3 – Practical Work: Assigned tasks; Implementation; Testing

Unit 4 – Industry Practices: Quality; Version control; Teamwork; Communication

Unit 5 – Report: Learning outcomes; Case study; Summary; Presentation

5. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

6. Professional Practice

Unit 1 – Professional Skills: Communication; Teamwork; Time management

Unit 2 – Ethics: Professional ethics; Responsible engineering; Safety

Unit 3 – Industry: Standards; Quality; Documentation

Unit 4 – Career Skills: Resume; Interview; Portfolio; Continuous learning

Unit 5 – Practice: Case studies; Problem solving; Professional conduct

Polytechnic – Chemical Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Programming Fundamentals
6	Workshop Practice

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

6. Workshop Practice

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Programming Fundamentals
3	Basic Electronics
4	Communication Skills
5	Engineering Mechanics
6	Laboratory

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

3. Basic Electronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Engineering Mechanics

Unit 1 – Statics: Forces and moments; Free-body diagrams; Equilibrium; Friction

Unit 2 – Structures: Trusses; Frames; Centroids; Moments of inertia

Unit 3 – Kinematics: Particle motion; Rigid body motion; Relative motion

Unit 4 – Dynamics: Newton's laws; Work-energy; Impulse-momentum

Unit 5 – Applications: Virtual work; Simple machines; Engineering applications

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 3

No.	Subject
1	Data Structures
2	Digital Electronics
3	Database Management Systems
4	Computer Networks
5	Core Technical Subject I
6	Laboratory

1. Data Structures

Unit 1 – Linear Structures: Arrays; Linked lists; Stacks; Queues; Circular queues

Unit 2 – Trees: Binary trees; BST; AVL trees; Heaps; Tree traversal

Unit 3 – Graphs: Graph representations; BFS; DFS; Shortest paths; Spanning trees

Unit 4 – Hashing: Hash functions; Collision handling; Hash tables

Unit 5 – Applications: Searching; Sorting; Priority queues; Complexity analysis

2. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates

Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders

Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers

Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic

Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

3. Database Management Systems

Unit 1 – Database Foundations: DBMS architecture; Data models; Schema and instances; Data independence

Unit 2 – Relational Model: Relations; Keys; Constraints; Relational algebra

Unit 3 – SQL: DDL; DML; Queries; Joins; Subqueries; Views

Unit 4 – Design: ER modeling; Functional dependencies; Normalization; Decomposition

Unit 5 – Transactions: ACID; Concurrency; Locking; Deadlocks; Recovery

4. Computer Networks

Unit 1 – Networking Basics: Network types; Topologies; OSI model; TCP/IP model

Unit 2 – Data Link: Framing; Error control; MAC; Ethernet

Unit 3 – Network Layer: IP addressing; Subnetting; Routing; IPv4/IPv6

Unit 4 – Transport: TCP; UDP; Flow control; Congestion control

Unit 5 – Applications: DNS; HTTP/HTTPS; Email; DHCP; Network security basics

5. Core Technical Subject I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Operating Systems
2	Software Engineering
3	Core Technical Subject II
4	Core Technical Subject III
5	Core Technical Subject IV
6	Laboratory

1. Operating Systems

Unit 1 – OS Fundamentals: OS services; System calls; Processes; Threads

Unit 2 – CPU Management: Scheduling; Synchronization; Critical sections; Deadlocks

Unit 3 – Memory: Allocation; Paging; Segmentation; Virtual memory

Unit 4 – Storage: File systems; Disk scheduling; I/O systems

Unit 5 – Security: Protection; Access control; Authentication; OS security

2. Software Engineering

Unit 1 – Software Process: Software characteristics; Process models; Agile; DevOps overview

Unit 2 – Requirements: Elicitation; SRS; Use cases; Validation

Unit 3 – Design: Architecture; Modularity; UML; Design principles

Unit 4 – Testing: Test levels; Black-box testing; White-box testing; Automation concepts

Unit 5 – Management: Estimation; Scheduling; Risk; Configuration management; Quality

3. Core Technical Subject II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Core Technical Subject III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Core Technical Subject IV

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Advanced Core I
2	Advanced Core II
3	Elective I
4	Elective II
5	Industry Skills
6	Project

1. Advanced Core I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Core II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Industry Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

Semester 6

No.	Subject
1	Advanced Core III
2	Elective III
3	Project
4	Industrial Training
5	Seminar
6	Professional Practice

1. Advanced Core III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Elective III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

4. Industrial Training

Unit 1 – Orientation: Organization; Safety; Tools; Professional conduct

Unit 2 – Technical Exposure: Workflow; Systems; Standards; Documentation

Unit 3 – Practical Work: Assigned tasks; Implementation; Testing

Unit 4 – Industry Practices: Quality; Version control; Teamwork; Communication

Unit 5 – Report: Learning outcomes; Case study; Summary; Presentation

5. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

6. Professional Practice

Unit 1 – Professional Skills: Communication; Teamwork; Time management

Unit 2 – Ethics: Professional ethics; Responsible engineering; Safety

Unit 3 – Industry: Standards; Quality; Documentation

Unit 4 – Career Skills: Resume; Interview; Portfolio; Continuous learning

Unit 5 – Practice: Case studies; Problem solving; Professional conduct

Polytechnic – Aeronautical Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Programming Fundamentals
6	Workshop Practice

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

6. Workshop Practice

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Programming Fundamentals
3	Basic Electronics
4	Communication Skills
5	Engineering Mechanics
6	Laboratory

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

3. Basic Electronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Engineering Mechanics

Unit 1 – Statics: Forces and moments; Free-body diagrams; Equilibrium; Friction

Unit 2 – Structures: Trusses; Frames; Centroids; Moments of inertia

Unit 3 – Kinematics: Particle motion; Rigid body motion; Relative motion

Unit 4 – Dynamics: Newton's laws; Work-energy; Impulse-momentum

Unit 5 – Applications: Virtual work; Simple machines; Engineering applications

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 3

No.	Subject
1	Data Structures
2	Digital Electronics
3	Database Management Systems
4	Computer Networks
5	Core Technical Subject I
6	Laboratory

1. Data Structures

Unit 1 – Linear Structures: Arrays; Linked lists; Stacks; Queues; Circular queues
Unit 2 – Trees: Binary trees; BST; AVL trees; Heaps; Tree traversal
Unit 3 – Graphs: Graph representations; BFS; DFS; Shortest paths; Spanning trees
Unit 4 – Hashing: Hash functions; Collision handling; Hash tables
Unit 5 – Applications: Searching; Sorting; Priority queues; Complexity analysis

2. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates
Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders
Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers
Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic
Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

3. Database Management Systems

Unit 1 – Database Foundations: DBMS architecture; Data models; Schema and instances; Data independence
Unit 2 – Relational Model: Relations; Keys; Constraints; Relational algebra
Unit 3 – SQL: DDL; DML; Queries; Joins; Subqueries; Views
Unit 4 – Design: ER modeling; Functional dependencies; Normalization; Decomposition
Unit 5 – Transactions: ACID; Concurrency; Locking; Deadlocks; Recovery

4. Computer Networks

Unit 1 – Networking Basics: Network types; Topologies; OSI model; TCP/IP model
Unit 2 – Data Link: Framing; Error control; MAC; Ethernet
Unit 3 – Network Layer: IP addressing; Subnetting; Routing; IPv4/IPv6
Unit 4 – Transport: TCP; UDP; Flow control; Congestion control
Unit 5 – Applications: DNS; HTTP/HTTPS; Email; DHCP; Network security basics

5. Core Technical Subject I

Unit 1 – Foundations: Core concepts; Terminology; Architecture
Unit 2 – Methods: Algorithms; Design methods; Implementation
Unit 3 – Applications: Use cases; Tools; Case studies
Unit 4 – Evaluation: Performance; Testing; Security/quality
Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice
Unit 2 – Fundamentals: Core experiments; Observation; Data collection
Unit 3 – Analysis: Calculations; Graphs; Error analysis
Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation
Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Operating Systems
2	Software Engineering
3	Core Technical Subject II
4	Core Technical Subject III
5	Core Technical Subject IV
6	Laboratory

1. Operating Systems

Unit 1 – OS Fundamentals: OS services; System calls; Processes; Threads

Unit 2 – CPU Management: Scheduling; Synchronization; Critical sections; Deadlocks

Unit 3 – Memory: Allocation; Paging; Segmentation; Virtual memory

Unit 4 – Storage: File systems; Disk scheduling; I/O systems

Unit 5 – Security: Protection; Access control; Authentication; OS security

2. Software Engineering

Unit 1 – Software Process: Software characteristics; Process models; Agile; DevOps overview

Unit 2 – Requirements: Elicitation; SRS; Use cases; Validation

Unit 3 – Design: Architecture; Modularity; UML; Design principles

Unit 4 – Testing: Test levels; Black-box testing; White-box testing; Automation concepts

Unit 5 – Management: Estimation; Scheduling; Risk; Configuration management; Quality

3. Core Technical Subject II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Core Technical Subject III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Core Technical Subject IV

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Advanced Core I
2	Advanced Core II
3	Elective I
4	Elective II
5	Industry Skills
6	Project

1. Advanced Core I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Core II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Industry Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

Semester 6

No.	Subject
1	Advanced Core III
2	Elective III
3	Project
4	Industrial Training
5	Seminar
6	Professional Practice

1. Advanced Core III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Elective III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

4. Industrial Training

Unit 1 – Orientation: Organization; Safety; Tools; Professional conduct

Unit 2 – Technical Exposure: Workflow; Systems; Standards; Documentation

Unit 3 – Practical Work: Assigned tasks; Implementation; Testing

Unit 4 – Industry Practices: Quality; Version control; Teamwork; Communication

Unit 5 – Report: Learning outcomes; Case study; Summary; Presentation

5. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

6. Professional Practice

Unit 1 – Professional Skills: Communication; Teamwork; Time management

Unit 2 – Ethics: Professional ethics; Responsible engineering; Safety

Unit 3 – Industry: Standards; Quality; Documentation

Unit 4 – Career Skills: Resume; Interview; Portfolio; Continuous learning

Unit 5 – Practice: Case studies; Problem solving; Professional conduct

Polytechnic – Agricultural Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Programming Fundamentals
6	Workshop Practice

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

6. Workshop Practice

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Programming Fundamentals
3	Basic Electronics
4	Communication Skills
5	Engineering Mechanics
6	Laboratory

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

3. Basic Electronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Engineering Mechanics

Unit 1 – Statics: Forces and moments; Free-body diagrams; Equilibrium; Friction

Unit 2 – Structures: Trusses; Frames; Centroids; Moments of inertia

Unit 3 – Kinematics: Particle motion; Rigid body motion; Relative motion

Unit 4 – Dynamics: Newton's laws; Work-energy; Impulse-momentum

Unit 5 – Applications: Virtual work; Simple machines; Engineering applications

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 3

No.	Subject
1	Data Structures
2	Digital Electronics
3	Database Management Systems
4	Computer Networks
5	Core Technical Subject I
6	Laboratory

1. Data Structures

Unit 1 – Linear Structures: Arrays; Linked lists; Stacks; Queues; Circular queues
Unit 2 – Trees: Binary trees; BST; AVL trees; Heaps; Tree traversal
Unit 3 – Graphs: Graph representations; BFS; DFS; Shortest paths; Spanning trees
Unit 4 – Hashing: Hash functions; Collision handling; Hash tables
Unit 5 – Applications: Searching; Sorting; Priority queues; Complexity analysis

2. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates
Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders
Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers
Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic
Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

3. Database Management Systems

Unit 1 – Database Foundations: DBMS architecture; Data models; Schema and instances; Data independence
Unit 2 – Relational Model: Relations; Keys; Constraints; Relational algebra
Unit 3 – SQL: DDL; DML; Queries; Joins; Subqueries; Views
Unit 4 – Design: ER modeling; Functional dependencies; Normalization; Decomposition
Unit 5 – Transactions: ACID; Concurrency; Locking; Deadlocks; Recovery

4. Computer Networks

Unit 1 – Networking Basics: Network types; Topologies; OSI model; TCP/IP model
Unit 2 – Data Link: Framing; Error control; MAC; Ethernet
Unit 3 – Network Layer: IP addressing; Subnetting; Routing; IPv4/IPv6
Unit 4 – Transport: TCP; UDP; Flow control; Congestion control
Unit 5 – Applications: DNS; HTTP/HTTPS; Email; DHCP; Network security basics

5. Core Technical Subject I

Unit 1 – Foundations: Core concepts; Terminology; Architecture
Unit 2 – Methods: Algorithms; Design methods; Implementation
Unit 3 – Applications: Use cases; Tools; Case studies
Unit 4 – Evaluation: Performance; Testing; Security/quality
Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice
Unit 2 – Fundamentals: Core experiments; Observation; Data collection
Unit 3 – Analysis: Calculations; Graphs; Error analysis
Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation
Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Operating Systems
2	Software Engineering
3	Core Technical Subject II
4	Core Technical Subject III
5	Core Technical Subject IV
6	Laboratory

1. Operating Systems

Unit 1 – OS Fundamentals: OS services; System calls; Processes; Threads

Unit 2 – CPU Management: Scheduling; Synchronization; Critical sections; Deadlocks

Unit 3 – Memory: Allocation; Paging; Segmentation; Virtual memory

Unit 4 – Storage: File systems; Disk scheduling; I/O systems

Unit 5 – Security: Protection; Access control; Authentication; OS security

2. Software Engineering

Unit 1 – Software Process: Software characteristics; Process models; Agile; DevOps overview

Unit 2 – Requirements: Elicitation; SRS; Use cases; Validation

Unit 3 – Design: Architecture; Modularity; UML; Design principles

Unit 4 – Testing: Test levels; Black-box testing; White-box testing; Automation concepts

Unit 5 – Management: Estimation; Scheduling; Risk; Configuration management; Quality

3. Core Technical Subject II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Core Technical Subject III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Core Technical Subject IV

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Advanced Core I
2	Advanced Core II
3	Elective I
4	Elective II
5	Industry Skills
6	Project

1. Advanced Core I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Core II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Industry Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

Semester 6

No.	Subject
1	Advanced Core III
2	Elective III
3	Project
4	Industrial Training
5	Seminar
6	Professional Practice

1. Advanced Core III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Elective III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

4. Industrial Training

Unit 1 – Orientation: Organization; Safety; Tools; Professional conduct

Unit 2 – Technical Exposure: Workflow; Systems; Standards; Documentation

Unit 3 – Practical Work: Assigned tasks; Implementation; Testing

Unit 4 – Industry Practices: Quality; Version control; Teamwork; Communication

Unit 5 – Report: Learning outcomes; Case study; Summary; Presentation

5. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

6. Professional Practice

Unit 1 – Professional Skills: Communication; Teamwork; Time management

Unit 2 – Ethics: Professional ethics; Responsible engineering; Safety

Unit 3 – Industry: Standards; Quality; Documentation

Unit 4 – Career Skills: Resume; Interview; Portfolio; Continuous learning

Unit 5 – Practice: Case studies; Problem solving; Professional conduct

Polytechnic – Mining Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Programming Fundamentals
6	Workshop Practice

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

6. Workshop Practice

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Programming Fundamentals
3	Basic Electronics
4	Communication Skills
5	Engineering Mechanics
6	Laboratory

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

3. Basic Electronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Engineering Mechanics

Unit 1 – Statics: Forces and moments; Free-body diagrams; Equilibrium; Friction

Unit 2 – Structures: Trusses; Frames; Centroids; Moments of inertia

Unit 3 – Kinematics: Particle motion; Rigid body motion; Relative motion

Unit 4 – Dynamics: Newton's laws; Work-energy; Impulse-momentum

Unit 5 – Applications: Virtual work; Simple machines; Engineering applications

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 3

No.	Subject
1	Data Structures
2	Digital Electronics
3	Database Management Systems
4	Computer Networks
5	Core Technical Subject I
6	Laboratory

1. Data Structures

Unit 1 – Linear Structures: Arrays; Linked lists; Stacks; Queues; Circular queues
Unit 2 – Trees: Binary trees; BST; AVL trees; Heaps; Tree traversal
Unit 3 – Graphs: Graph representations; BFS; DFS; Shortest paths; Spanning trees
Unit 4 – Hashing: Hash functions; Collision handling; Hash tables
Unit 5 – Applications: Searching; Sorting; Priority queues; Complexity analysis

2. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates
Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders
Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers
Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic
Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

3. Database Management Systems

Unit 1 – Database Foundations: DBMS architecture; Data models; Schema and instances; Data independence
Unit 2 – Relational Model: Relations; Keys; Constraints; Relational algebra
Unit 3 – SQL: DDL; DML; Queries; Joins; Subqueries; Views
Unit 4 – Design: ER modeling; Functional dependencies; Normalization; Decomposition
Unit 5 – Transactions: ACID; Concurrency; Locking; Deadlocks; Recovery

4. Computer Networks

Unit 1 – Networking Basics: Network types; Topologies; OSI model; TCP/IP model
Unit 2 – Data Link: Framing; Error control; MAC; Ethernet
Unit 3 – Network Layer: IP addressing; Subnetting; Routing; IPv4/IPv6
Unit 4 – Transport: TCP; UDP; Flow control; Congestion control
Unit 5 – Applications: DNS; HTTP/HTTPS; Email; DHCP; Network security basics

5. Core Technical Subject I

Unit 1 – Foundations: Core concepts; Terminology; Architecture
Unit 2 – Methods: Algorithms; Design methods; Implementation
Unit 3 – Applications: Use cases; Tools; Case studies
Unit 4 – Evaluation: Performance; Testing; Security/quality
Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice
Unit 2 – Fundamentals: Core experiments; Observation; Data collection
Unit 3 – Analysis: Calculations; Graphs; Error analysis
Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation
Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Operating Systems
2	Software Engineering
3	Core Technical Subject II
4	Core Technical Subject III
5	Core Technical Subject IV
6	Laboratory

1. Operating Systems

Unit 1 – OS Fundamentals: OS services; System calls; Processes; Threads

Unit 2 – CPU Management: Scheduling; Synchronization; Critical sections; Deadlocks

Unit 3 – Memory: Allocation; Paging; Segmentation; Virtual memory

Unit 4 – Storage: File systems; Disk scheduling; I/O systems

Unit 5 – Security: Protection; Access control; Authentication; OS security

2. Software Engineering

Unit 1 – Software Process: Software characteristics; Process models; Agile; DevOps overview

Unit 2 – Requirements: Elicitation; SRS; Use cases; Validation

Unit 3 – Design: Architecture; Modularity; UML; Design principles

Unit 4 – Testing: Test levels; Black-box testing; White-box testing; Automation concepts

Unit 5 – Management: Estimation; Scheduling; Risk; Configuration management; Quality

3. Core Technical Subject II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Core Technical Subject III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Core Technical Subject IV

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Advanced Core I
2	Advanced Core II
3	Elective I
4	Elective II
5	Industry Skills
6	Project

1. Advanced Core I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Core II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Industry Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

Semester 6

No.	Subject
1	Advanced Core III
2	Elective III
3	Project
4	Industrial Training
5	Seminar
6	Professional Practice

1. Advanced Core III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Elective III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

4. Industrial Training

Unit 1 – Orientation: Organization; Safety; Tools; Professional conduct

Unit 2 – Technical Exposure: Workflow; Systems; Standards; Documentation

Unit 3 – Practical Work: Assigned tasks; Implementation; Testing

Unit 4 – Industry Practices: Quality; Version control; Teamwork; Communication

Unit 5 – Report: Learning outcomes; Case study; Summary; Presentation

5. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

6. Professional Practice

Unit 1 – Professional Skills: Communication; Teamwork; Time management

Unit 2 – Ethics: Professional ethics; Responsible engineering; Safety

Unit 3 – Industry: Standards; Quality; Documentation

Unit 4 – Career Skills: Resume; Interview; Portfolio; Continuous learning

Unit 5 – Practice: Case studies; Problem solving; Professional conduct

Polytechnic – Food Technology

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Programming Fundamentals
6	Workshop Practice

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

6. Workshop Practice

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Programming Fundamentals
3	Basic Electronics
4	Communication Skills
5	Engineering Mechanics
6	Laboratory

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

3. Basic Electronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Engineering Mechanics

Unit 1 – Statics: Forces and moments; Free-body diagrams; Equilibrium; Friction

Unit 2 – Structures: Trusses; Frames; Centroids; Moments of inertia

Unit 3 – Kinematics: Particle motion; Rigid body motion; Relative motion

Unit 4 – Dynamics: Newton's laws; Work-energy; Impulse-momentum

Unit 5 – Applications: Virtual work; Simple machines; Engineering applications

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 3

No.	Subject
1	Data Structures
2	Digital Electronics
3	Database Management Systems
4	Computer Networks
5	Core Technical Subject I
6	Laboratory

1. Data Structures

Unit 1 – Linear Structures: Arrays; Linked lists; Stacks; Queues; Circular queues
Unit 2 – Trees: Binary trees; BST; AVL trees; Heaps; Tree traversal
Unit 3 – Graphs: Graph representations; BFS; DFS; Shortest paths; Spanning trees
Unit 4 – Hashing: Hash functions; Collision handling; Hash tables
Unit 5 – Applications: Searching; Sorting; Priority queues; Complexity analysis

2. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates
Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders
Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers
Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic
Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

3. Database Management Systems

Unit 1 – Database Foundations: DBMS architecture; Data models; Schema and instances; Data independence
Unit 2 – Relational Model: Relations; Keys; Constraints; Relational algebra
Unit 3 – SQL: DDL; DML; Queries; Joins; Subqueries; Views
Unit 4 – Design: ER modeling; Functional dependencies; Normalization; Decomposition
Unit 5 – Transactions: ACID; Concurrency; Locking; Deadlocks; Recovery

4. Computer Networks

Unit 1 – Networking Basics: Network types; Topologies; OSI model; TCP/IP model
Unit 2 – Data Link: Framing; Error control; MAC; Ethernet
Unit 3 – Network Layer: IP addressing; Subnetting; Routing; IPv4/IPv6
Unit 4 – Transport: TCP; UDP; Flow control; Congestion control
Unit 5 – Applications: DNS; HTTP/HTTPS; Email; DHCP; Network security basics

5. Core Technical Subject I

Unit 1 – Foundations: Core concepts; Terminology; Architecture
Unit 2 – Methods: Algorithms; Design methods; Implementation
Unit 3 – Applications: Use cases; Tools; Case studies
Unit 4 – Evaluation: Performance; Testing; Security/quality
Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice
Unit 2 – Fundamentals: Core experiments; Observation; Data collection
Unit 3 – Analysis: Calculations; Graphs; Error analysis
Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation
Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Operating Systems
2	Software Engineering
3	Core Technical Subject II
4	Core Technical Subject III
5	Core Technical Subject IV
6	Laboratory

1. Operating Systems

Unit 1 – OS Fundamentals: OS services; System calls; Processes; Threads

Unit 2 – CPU Management: Scheduling; Synchronization; Critical sections; Deadlocks

Unit 3 – Memory: Allocation; Paging; Segmentation; Virtual memory

Unit 4 – Storage: File systems; Disk scheduling; I/O systems

Unit 5 – Security: Protection; Access control; Authentication; OS security

2. Software Engineering

Unit 1 – Software Process: Software characteristics; Process models; Agile; DevOps overview

Unit 2 – Requirements: Elicitation; SRS; Use cases; Validation

Unit 3 – Design: Architecture; Modularity; UML; Design principles

Unit 4 – Testing: Test levels; Black-box testing; White-box testing; Automation concepts

Unit 5 – Management: Estimation; Scheduling; Risk; Configuration management; Quality

3. Core Technical Subject II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Core Technical Subject III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Core Technical Subject IV

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Advanced Core I
2	Advanced Core II
3	Elective I
4	Elective II
5	Industry Skills
6	Project

1. Advanced Core I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Core II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Industry Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

Semester 6

No.	Subject
1	Advanced Core III
2	Elective III
3	Project
4	Industrial Training
5	Seminar
6	Professional Practice

1. Advanced Core III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Elective III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

4. Industrial Training

Unit 1 – Orientation: Organization; Safety; Tools; Professional conduct

Unit 2 – Technical Exposure: Workflow; Systems; Standards; Documentation

Unit 3 – Practical Work: Assigned tasks; Implementation; Testing

Unit 4 – Industry Practices: Quality; Version control; Teamwork; Communication

Unit 5 – Report: Learning outcomes; Case study; Summary; Presentation

5. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

6. Professional Practice

Unit 1 – Professional Skills: Communication; Teamwork; Time management

Unit 2 – Ethics: Professional ethics; Responsible engineering; Safety

Unit 3 – Industry: Standards; Quality; Documentation

Unit 4 – Career Skills: Resume; Interview; Portfolio; Continuous learning

Unit 5 – Practice: Case studies; Problem solving; Professional conduct

Polytechnic – Textile Technology

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Programming Fundamentals
6	Workshop Practice

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

6. Workshop Practice

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Programming Fundamentals
3	Basic Electronics
4	Communication Skills
5	Engineering Mechanics
6	Laboratory

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

3. Basic Electronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Engineering Mechanics

Unit 1 – Statics: Forces and moments; Free-body diagrams; Equilibrium; Friction

Unit 2 – Structures: Trusses; Frames; Centroids; Moments of inertia

Unit 3 – Kinematics: Particle motion; Rigid body motion; Relative motion

Unit 4 – Dynamics: Newton's laws; Work-energy; Impulse-momentum

Unit 5 – Applications: Virtual work; Simple machines; Engineering applications

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 3

No.	Subject
1	Data Structures
2	Digital Electronics
3	Database Management Systems
4	Computer Networks
5	Core Technical Subject I
6	Laboratory

1. Data Structures

Unit 1 – Linear Structures: Arrays; Linked lists; Stacks; Queues; Circular queues
Unit 2 – Trees: Binary trees; BST; AVL trees; Heaps; Tree traversal
Unit 3 – Graphs: Graph representations; BFS; DFS; Shortest paths; Spanning trees
Unit 4 – Hashing: Hash functions; Collision handling; Hash tables
Unit 5 – Applications: Searching; Sorting; Priority queues; Complexity analysis

2. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates
Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders
Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers
Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic
Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

3. Database Management Systems

Unit 1 – Database Foundations: DBMS architecture; Data models; Schema and instances; Data independence
Unit 2 – Relational Model: Relations; Keys; Constraints; Relational algebra
Unit 3 – SQL: DDL; DML; Queries; Joins; Subqueries; Views
Unit 4 – Design: ER modeling; Functional dependencies; Normalization; Decomposition
Unit 5 – Transactions: ACID; Concurrency; Locking; Deadlocks; Recovery

4. Computer Networks

Unit 1 – Networking Basics: Network types; Topologies; OSI model; TCP/IP model
Unit 2 – Data Link: Framing; Error control; MAC; Ethernet
Unit 3 – Network Layer: IP addressing; Subnetting; Routing; IPv4/IPv6
Unit 4 – Transport: TCP; UDP; Flow control; Congestion control
Unit 5 – Applications: DNS; HTTP/HTTPS; Email; DHCP; Network security basics

5. Core Technical Subject I

Unit 1 – Foundations: Core concepts; Terminology; Architecture
Unit 2 – Methods: Algorithms; Design methods; Implementation
Unit 3 – Applications: Use cases; Tools; Case studies
Unit 4 – Evaluation: Performance; Testing; Security/quality
Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice
Unit 2 – Fundamentals: Core experiments; Observation; Data collection
Unit 3 – Analysis: Calculations; Graphs; Error analysis
Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation
Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Operating Systems
2	Software Engineering
3	Core Technical Subject II
4	Core Technical Subject III
5	Core Technical Subject IV
6	Laboratory

1. Operating Systems

Unit 1 – OS Fundamentals: OS services; System calls; Processes; Threads

Unit 2 – CPU Management: Scheduling; Synchronization; Critical sections; Deadlocks

Unit 3 – Memory: Allocation; Paging; Segmentation; Virtual memory

Unit 4 – Storage: File systems; Disk scheduling; I/O systems

Unit 5 – Security: Protection; Access control; Authentication; OS security

2. Software Engineering

Unit 1 – Software Process: Software characteristics; Process models; Agile; DevOps overview

Unit 2 – Requirements: Elicitation; SRS; Use cases; Validation

Unit 3 – Design: Architecture; Modularity; UML; Design principles

Unit 4 – Testing: Test levels; Black-box testing; White-box testing; Automation concepts

Unit 5 – Management: Estimation; Scheduling; Risk; Configuration management; Quality

3. Core Technical Subject II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Core Technical Subject III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Core Technical Subject IV

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Advanced Core I
2	Advanced Core II
3	Elective I
4	Elective II
5	Industry Skills
6	Project

1. Advanced Core I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Core II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Industry Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

Semester 6

No.	Subject
1	Advanced Core III
2	Elective III
3	Project
4	Industrial Training
5	Seminar
6	Professional Practice

1. Advanced Core III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Elective III

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

4. Industrial Training

Unit 1 – Orientation: Organization; Safety; Tools; Professional conduct

Unit 2 – Technical Exposure: Workflow; Systems; Standards; Documentation

Unit 3 – Practical Work: Assigned tasks; Implementation; Testing

Unit 4 – Industry Practices: Quality; Version control; Teamwork; Communication

Unit 5 – Report: Learning outcomes; Case study; Summary; Presentation

5. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

6. Professional Practice

Unit 1 – Professional Skills: Communication; Teamwork; Time management

Unit 2 – Ethics: Professional ethics; Responsible engineering; Safety

Unit 3 – Industry: Standards; Quality; Documentation

Unit 4 – Career Skills: Resume; Interview; Portfolio; Continuous learning

Unit 5 – Practice: Case studies; Problem solving; Professional conduct

PART II — B.TECH / B.E.

The UG section uses an eight-semester, four-year structure. AICTE's UG model framework uses a broad 150–160 credit range and permits discipline-specific variation. ■cite■turn0search33■turn0search35■

B.Tech – Computer Science & Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Programming Fundamentals
5	Engineering Graphics
6	Basic Electrical Engineering

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

5. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

6. Basic Electrical Engineering

Unit 1 – DC Circuits: Voltage and current; Ohm's law; KCL/KVL; Network reduction

Unit 2 – AC Circuits: Sinusoids; Phasors; Impedance; Power factor; Three-phase systems

Unit 3 – Machines: Transformers; DC machines; Induction motors; Synchronous machines

Unit 4 – Measurements: Meters; Bridges; Errors; Power and energy measurement

Unit 5 – Safety and Power: Earthing; Protection; Energy systems; Electrical safety

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Data Structures
3	Engineering Physics/Chemistry
4	Basic Electronics
5	Communication Skills
6	Engineering Workshop

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Data Structures

Unit 1 – Linear Structures: Arrays; Linked lists; Stacks; Queues; Circular queues

Unit 2 – Trees: Binary trees; BST; AVL trees; Heaps; Tree traversal

Unit 3 – Graphs: Graph representations; BFS; DFS; Shortest paths; Spanning trees

Unit 4 – Hashing: Hash functions; Collision handling; Hash tables

Unit 5 – Applications: Searching; Sorting; Priority queues; Complexity analysis

3. Engineering Physics/Chemistry

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Basic Electronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Engineering Workshop

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 3

No.	Subject
1	Discrete Mathematics
2	Data Structures
3	Object-Oriented Programming
4	Digital Logic
5	Computer Organization
6	Probability & Statistics

1. Discrete Mathematics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Data Structures

Unit 1 – Linear Structures: Arrays; Linked lists; Stacks; Queues; Circular queues

Unit 2 – Trees: Binary trees; BST; AVL trees; Heaps; Tree traversal

Unit 3 – Graphs: Graph representations; BFS; DFS; Shortest paths; Spanning trees

Unit 4 – Hashing: Hash functions; Collision handling; Hash tables

Unit 5 – Applications: Searching; Sorting; Priority queues; Complexity analysis

3. Object-Oriented Programming

Unit 1 – OOP Foundations: Classes and objects; Constructors; Methods; Encapsulation

Unit 2 – Inheritance: Inheritance models; Method overriding; Interfaces

Unit 3 – Polymorphism: Overloading; Overriding; Dynamic binding

Unit 4 – Exceptions and Files: Exception handling; Streams; File I/O

Unit 5 – Modern OOP: Generics; Collections; Packages; Basic design principles

4. Digital Logic

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Computer Organization

Unit 1 – Computer Basics: Instruction execution; CPU organization; Performance; Data representation

Unit 2 – Digital Arithmetic: Integer arithmetic; Floating point; ALU

Unit 3 – Processor: Instruction formats; Addressing modes; Control unit; Pipelining

Unit 4 – Memory: Cache; Main memory; Virtual memory; Memory hierarchy

Unit 5 – I/O: I/O interfaces; Interrupts; DMA; Buses

6. Probability & Statistics

Unit 1: Probability axioms; Conditional probability; Bayes theorem

Unit 2: Random variables; Distributions

Unit 3: Expectation; Variance; Moments

Unit 4: Sampling; Estimation

Unit 5: Hypothesis testing; Regression; Correlation

Semester 4

No.	Subject
1	Algorithms
2	Operating Systems
3	Database Management Systems
4	Computer Networks
5	Theory of Computation
6	Software Engineering

1. Algorithms

Unit 1 – Algorithm Analysis: Asymptotic notation; Time complexity; Space complexity; Recurrences

Unit 2 – Divide and Conquer: Binary search; Merge sort; Quick sort; Strassen concept

Unit 3 – Greedy Methods: Activity selection; Knapsack; Huffman coding; MST

Unit 4 – Dynamic Programming: Optimal substructure; LCS; Knapsack; Matrix-chain multiplication

Unit 5 – Graph and Advanced Algorithms: Shortest paths; Network flow; Backtracking; Branch and bound

2. Operating Systems

Unit 1 – OS Fundamentals: OS services; System calls; Processes; Threads

Unit 2 – CPU Management: Scheduling; Synchronization; Critical sections; Deadlocks

Unit 3 – Memory: Allocation; Paging; Segmentation; Virtual memory

Unit 4 – Storage: File systems; Disk scheduling; I/O systems

Unit 5 – Security: Protection; Access control; Authentication; OS security

3. Database Management Systems

Unit 1 – Database Foundations: DBMS architecture; Data models; Schema and instances; Data independence

Unit 2 – Relational Model: Relations; Keys; Constraints; Relational algebra

Unit 3 – SQL: DDL; DML; Queries; Joins; Subqueries; Views

Unit 4 – Design: ER modeling; Functional dependencies; Normalization; Decomposition

Unit 5 – Transactions: ACID; Concurrency; Locking; Deadlocks; Recovery

4. Computer Networks

Unit 1 – Networking Basics: Network types; Topologies; OSI model; TCP/IP model

Unit 2 – Data Link: Framing; Error control; MAC; Ethernet

Unit 3 – Network Layer: IP addressing; Subnetting; Routing; IPv4/IPv6

Unit 4 – Transport: TCP; UDP; Flow control; Congestion control

Unit 5 – Applications: DNS; HTTP/HTTPS; Email; DHCP; Network security basics

5. Theory of Computation

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Software Engineering

Unit 1 – Software Process: Software characteristics; Process models; Agile; DevOps overview

Unit 2 – Requirements: Elicitation; SRS; Use cases; Validation

Unit 3 – Design: Architecture; Modularity; UML; Design principles

Unit 4 – Testing: Test levels; Black-box testing; White-box testing; Automation concepts

Unit 5 – Management: Estimation; Scheduling; Risk; Configuration management; Quality

Semester 5

No.	Subject
1	Compiler Design
2	Artificial Intelligence
3	Machine Learning
4	Web Technologies
5	Computer Architecture
6	Professional Elective

1. Compiler Design

Unit 1 – Language Processing: Compiler phases; Lexical analysis; Tokens; Regular expressions

Unit 2 – Parsing: CFG; Top-down parsing; Bottom-up parsing; LR parsing

Unit 3 – Semantic Analysis: Type checking; Symbol tables; Syntax-directed translation

Unit 4 – Intermediate Code: Three-address code; Control-flow graphs; Runtime environment

Unit 5 – Optimization: Local optimization; Data-flow analysis; Loop optimization; Code generation

2. Artificial Intelligence

Unit 1 – AI Foundations: Intelligent agents; Problem formulation; State spaces; Search

Unit 2 – Knowledge: Logic; Knowledge representation; Inference; Uncertainty

Unit 3 – Search: BFS/DFS; A*; Game playing; Constraint satisfaction

Unit 4 – Learning: Supervised learning; Unsupervised learning; Decision trees; Basic neural networks

Unit 5 – Applications: Expert systems; Planning; NLP overview; Robotics

3. Machine Learning

Unit 1 – ML Foundations: Learning paradigms; Data preprocessing; Feature engineering; Train/test split

Unit 2 – Regression: Linear regression; Polynomial regression; Regularization; Metrics

Unit 3 – Classification: Logistic regression; KNN; Naive Bayes; SVM

Unit 4 – Trees and Ensembles: Decision trees; Random forests; Boosting; Bagging

Unit 5 – Unsupervised Learning: K-means; Hierarchical clustering; PCA; Anomaly detection

4. Web Technologies

Unit 1 – Web Basics: HTTP; Browsers; HTML; Semantic markup

Unit 2 – Styling: CSS; Layouts; Responsive design; Accessibility

Unit 3 – Client Programming: JavaScript; DOM; Events; Fetch/API

Unit 4 – Server Programming: REST; Authentication; Sessions; Database integration

Unit 5 – Modern Web: Framework concepts; Security; Testing; Deployment

5. Computer Architecture

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Professional Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 6

No.	Subject
1	Cloud Computing
2	Cyber Security
3	Big Data Analytics
4	Internet of Things
5	Distributed Systems
6	Professional Elective

1. Cloud Computing

Unit 1 – Cloud Foundations: Service models; Deployment models; Virtualization; Elasticity

Unit 2 – Cloud Architecture: Compute; Storage; Networking; Identity

Unit 3 – Cloud Platforms: IaaS; PaaS; SaaS; Containers

Unit 4 – Cloud Security: IAM; Encryption; Security monitoring; Compliance

Unit 5 – Cloud Applications: Serverless; Autoscaling; DevOps; Cost management

2. Cyber Security

Unit 1 – Security Foundations: CIA triad; Threats; Vulnerabilities; Risk

Unit 2 – Cryptography: Symmetric encryption; Asymmetric encryption; Hashing; Digital signatures

Unit 3 – Network Security: Firewalls; IDS/IPS; VPN; Secure protocols

Unit 4 – Application Security: Authentication; Authorization; Web vulnerabilities; Secure coding

Unit 5 – Governance: Incident response; Forensics basics; Policies; Security standards

3. Big Data Analytics

Unit 1 – Big Data: Characteristics; Data sources; Distributed storage; Batch vs stream

Unit 2 – Distributed Processing: MapReduce; Spark concepts; Data partitioning

Unit 3 – NoSQL: Key-value; Document; Column; Graph databases

Unit 4 – Analytics: Data preparation; Descriptive analytics; Predictive analytics; Visualization

Unit 5 – Platforms: Data lakes; Pipelines; Governance; Security

4. Internet of Things

Unit 1 – IoT Foundations: IoT architecture; Sensors; Actuators; Edge devices

Unit 2 – Connectivity: Wi-Fi; Bluetooth; Zigbee; LPWAN; MQTT

Unit 3 – IoT Platforms: Gateways; Cloud integration; Data pipelines; Device management

Unit 4 – Analytics: Stream processing; Time-series data; Edge analytics; AI at edge

Unit 5 – Security: Device identity; Secure boot; Encryption; OTA updates

5. Distributed Systems

Unit 1 – Foundations: Distributed models; Communication; Clocks; Consistency

Unit 2 – Coordination: Mutual exclusion; Leader election; Consensus

Unit 3 – Replication: Replication models; Consistency models; Fault tolerance

Unit 4 – Distributed Storage: Distributed file systems; Distributed databases; Transactions

Unit 5 – Modern Systems: Microservices; Containers; Service discovery; Observability

6. Professional Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 7

No.	Subject
1	Deep Learning
2	Computer Vision
3	Natural Language Processing
4	DevOps
5	Advanced AI
6	Project Phase I

1. Deep Learning

Unit 1 – Neural Networks: Perceptron; Activation functions; Loss functions; Backpropagation

Unit 2 – Optimization: Gradient descent; Learning rates; Regularization; Batch normalization

Unit 3 – CNN: Convolution; Pooling; Architectures; Transfer learning

Unit 4 – Sequence Models: RNN; LSTM; GRU; Sequence prediction

Unit 5 – Generative Models: Autoencoders; GANs; Transformers overview; Generative applications

2. Computer Vision

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Natural Language Processing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. DevOps

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Advanced AI

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Project Phase I

Unit 1 – Problem Definition: Problem statement; Objectives; Requirements

Unit 2 – Literature Review: Research papers; Existing systems; Gap analysis

Unit 3 – System Design: Architecture; Modules; Technology selection

Unit 4 – Initial Implementation: Prototype; Testing; Initial results

Unit 5 – Project Report: Documentation; Presentation; Future plan

Semester 8

No.	Subject
1	Major Project
2	Internship
3	Seminar
4	Technical Elective
5	Professional Practice
6	Project Viva

1. Major Project

Unit 1 – Problem and Requirements: Problem statement; Objectives; Scope; Requirements
Unit 2 – Architecture and Design: System architecture; Modules; Data flow; Technology stack
Unit 3 – Implementation: Development; Integration; Testing; Deployment
Unit 4 – Results: Evaluation; Metrics; Comparison; Limitations
Unit 5 – Documentation and Viva: Final report; Presentation; Demonstration; Future work

2. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics
Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation
Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment
Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting
Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

3. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope
Unit 2 – Literature: Research papers; Comparison; Gap identification
Unit 3 – Technical Study: Methodology; Architecture; Results
Unit 4 – Presentation: Slides; Technical explanation; Questions
Unit 5 – Report: Citation; References; Summary; Future scope

4. Technical Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture
Unit 2 – Methods: Algorithms; Design methods; Implementation
Unit 3 – Applications: Use cases; Tools; Case studies
Unit 4 – Evaluation: Performance; Testing; Security/quality
Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Professional Practice

Unit 1 – Professional Skills: Communication; Teamwork; Time management
Unit 2 – Ethics: Professional ethics; Responsible engineering; Safety
Unit 3 – Industry: Standards; Quality; Documentation
Unit 4 – Career Skills: Resume; Interview; Portfolio; Continuous learning
Unit 5 – Practice: Case studies; Problem solving; Professional conduct

6. Project Viva

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements
Unit 2 – Design: Architecture; Technology selection; Methodology; Planning
Unit 3 – Implementation: Development; Integration; Testing; Iteration
Unit 4 – Evaluation: Results; Metrics; Validation; Limitations
Unit 5 – Documentation: Report; Presentation; Future work; Viva

B.Tech – Information Technology

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Programming Fundamentals
5	Engineering Graphics
6	Basic Electrical Engineering

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

5. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

6. Basic Electrical Engineering

Unit 1 – DC Circuits: Voltage and current; Ohm's law; KCL/KVL; Network reduction

Unit 2 – AC Circuits: Sinusoids; Phasors; Impedance; Power factor; Three-phase systems

Unit 3 – Machines: Transformers; DC machines; Induction motors; Synchronous machines

Unit 4 – Measurements: Meters; Bridges; Errors; Power and energy measurement

Unit 5 – Safety and Power: Earthing; Protection; Energy systems; Electrical safety

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Data Structures
3	Digital Electronics
4	Object-Oriented Programming
5	Communication Skills
6	Workshop

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Data Structures

Unit 1 – Linear Structures: Arrays; Linked lists; Stacks; Queues; Circular queues

Unit 2 – Trees: Binary trees; BST; AVL trees; Heaps; Tree traversal

Unit 3 – Graphs: Graph representations; BFS; DFS; Shortest paths; Spanning trees

Unit 4 – Hashing: Hash functions; Collision handling; Hash tables

Unit 5 – Applications: Searching; Sorting; Priority queues; Complexity analysis

3. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates

Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders

Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers

Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic

Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

4. Object-Oriented Programming

Unit 1 – OOP Foundations: Classes and objects; Constructors; Methods; Encapsulation

Unit 2 – Inheritance: Inheritance models; Method overriding; Interfaces

Unit 3 – Polymorphism: Overloading; Overriding; Dynamic binding

Unit 4 – Exceptions and Files: Exception handling; Streams; File I/O

Unit 5 – Modern OOP: Generics; Collections; Packages; Basic design principles

5. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Workshop

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 3

No.	Subject
1	Discrete Mathematics
2	Data Structures
3	OOP
4	DBMS
5	Computer Organization
6	Web Technologies

1. Discrete Mathematics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Data Structures

Unit 1 – Linear Structures: Arrays; Linked lists; Stacks; Queues; Circular queues

Unit 2 – Trees: Binary trees; BST; AVL trees; Heaps; Tree traversal

Unit 3 – Graphs: Graph representations; BFS; DFS; Shortest paths; Spanning trees

Unit 4 – Hashing: Hash functions; Collision handling; Hash tables

Unit 5 – Applications: Searching; Sorting; Priority queues; Complexity analysis

3. OOP

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. DBMS

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Computer Organization

Unit 1 – Computer Basics: Instruction execution; CPU organization; Performance; Data representation

Unit 2 – Digital Arithmetic: Integer arithmetic; Floating point; ALU

Unit 3 – Processor: Instruction formats; Addressing modes; Control unit; Pipelining

Unit 4 – Memory: Cache; Main memory; Virtual memory; Memory hierarchy

Unit 5 – I/O: I/O interfaces; Interrupts; DMA; Buses

6. Web Technologies

Unit 1 – Web Basics: HTTP; Browsers; HTML; Semantic markup

Unit 2 – Styling: CSS; Layouts; Responsive design; Accessibility

Unit 3 – Client Programming: JavaScript; DOM; Events; Fetch/API

Unit 4 – Server Programming: REST; Authentication; Sessions; Database integration

Unit 5 – Modern Web: Framework concepts; Security; Testing; Deployment

Semester 4

No.	Subject
1	Algorithms
2	Operating Systems
3	Computer Networks
4	Software Engineering
5	Theory of Computation
6	DBMS

1. Algorithms

Unit 1 – Algorithm Analysis: Asymptotic notation; Time complexity; Space complexity; Recurrences

Unit 2 – Divide and Conquer: Binary search; Merge sort; Quick sort; Strassen concept

Unit 3 – Greedy Methods: Activity selection; Knapsack; Huffman coding; MST

Unit 4 – Dynamic Programming: Optimal substructure; LCS; Knapsack; Matrix-chain multiplication

Unit 5 – Graph and Advanced Algorithms: Shortest paths; Network flow; Backtracking; Branch and bound

2. Operating Systems

Unit 1 – OS Fundamentals: OS services; System calls; Processes; Threads

Unit 2 – CPU Management: Scheduling; Synchronization; Critical sections; Deadlocks

Unit 3 – Memory: Allocation; Paging; Segmentation; Virtual memory

Unit 4 – Storage: File systems; Disk scheduling; I/O systems

Unit 5 – Security: Protection; Access control; Authentication; OS security

3. Computer Networks

Unit 1 – Networking Basics: Network types; Topologies; OSI model; TCP/IP model

Unit 2 – Data Link: Framing; Error control; MAC; Ethernet

Unit 3 – Network Layer: IP addressing; Subnetting; Routing; IPv4/IPv6

Unit 4 – Transport: TCP; UDP; Flow control; Congestion control

Unit 5 – Applications: DNS; HTTP/HTTPS; Email; DHCP; Network security basics

4. Software Engineering

Unit 1 – Software Process: Software characteristics; Process models; Agile; DevOps overview

Unit 2 – Requirements: Elicitation; SRS; Use cases; Validation

Unit 3 – Design: Architecture; Modularity; UML; Design principles

Unit 4 – Testing: Test levels; Black-box testing; White-box testing; Automation concepts

Unit 5 – Management: Estimation; Scheduling; Risk; Configuration management; Quality

5. Theory of Computation

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. DBMS

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 5

No.	Subject
1	Cloud Computing
2	AI
3	Machine Learning
4	Web Technologies
5	Cyber Security
6	Elective

1. Cloud Computing

Unit 1 – Cloud Foundations: Service models; Deployment models; Virtualization; Elasticity

Unit 2 – Cloud Architecture: Compute; Storage; Networking; Identity

Unit 3 – Cloud Platforms: IaaS; PaaS; SaaS; Containers

Unit 4 – Cloud Security: IAM; Encryption; Security monitoring; Compliance

Unit 5 – Cloud Applications: Serverless; Autoscaling; DevOps; Cost management

2. AI

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Machine Learning

Unit 1 – ML Foundations: Learning paradigms; Data preprocessing; Feature engineering; Train/test split

Unit 2 – Regression: Linear regression; Polynomial regression; Regularization; Metrics

Unit 3 – Classification: Logistic regression; KNN; Naive Bayes; SVM

Unit 4 – Trees and Ensembles: Decision trees; Random forests; Boosting; Bagging

Unit 5 – Unsupervised Learning: K-means; Hierarchical clustering; PCA; Anomaly detection

4. Web Technologies

Unit 1 – Web Basics: HTTP; Browsers; HTML; Semantic markup

Unit 2 – Styling: CSS; Layouts; Responsive design; Accessibility

Unit 3 – Client Programming: JavaScript; DOM; Events; Fetch/API

Unit 4 – Server Programming: REST; Authentication; Sessions; Database integration

Unit 5 – Modern Web: Framework concepts; Security; Testing; Deployment

5. Cyber Security

Unit 1 – Security Foundations: CIA triad; Threats; Vulnerabilities; Risk

Unit 2 – Cryptography: Symmetric encryption; Asymmetric encryption; Hashing; Digital signatures

Unit 3 – Network Security: Firewalls; IDS/IPS; VPN; Secure protocols

Unit 4 – Application Security: Authentication; Authorization; Web vulnerabilities; Secure coding

Unit 5 – Governance: Incident response; Forensics basics; Policies; Security standards

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 6

No.	Subject
1	Big Data Analytics
2	IoT
3	Mobile Computing
4	Distributed Systems
5	DevOps
6	Elective

1. Big Data Analytics

Unit 1 – Big Data: Characteristics; Data sources; Distributed storage; Batch vs stream

Unit 2 – Distributed Processing: MapReduce; Spark concepts; Data partitioning

Unit 3 – NoSQL: Key-value; Document; Column; Graph databases

Unit 4 – Analytics: Data preparation; Descriptive analytics; Predictive analytics; Visualization

Unit 5 – Platforms: Data lakes; Pipelines; Governance; Security

2. IoT

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Mobile Computing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Distributed Systems

Unit 1 – Foundations: Distributed models; Communication; Clocks; Consistency

Unit 2 – Coordination: Mutual exclusion; Leader election; Consensus

Unit 3 – Replication: Replication models; Consistency models; Fault tolerance

Unit 4 – Distributed Storage: Distributed file systems; Distributed databases; Transactions

Unit 5 – Modern Systems: Microservices; Containers; Service discovery; Observability

5. DevOps

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 7

No.	Subject
1	Advanced AI/ML
2	Cloud Security
3	Data Engineering
4	Project Phase I
5	Elective
6	Internship

1. Advanced AI/ML

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Cloud Security

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Data Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Project Phase I

Unit 1 – Problem Definition: Problem statement; Objectives; Requirements

Unit 2 – Literature Review: Research papers; Existing systems; Gap analysis

Unit 3 – System Design: Architecture; Modules; Technology selection

Unit 4 – Initial Implementation: Prototype; Testing; Initial results

Unit 5 – Project Report: Documentation; Presentation; Future plan

5. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics

Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation

Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment

Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting

Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

Semester 8

No.	Subject
1	Major Project
2	Internship
3	Seminar
4	Elective
5	Professional Practice
6	Viva

1. Major Project

Unit 1 – Problem and Requirements: Problem statement; Objectives; Scope; Requirements
Unit 2 – Architecture and Design: System architecture; Modules; Data flow; Technology stack
Unit 3 – Implementation: Development; Integration; Testing; Deployment
Unit 4 – Results: Evaluation; Metrics; Comparison; Limitations
Unit 5 – Documentation and Viva: Final report; Presentation; Demonstration; Future work

2. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics
Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation
Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment
Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting
Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

3. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope
Unit 2 – Literature: Research papers; Comparison; Gap identification
Unit 3 – Technical Study: Methodology; Architecture; Results
Unit 4 – Presentation: Slides; Technical explanation; Questions
Unit 5 – Report: Citation; References; Summary; Future scope

4. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture
Unit 2 – Methods: Algorithms; Design methods; Implementation
Unit 3 – Applications: Use cases; Tools; Case studies
Unit 4 – Evaluation: Performance; Testing; Security/quality
Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Professional Practice

Unit 1 – Professional Skills: Communication; Teamwork; Time management
Unit 2 – Ethics: Professional ethics; Responsible engineering; Safety
Unit 3 – Industry: Standards; Quality; Documentation
Unit 4 – Career Skills: Resume; Interview; Portfolio; Continuous learning
Unit 5 – Practice: Case studies; Problem solving; Professional conduct

6. Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications
Unit 2 – Specialization: Advanced concepts; Problem solving
Unit 3 – Project: Architecture; Implementation; Testing; Results
Unit 4 – Professional Practice: Standards; Ethics; Industry tools
Unit 5 – Defense: Technical questions; Future scope; Communication

B.Tech – Artificial Intelligence & Machine Learning

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Programming Fundamentals
5	Engineering Graphics
6	Basic Electrical Engineering

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

5. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

6. Basic Electrical Engineering

Unit 1 – DC Circuits: Voltage and current; Ohm's law; KCL/KVL; Network reduction

Unit 2 – AC Circuits: Sinusoids; Phasors; Impedance; Power factor; Three-phase systems

Unit 3 – Machines: Transformers; DC machines; Induction motors; Synchronous machines

Unit 4 – Measurements: Meters; Bridges; Errors; Power and energy measurement

Unit 5 – Safety and Power: Earthing; Protection; Energy systems; Electrical safety

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Data Structures
3	OOP
4	Probability & Statistics
5	Communication Skills
6	Workshop

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Data Structures

Unit 1 – Linear Structures: Arrays; Linked lists; Stacks; Queues; Circular queues

Unit 2 – Trees: Binary trees; BST; AVL trees; Heaps; Tree traversal

Unit 3 – Graphs: Graph representations; BFS; DFS; Shortest paths; Spanning trees

Unit 4 – Hashing: Hash functions; Collision handling; Hash tables

Unit 5 – Applications: Searching; Sorting; Priority queues; Complexity analysis

3. OOP

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Probability & Statistics

Unit 1: Probability axioms; Conditional probability; Bayes theorem

Unit 2: Random variables; Distributions

Unit 3: Expectation; Variance; Moments

Unit 4: Sampling; Estimation

Unit 5: Hypothesis testing; Regression; Correlation

5. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Workshop

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 3

No.	Subject
1	Discrete Mathematics
2	Data Structures
3	Linear Algebra
4	Probability & Statistics
5	Digital Logic
6	Computer Organization

1. Discrete Mathematics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Data Structures

Unit 1 – Linear Structures: Arrays; Linked lists; Stacks; Queues; Circular queues

Unit 2 – Trees: Binary trees; BST; AVL trees; Heaps; Tree traversal

Unit 3 – Graphs: Graph representations; BFS; DFS; Shortest paths; Spanning trees

Unit 4 – Hashing: Hash functions; Collision handling; Hash tables

Unit 5 – Applications: Searching; Sorting; Priority queues; Complexity analysis

3. Linear Algebra

Unit 1: Matrices; Vector spaces; Linear transformations

Unit 2: Eigenvalues; Eigenvectors; Diagonalization

Unit 3: Inner products; Orthogonality

Unit 4: Numerical linear algebra

Unit 5: Applications to ML

4. Probability & Statistics

Unit 1: Probability axioms; Conditional probability; Bayes theorem

Unit 2: Random variables; Distributions

Unit 3: Expectation; Variance; Moments

Unit 4: Sampling; Estimation

Unit 5: Hypothesis testing; Regression; Correlation

5. Digital Logic

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Computer Organization

Unit 1 – Computer Basics: Instruction execution; CPU organization; Performance; Data representation

Unit 2 – Digital Arithmetic: Integer arithmetic; Floating point; ALU

Unit 3 – Processor: Instruction formats; Addressing modes; Control unit; Pipelining

Unit 4 – Memory: Cache; Main memory; Virtual memory; Memory hierarchy

Unit 5 – I/O: I/O interfaces; Interrupts; DMA; Buses

Semester 4

No.	Subject
1	Algorithms
2	Artificial Intelligence
3	Database Management Systems
4	Machine Learning
5	Computer Networks
6	Software Engineering

1. Algorithms

Unit 1 – Algorithm Analysis: Asymptotic notation; Time complexity; Space complexity; Recurrences

Unit 2 – Divide and Conquer: Binary search; Merge sort; Quick sort; Strassen concept

Unit 3 – Greedy Methods: Activity selection; Knapsack; Huffman coding; MST

Unit 4 – Dynamic Programming: Optimal substructure; LCS; Knapsack; Matrix-chain multiplication

Unit 5 – Graph and Advanced Algorithms: Shortest paths; Network flow; Backtracking; Branch and bound

2. Artificial Intelligence

Unit 1 – AI Foundations: Intelligent agents; Problem formulation; State spaces; Search

Unit 2 – Knowledge: Logic; Knowledge representation; Inference; Uncertainty

Unit 3 – Search: BFS/DFS; A*; Game playing; Constraint satisfaction

Unit 4 – Learning: Supervised learning; Unsupervised learning; Decision trees; Basic neural networks

Unit 5 – Applications: Expert systems; Planning; NLP overview; Robotics

3. Database Management Systems

Unit 1 – Database Foundations: DBMS architecture; Data models; Schema and instances; Data independence

Unit 2 – Relational Model: Relations; Keys; Constraints; Relational algebra

Unit 3 – SQL: DDL; DML; Queries; Joins; Subqueries; Views

Unit 4 – Design: ER modeling; Functional dependencies; Normalization; Decomposition

Unit 5 – Transactions: ACID; Concurrency; Locking; Deadlocks; Recovery

4. Machine Learning

Unit 1 – ML Foundations: Learning paradigms; Data preprocessing; Feature engineering; Train/test split

Unit 2 – Regression: Linear regression; Polynomial regression; Regularization; Metrics

Unit 3 – Classification: Logistic regression; KNN; Naive Bayes; SVM

Unit 4 – Trees and Ensembles: Decision trees; Random forests; Boosting; Bagging

Unit 5 – Unsupervised Learning: K-means; Hierarchical clustering; PCA; Anomaly detection

5. Computer Networks

Unit 1 – Networking Basics: Network types; Topologies; OSI model; TCP/IP model

Unit 2 – Data Link: Framing; Error control; MAC; Ethernet

Unit 3 – Network Layer: IP addressing; Subnetting; Routing; IPv4/IPv6

Unit 4 – Transport: TCP; UDP; Flow control; Congestion control

Unit 5 – Applications: DNS; HTTP/HTTPS; Email; DHCP; Network security basics

6. Software Engineering

Unit 1 – Software Process: Software characteristics; Process models; Agile; DevOps overview

Unit 2 – Requirements: Elicitation; SRS; Use cases; Validation

Unit 3 – Design: Architecture; Modularity; UML; Design principles

Unit 4 – Testing: Test levels; Black-box testing; White-box testing; Automation concepts

Unit 5 – Management: Estimation; Scheduling; Risk; Configuration management; Quality

Semester 5

No.	Subject
1	Deep Learning
2	Computer Vision
3	Natural Language Processing
4	Data Mining
5	Optimization
6	Elective

1. Deep Learning

Unit 1 – Neural Networks: Perceptron; Activation functions; Loss functions; Backpropagation

Unit 2 – Optimization: Gradient descent; Learning rates; Regularization; Batch normalization

Unit 3 – CNN: Convolution; Pooling; Architectures; Transfer learning

Unit 4 – Sequence Models: RNN; LSTM; GRU; Sequence prediction

Unit 5 – Generative Models: Autoencoders; GANs; Transformers overview; Generative applications

2. Computer Vision

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Natural Language Processing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Data Mining

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Optimization

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 6

No.	Subject
1	Reinforcement Learning
2	Generative AI
3	Big Data Analytics
4	Cloud AI
5	IoT/Edge AI
6	Elective

1. Reinforcement Learning

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Generative AI

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Big Data Analytics

Unit 1 – Big Data: Characteristics; Data sources; Distributed storage; Batch vs stream

Unit 2 – Distributed Processing: MapReduce; Spark concepts; Data partitioning

Unit 3 – NoSQL: Key-value; Document; Column; Graph databases

Unit 4 – Analytics: Data preparation; Descriptive analytics; Predictive analytics; Visualization

Unit 5 – Platforms: Data lakes; Pipelines; Governance; Security

4. Cloud AI

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. IoT/Edge AI

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 7

No.	Subject
1	Advanced Deep Learning
2	Transformers
3	MLOps
4	Responsible AI
5	Research Project I
6	Elective

1. Advanced Deep Learning

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Transformers

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. MLOps

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Responsible AI

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Research Project I

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 8

No.	Subject
1	Major AI Project
2	Internship
3	Seminar
4	Elective
5	Thesis/Report
6	Viva

1. Major AI Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

2. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics

Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation

Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment

Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting

Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

3. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

4. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Thesis/Report

Unit 1 – Research Problem: Problem; Objectives; Scope

Unit 2 – Literature and Method: Literature review; Methodology; Experimental setup

Unit 3 – Results: Results; Analysis; Comparison

Unit 4 – Conclusions: Conclusions; Contributions; Limitations

Unit 5 – Future Work: Extensions; Applications; Publication plan

6. Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications

Unit 2 – Specialization: Advanced concepts; Problem solving

Unit 3 – Project: Architecture; Implementation; Testing; Results

Unit 4 – Professional Practice: Standards; Ethics; Industry tools

Unit 5 – Defense: Technical questions; Future scope; Communication

B.Tech – Data Science

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Programming Fundamentals
5	Engineering Graphics
6	Basic Electrical Engineering

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

5. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

6. Basic Electrical Engineering

Unit 1 – DC Circuits: Voltage and current; Ohm's law; KCL/KVL; Network reduction

Unit 2 – AC Circuits: Sinusoids; Phasors; Impedance; Power factor; Three-phase systems

Unit 3 – Machines: Transformers; DC machines; Induction motors; Synchronous machines

Unit 4 – Measurements: Meters; Bridges; Errors; Power and energy measurement

Unit 5 – Safety and Power: Earthing; Protection; Energy systems; Electrical safety

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Data Structures
3	OOP
4	Probability & Statistics
5	Communication Skills
6	Workshop

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Data Structures

Unit 1 – Linear Structures: Arrays; Linked lists; Stacks; Queues; Circular queues

Unit 2 – Trees: Binary trees; BST; AVL trees; Heaps; Tree traversal

Unit 3 – Graphs: Graph representations; BFS; DFS; Shortest paths; Spanning trees

Unit 4 – Hashing: Hash functions; Collision handling; Hash tables

Unit 5 – Applications: Searching; Sorting; Priority queues; Complexity analysis

3. OOP

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Probability & Statistics

Unit 1: Probability axioms; Conditional probability; Bayes theorem

Unit 2: Random variables; Distributions

Unit 3: Expectation; Variance; Moments

Unit 4: Sampling; Estimation

Unit 5: Hypothesis testing; Regression; Correlation

5. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Workshop

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 3

No.	Subject
1	Linear Algebra
2	Probability & Statistics
3	Data Structures
4	DBMS
5	Python for Data Science
6	Data Visualization

1. Linear Algebra

Unit 1: Matrices; Vector spaces; Linear transformations

Unit 2: Eigenvalues; Eigenvectors; Diagonalization

Unit 3: Inner products; Orthogonality

Unit 4: Numerical linear algebra

Unit 5: Applications to ML

2. Probability & Statistics

Unit 1: Probability axioms; Conditional probability; Bayes theorem

Unit 2: Random variables; Distributions

Unit 3: Expectation; Variance; Moments

Unit 4: Sampling; Estimation

Unit 5: Hypothesis testing; Regression; Correlation

3. Data Structures

Unit 1 – Linear Structures: Arrays; Linked lists; Stacks; Queues; Circular queues

Unit 2 – Trees: Binary trees; BST; AVL trees; Heaps; Tree traversal

Unit 3 – Graphs: Graph representations; BFS; DFS; Shortest paths; Spanning trees

Unit 4 – Hashing: Hash functions; Collision handling; Hash tables

Unit 5 – Applications: Searching; Sorting; Priority queues; Complexity analysis

4. DBMS

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Python for Data Science

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Data Visualization

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 4

No.	Subject
1	Algorithms
2	Machine Learning
3	Data Warehousing
4	Computer Networks
5	Operating Systems
6	Software Engineering

1. Algorithms

Unit 1 – Algorithm Analysis: Asymptotic notation; Time complexity; Space complexity; Recurrences

Unit 2 – Divide and Conquer: Binary search; Merge sort; Quick sort; Strassen concept

Unit 3 – Greedy Methods: Activity selection; Knapsack; Huffman coding; MST

Unit 4 – Dynamic Programming: Optimal substructure; LCS; Knapsack; Matrix-chain multiplication

Unit 5 – Graph and Advanced Algorithms: Shortest paths; Network flow; Backtracking; Branch and bound

2. Machine Learning

Unit 1 – ML Foundations: Learning paradigms; Data preprocessing; Feature engineering; Train/test split

Unit 2 – Regression: Linear regression; Polynomial regression; Regularization; Metrics

Unit 3 – Classification: Logistic regression; KNN; Naive Bayes; SVM

Unit 4 – Trees and Ensembles: Decision trees; Random forests; Boosting; Bagging

Unit 5 – Unsupervised Learning: K-means; Hierarchical clustering; PCA; Anomaly detection

3. Data Warehousing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Computer Networks

Unit 1 – Networking Basics: Network types; Topologies; OSI model; TCP/IP model

Unit 2 – Data Link: Framing; Error control; MAC; Ethernet

Unit 3 – Network Layer: IP addressing; Subnetting; Routing; IPv4/IPv6

Unit 4 – Transport: TCP; UDP; Flow control; Congestion control

Unit 5 – Applications: DNS; HTTP/HTTPS; Email; DHCP; Network security basics

5. Operating Systems

Unit 1 – OS Fundamentals: OS services; System calls; Processes; Threads

Unit 2 – CPU Management: Scheduling; Synchronization; Critical sections; Deadlocks

Unit 3 – Memory: Allocation; Paging; Segmentation; Virtual memory

Unit 4 – Storage: File systems; Disk scheduling; I/O systems

Unit 5 – Security: Protection; Access control; Authentication; OS security

6. Software Engineering

Unit 1 – Software Process: Software characteristics; Process models; Agile; DevOps overview

Unit 2 – Requirements: Elicitation; SRS; Use cases; Validation

Unit 3 – Design: Architecture; Modularity; UML; Design principles

Unit 4 – Testing: Test levels; Black-box testing; White-box testing; Automation concepts

Unit 5 – Management: Estimation; Scheduling; Risk; Configuration management; Quality

Semester 5

No.	Subject
1	Data Mining
2	Big Data Analytics
3	Deep Learning
4	Cloud Computing
5	NLP
6	Elective

1. Data Mining

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Big Data Analytics

Unit 1 – Big Data: Characteristics; Data sources; Distributed storage; Batch vs stream

Unit 2 – Distributed Processing: MapReduce; Spark concepts; Data partitioning

Unit 3 – NoSQL: Key-value; Document; Column; Graph databases

Unit 4 – Analytics: Data preparation; Descriptive analytics; Predictive analytics; Visualization

Unit 5 – Platforms: Data lakes; Pipelines; Governance; Security

3. Deep Learning

Unit 1 – Neural Networks: Perceptron; Activation functions; Loss functions; Backpropagation

Unit 2 – Optimization: Gradient descent; Learning rates; Regularization; Batch normalization

Unit 3 – CNN: Convolution; Pooling; Architectures; Transfer learning

Unit 4 – Sequence Models: RNN; LSTM; GRU; Sequence prediction

Unit 5 – Generative Models: Autoencoders; GANs; Transformers overview; Generative applications

4. Cloud Computing

Unit 1 – Cloud Foundations: Service models; Deployment models; Virtualization; Elasticity

Unit 2 – Cloud Architecture: Compute; Storage; Networking; Identity

Unit 3 – Cloud Platforms: IaaS; PaaS; SaaS; Containers

Unit 4 – Cloud Security: IAM; Encryption; Security monitoring; Compliance

Unit 5 – Cloud Applications: Serverless; Autoscaling; DevOps; Cost management

5. NLP

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 6

No.	Subject
1	Advanced ML
2	Time Series
3	Business Analytics
4	Distributed Data Processing
5	Data Engineering
6	Elective

1. Advanced ML

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Time Series

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Business Analytics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Distributed Data Processing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Data Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 7

No.	Subject
1	MLOps
2	Generative AI
3	Advanced Analytics
4	Project Phase I
5	Elective
6	Internship

1. MLOps

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Generative AI

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Advanced Analytics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Project Phase I

Unit 1 – Problem Definition: Problem statement; Objectives; Requirements

Unit 2 – Literature Review: Research papers; Existing systems; Gap analysis

Unit 3 – System Design: Architecture; Modules; Technology selection

Unit 4 – Initial Implementation: Prototype; Testing; Initial results

Unit 5 – Project Report: Documentation; Presentation; Future plan

5. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics

Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation

Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment

Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting

Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

Semester 8

No.	Subject
1	Major Data Science Project
2	Internship
3	Seminar
4	Elective
5	Report
6	Viva

1. Major Data Science Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

2. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics

Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation

Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment

Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting

Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

3. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

4. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Report

Unit 1 – Structure: Title; Abstract; Introduction

Unit 2 – Literature: Related work; References; Research gap

Unit 3 – Methodology: Design; Implementation; Evaluation

Unit 4 – Results: Results; Discussion; Limitations

Unit 5 – Conclusion: Conclusions; Future work; References

6. Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications

Unit 2 – Specialization: Advanced concepts; Problem solving

Unit 3 – Project: Architecture; Implementation; Testing; Results

Unit 4 – Professional Practice: Standards; Ethics; Industry tools

Unit 5 – Defense: Technical questions; Future scope; Communication

B.Tech – Electronics & Communication Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Basic Electrical Engineering
5	Engineering Graphics
6	Programming Fundamentals

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Basic Electrical Engineering

Unit 1 – DC Circuits: Voltage and current; Ohm's law; KCL/KVL; Network reduction

Unit 2 – AC Circuits: Sinusoids; Phasors; Impedance; Power factor; Three-phase systems

Unit 3 – Machines: Transformers; DC machines; Induction motors; Synchronous machines

Unit 4 – Measurements: Meters; Bridges; Errors; Power and energy measurement

Unit 5 – Safety and Power: Earthing; Protection; Energy systems; Electrical safety

5. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

6. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Circuit Theory
3	Electronic Devices
4	Digital Electronics
5	Communication Skills
6	Workshop

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Circuit Theory

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Electronic Devices

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates

Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders

Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers

Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic

Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

5. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Workshop

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 3

No.	Subject
1	Signals and Systems
2	Analog Electronics
3	Digital Electronics
4	Network Theory
5	Microprocessors
6	Electromagnetic Theory

1. Signals and Systems

Unit 1 – Signals: Continuous/discrete signals; Energy and power; Basic operations

Unit 2 – Systems: LTI systems; Convolution; Causality; Stability

Unit 3 – Fourier Analysis: Fourier series; Fourier transform; Properties

Unit 4 – Laplace/Z Transform: Laplace transform; Z transform; ROC; Applications

Unit 5 – Sampling: Sampling theorem; Reconstruction; Aliasing; Practical systems

2. Analog Electronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates

Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders

Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers

Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic

Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

4. Network Theory

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Microprocessors

Unit 1: Architecture; Instruction set

Unit 2: Assembly programming

Unit 3: Memory and I/O

Unit 4: Interrupts; Timers

Unit 5: Interfacing; Applications

6. Electromagnetic Theory

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 4

No.	Subject
1	Communication Systems
2	Microcontrollers
3	Control Systems
4	Digital Signal Processing
5	Electronic Measurements
6	Probability & Statistics

1. Communication Systems

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Microcontrollers

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Control Systems

Unit 1: Modeling; Transfer functions; Block diagrams

Unit 2: Time response; Stability

Unit 3: Root locus; Frequency response

Unit 4: State-space methods

Unit 5: Controllers; Industrial applications

4. Digital Signal Processing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Electronic Measurements

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Probability & Statistics

Unit 1: Probability axioms; Conditional probability; Bayes theorem

Unit 2: Random variables; Distributions

Unit 3: Expectation; Variance; Moments

Unit 4: Sampling; Estimation

Unit 5: Hypothesis testing; Regression; Correlation

Semester 5

No.	Subject
1	VLSI Design
2	Embedded Systems
3	Wireless Communication
4	Antenna Theory
5	Computer Networks
6	Elective

1. VLSI Design

Unit 1: CMOS basics; Logic design

Unit 2: Combinational/sequential VLSI

Unit 3: Physical design

Unit 4: Timing and power

Unit 5: Verification; FPGA/ASIC flow

2. Embedded Systems

Unit 1: Embedded architecture; Microcontrollers

Unit 2: Real-time concepts; RTOS

Unit 3: Interfaces; Peripherals

Unit 4: Embedded networking

Unit 5: Testing; Deployment

3. Wireless Communication

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Antenna Theory

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Computer Networks

Unit 1 – Networking Basics: Network types; Topologies; OSI model; TCP/IP model

Unit 2 – Data Link: Framing; Error control; MAC; Ethernet

Unit 3 – Network Layer: IP addressing; Subnetting; Routing; IPv4/IPv6

Unit 4 – Transport: TCP; UDP; Flow control; Congestion control

Unit 5 – Applications: DNS; HTTP/HTTPS; Email; DHCP; Network security basics

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 6

No.	Subject
1	Advanced Communication
2	Optical Communication
3	RF/Microwave
4	IoT
5	Image Processing
6	Elective

1. Advanced Communication

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Optical Communication

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. RF/Microwave

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. IoT

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Image Processing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 7

No.	Subject
1	5G/6G Concepts
2	Advanced VLSI/FPGA
3	Embedded AI
4	Project Phase I
5	Elective
6	Internship

1. 5G/6G Concepts

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced VLSI/FPGA

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Embedded AI

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Project Phase I

Unit 1 – Problem Definition: Problem statement; Objectives; Requirements

Unit 2 – Literature Review: Research papers; Existing systems; Gap analysis

Unit 3 – System Design: Architecture; Modules; Technology selection

Unit 4 – Initial Implementation: Prototype; Testing; Initial results

Unit 5 – Project Report: Documentation; Presentation; Future plan

5. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics

Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation

Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment

Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting

Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

Semester 8

No.	Subject
1	Major Project
2	Internship
3	Seminar
4	Elective
5	Report
6	Viva

1. Major Project

Unit 1 – Problem and Requirements: Problem statement; Objectives; Scope; Requirements
Unit 2 – Architecture and Design: System architecture; Modules; Data flow; Technology stack
Unit 3 – Implementation: Development; Integration; Testing; Deployment
Unit 4 – Results: Evaluation; Metrics; Comparison; Limitations
Unit 5 – Documentation and Viva: Final report; Presentation; Demonstration; Future work

2. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics
Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation
Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment
Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting
Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

3. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope
Unit 2 – Literature: Research papers; Comparison; Gap identification
Unit 3 – Technical Study: Methodology; Architecture; Results
Unit 4 – Presentation: Slides; Technical explanation; Questions
Unit 5 – Report: Citation; References; Summary; Future scope

4. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture
Unit 2 – Methods: Algorithms; Design methods; Implementation
Unit 3 – Applications: Use cases; Tools; Case studies
Unit 4 – Evaluation: Performance; Testing; Security/quality
Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Report

Unit 1 – Structure: Title; Abstract; Introduction
Unit 2 – Literature: Related work; References; Research gap
Unit 3 – Methodology: Design; Implementation; Evaluation
Unit 4 – Results: Results; Discussion; Limitations
Unit 5 – Conclusion: Conclusions; Future work; References

6. Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications
Unit 2 – Specialization: Advanced concepts; Problem solving
Unit 3 – Project: Architecture; Implementation; Testing; Results
Unit 4 – Professional Practice: Standards; Ethics; Industry tools
Unit 5 – Defense: Technical questions; Future scope; Communication

B.Tech – Electrical & Electronics Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Basic Electrical Engineering
5	Engineering Graphics
6	Programming Fundamentals

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Basic Electrical Engineering

Unit 1 – DC Circuits: Voltage and current; Ohm's law; KCL/KVL; Network reduction

Unit 2 – AC Circuits: Sinusoids; Phasors; Impedance; Power factor; Three-phase systems

Unit 3 – Machines: Transformers; DC machines; Induction motors; Synchronous machines

Unit 4 – Measurements: Meters; Bridges; Errors; Power and energy measurement

Unit 5 – Safety and Power: Earthing; Protection; Energy systems; Electrical safety

5. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

6. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Circuit Theory
3	Digital Electronics
4	Electrical Measurements
5	Communication Skills
6	Workshop

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Circuit Theory

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates

Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders

Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers

Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic

Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

4. Electrical Measurements

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Workshop

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 3

No.	Subject
1	Electrical Machines I
2	Power Systems I
3	Analog Electronics
4	Signals and Systems
5	Electromagnetic Fields
6	Control Basics

1. Electrical Machines I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Power Systems I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Analog Electronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Signals and Systems

Unit 1 – Signals: Continuous/discrete signals; Energy and power; Basic operations

Unit 2 – Systems: LTI systems; Convolution; Causality; Stability

Unit 3 – Fourier Analysis: Fourier series; Fourier transform; Properties

Unit 4 – Laplace/Z Transform: Laplace transform; Z transform; ROC; Applications

Unit 5 – Sampling: Sampling theorem; Reconstruction; Aliasing; Practical systems

5. Electromagnetic Fields

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Control Basics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 4

No.	Subject
1	Electrical Machines II
2	Power Systems II
3	Power Electronics
4	Control Systems
5	Microprocessors
6	Measurements

1. Electrical Machines II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Power Systems II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Power Electronics

Unit 1: Power semiconductor devices

Unit 2: Controlled rectifiers

Unit 3: DC-DC converters

Unit 4: Inverters

Unit 5: Drives; Applications

4. Control Systems

Unit 1: Modeling; Transfer functions; Block diagrams

Unit 2: Time response; Stability

Unit 3: Root locus; Frequency response

Unit 4: State-space methods

Unit 5: Controllers; Industrial applications

5. Microprocessors

Unit 1: Architecture; Instruction set

Unit 2: Assembly programming

Unit 3: Memory and I/O

Unit 4: Interrupts; Timers

Unit 5: Interfacing; Applications

6. Measurements

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 5

No.	Subject
1	Power System Analysis
2	Renewable Energy
3	Industrial Drives
4	Electrical Protection
5	Embedded Systems
6	Elective

1. Power System Analysis

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Renewable Energy

Unit 1: Solar energy

Unit 2: Wind energy

Unit 3: Hydro and biomass

Unit 4: Fuel cells; Geothermal

Unit 5: Grid integration; Storage

3. Industrial Drives

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Electrical Protection

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Embedded Systems

Unit 1: Embedded architecture; Microcontrollers

Unit 2: Real-time concepts; RTOS

Unit 3: Interfaces; Peripherals

Unit 4: Embedded networking

Unit 5: Testing; Deployment

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 6

No.	Subject
1	Smart Grid
2	Electric Vehicles
3	Power Quality
4	High Voltage Engineering
5	IoT for Energy
6	Elective

1. Smart Grid

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Electric Vehicles

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Power Quality

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. High Voltage Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. IoT for Energy

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 7

No.	Subject
1	Advanced Power Electronics
2	Energy Storage
3	Grid Integration
4	Project Phase I
5	Elective
6	Internship

1. Advanced Power Electronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Energy Storage

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Grid Integration

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Project Phase I

Unit 1 – Problem Definition: Problem statement; Objectives; Requirements

Unit 2 – Literature Review: Research papers; Existing systems; Gap analysis

Unit 3 – System Design: Architecture; Modules; Technology selection

Unit 4 – Initial Implementation: Prototype; Testing; Initial results

Unit 5 – Project Report: Documentation; Presentation; Future plan

5. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics

Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation

Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment

Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting

Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

Semester 8

No.	Subject
1	Major Project
2	Internship
3	Seminar
4	Elective
5	Report
6	Viva

1. Major Project

Unit 1 – Problem and Requirements: Problem statement; Objectives; Scope; Requirements
Unit 2 – Architecture and Design: System architecture; Modules; Data flow; Technology stack
Unit 3 – Implementation: Development; Integration; Testing; Deployment
Unit 4 – Results: Evaluation; Metrics; Comparison; Limitations
Unit 5 – Documentation and Viva: Final report; Presentation; Demonstration; Future work

2. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics
Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation
Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment
Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting
Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

3. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope
Unit 2 – Literature: Research papers; Comparison; Gap identification
Unit 3 – Technical Study: Methodology; Architecture; Results
Unit 4 – Presentation: Slides; Technical explanation; Questions
Unit 5 – Report: Citation; References; Summary; Future scope

4. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture
Unit 2 – Methods: Algorithms; Design methods; Implementation
Unit 3 – Applications: Use cases; Tools; Case studies
Unit 4 – Evaluation: Performance; Testing; Security/quality
Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Report

Unit 1 – Structure: Title; Abstract; Introduction
Unit 2 – Literature: Related work; References; Research gap
Unit 3 – Methodology: Design; Implementation; Evaluation
Unit 4 – Results: Results; Discussion; Limitations
Unit 5 – Conclusion: Conclusions; Future work; References

6. Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications
Unit 2 – Specialization: Advanced concepts; Problem solving
Unit 3 – Project: Architecture; Implementation; Testing; Results
Unit 4 – Professional Practice: Standards; Ethics; Industry tools
Unit 5 – Defense: Technical questions; Future scope; Communication

B.Tech – Mechanical Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Engineering Mechanics
6	Programming Fundamentals

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Engineering Mechanics

Unit 1 – Statics: Forces and moments; Free-body diagrams; Equilibrium; Friction

Unit 2 – Structures: Trusses; Frames; Centroids; Moments of inertia

Unit 3 – Kinematics: Particle motion; Rigid body motion; Relative motion

Unit 4 – Dynamics: Newton's laws; Work-energy; Impulse-momentum

Unit 5 – Applications: Virtual work; Simple machines; Engineering applications

6. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Thermodynamics
3	Strength of Materials
4	Basic Electrical
5	Communication Skills
6	Workshop

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Thermodynamics

Unit 1 – Fundamentals: Systems and properties; Processes; Zeroth law; First law

Unit 2 – Second Law: Entropy; Reversibility; Carnot cycle; Clausius inequality

Unit 3 – Properties: Pure substances; Steam tables; Ideal gases; Real gases

Unit 4 – Cycles: Otto; Diesel; Dual; Rankine cycles

Unit 5 – Applications: Nozzles; Compressors; Turbines; Heat engines

3. Strength of Materials

Unit 1 – Stress and Strain: Normal stress; Shear stress; Strain; Hooke's law; Elastic constants

Unit 2 – Bending: Shear force; Bending moment; Bending stress; Beam deflection

Unit 3 – Torsion: Torsion equation; Shafts; Power transmission

Unit 4 – Columns: Buckling; Euler theory; Rankine theory

Unit 5 – Failure: Principal stresses; Mohr's circle; Failure theories

4. Basic Electrical

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Workshop

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 3

No.	Subject
1	Fluid Mechanics
2	Theory of Machines
3	Manufacturing Processes
4	Materials Science
5	Machine Drawing
6	Thermal Engineering

1. Fluid Mechanics

Unit 1: Fluid properties; Pressure

Unit 2: Fluid kinematics

Unit 3: Bernoulli equation

Unit 4: Pipe flow

Unit 5: Boundary layers; Turbomachinery basics

2. Theory of Machines

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Manufacturing Processes

Unit 1: Casting

Unit 2: Forming

Unit 3: Machining

Unit 4: Joining

Unit 5: Advanced manufacturing

4. Materials Science

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Machine Drawing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Thermal Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 4

No.	Subject
1	Heat Transfer
2	Machine Design
3	Metrology
4	Dynamics of Machines
5	CAD/CAM
6	Manufacturing Technology

1. Heat Transfer

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Machine Design

Unit 1: Design principles; Failure theories

Unit 2: Shafts; Keys; Couplings

Unit 3: Gears

Unit 4: Bearings; Springs

Unit 5: Design optimization; CAD

3. Metrology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Dynamics of Machines

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. CAD/CAM

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Manufacturing Technology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 5

No.	Subject
1	IC Engines
2	Refrigeration and Air Conditioning
3	Finite Element Basics
4	Industrial Engineering
5	Robotics/Mechatronics
6	Elective

1. IC Engines

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Refrigeration and Air Conditioning

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Finite Element Basics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Industrial Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Robotics/Mechatronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 6

No.	Subject
1	CNC and Automation
2	Advanced Manufacturing
3	Additive Manufacturing
4	Computational Mechanics
5	Automobile Engineering
6	Elective

1. CNC and Automation

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Manufacturing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Additive Manufacturing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Computational Mechanics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Automobile Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 7

No.	Subject
1	Smart Manufacturing
2	Industrial IoT
3	Advanced CAD/CAE
4	Project Phase I
5	Elective
6	Internship

1. Smart Manufacturing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Industrial IoT

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Advanced CAD/CAE

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Project Phase I

Unit 1 – Problem Definition: Problem statement; Objectives; Requirements

Unit 2 – Literature Review: Research papers; Existing systems; Gap analysis

Unit 3 – System Design: Architecture; Modules; Technology selection

Unit 4 – Initial Implementation: Prototype; Testing; Initial results

Unit 5 – Project Report: Documentation; Presentation; Future plan

5. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics

Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation

Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment

Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting

Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

Semester 8

No.	Subject
1	Major Project
2	Internship
3	Seminar
4	Elective
5	Report
6	Viva

1. Major Project

Unit 1 – Problem and Requirements: Problem statement; Objectives; Scope; Requirements
Unit 2 – Architecture and Design: System architecture; Modules; Data flow; Technology stack
Unit 3 – Implementation: Development; Integration; Testing; Deployment
Unit 4 – Results: Evaluation; Metrics; Comparison; Limitations
Unit 5 – Documentation and Viva: Final report; Presentation; Demonstration; Future work

2. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics
Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation
Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment
Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting
Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

3. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope
Unit 2 – Literature: Research papers; Comparison; Gap identification
Unit 3 – Technical Study: Methodology; Architecture; Results
Unit 4 – Presentation: Slides; Technical explanation; Questions
Unit 5 – Report: Citation; References; Summary; Future scope

4. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture
Unit 2 – Methods: Algorithms; Design methods; Implementation
Unit 3 – Applications: Use cases; Tools; Case studies
Unit 4 – Evaluation: Performance; Testing; Security/quality
Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Report

Unit 1 – Structure: Title; Abstract; Introduction
Unit 2 – Literature: Related work; References; Research gap
Unit 3 – Methodology: Design; Implementation; Evaluation
Unit 4 – Results: Results; Discussion; Limitations
Unit 5 – Conclusion: Conclusions; Future work; References

6. Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications
Unit 2 – Specialization: Advanced concepts; Problem solving
Unit 3 – Project: Architecture; Implementation; Testing; Results
Unit 4 – Professional Practice: Standards; Ethics; Industry tools
Unit 5 – Defense: Technical questions; Future scope; Communication

B.Tech – Civil Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Engineering Mechanics
6	Basic Electrical Engineering

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Engineering Mechanics

Unit 1 – Statics: Forces and moments; Free-body diagrams; Equilibrium; Friction

Unit 2 – Structures: Trusses; Frames; Centroids; Moments of inertia

Unit 3 – Kinematics: Particle motion; Rigid body motion; Relative motion

Unit 4 – Dynamics: Newton's laws; Work-energy; Impulse-momentum

Unit 5 – Applications: Virtual work; Simple machines; Engineering applications

6. Basic Electrical Engineering

Unit 1 – DC Circuits: Voltage and current; Ohm's law; KCL/KVL; Network reduction

Unit 2 – AC Circuits: Sinusoids; Phasors; Impedance; Power factor; Three-phase systems

Unit 3 – Machines: Transformers; DC machines; Induction motors; Synchronous machines

Unit 4 – Measurements: Meters; Bridges; Errors; Power and energy measurement

Unit 5 – Safety and Power: Earthing; Protection; Energy systems; Electrical safety

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Surveying
3	Building Materials
4	Strength of Materials
5	Communication Skills
6	Workshop

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Surveying

Unit 1: Chain and compass surveying

Unit 2: Leveling

Unit 3: Theodolite

Unit 4: Tacheometry

Unit 5: Total station; GPS/GNSS

3. Building Materials

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Strength of Materials

Unit 1 – Stress and Strain: Normal stress; Shear stress; Strain; Hooke's law; Elastic constants

Unit 2 – Bending: Shear force; Bending moment; Bending stress; Beam deflection

Unit 3 – Torsion: Torsion equation; Shafts; Power transmission

Unit 4 – Columns: Buckling; Euler theory; Rankine theory

Unit 5 – Failure: Principal stresses; Mohr's circle; Failure theories

5. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Workshop

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 3

No.	Subject
1	Structural Analysis
2	Fluid Mechanics
3	Concrete Technology
4	Geotechnical Engineering
5	Surveying II
6	Transportation Engineering

1. Structural Analysis

Unit 1: Determinate structures

Unit 2: Deflection

Unit 3: Indeterminate structures

Unit 4: Matrix methods

Unit 5: Influence lines; Applications

2. Fluid Mechanics

Unit 1: Fluid properties; Pressure

Unit 2: Fluid kinematics

Unit 3: Bernoulli equation

Unit 4: Pipe flow

Unit 5: Boundary layers; Turbomachinery basics

3. Concrete Technology

Unit 1: Cement; Aggregates

Unit 2: Fresh concrete

Unit 3: Mix design

Unit 4: Strength and durability

Unit 5: Special concretes

4. Geotechnical Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Surveying II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Transportation Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 4

No.	Subject
1	RCC Design
2	Steel Structures
3	Hydraulics
4	Environmental Engineering
5	Geotechnical Engineering II
6	Construction Materials

1. RCC Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Steel Structures

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Hydraulics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Environmental Engineering

Unit 1: Water quality

Unit 2: Water treatment

Unit 3: Wastewater treatment

Unit 4: Air pollution

Unit 5: Solid waste management

5. Geotechnical Engineering II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Construction Materials

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 5

No.	Subject
1	Structural Design
2	Water Resources
3	Highway Engineering
4	Foundation Engineering
5	Construction Management
6	Elective

1. Structural Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Water Resources

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Highway Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Foundation Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Construction Management

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 6

No.	Subject
1	Advanced RCC/Steel
2	Earthquake Engineering
3	Urban Infrastructure
4	Environmental Management
5	GIS/Remote Sensing
6	Elective

1. Advanced RCC/Steel

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Earthquake Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Urban Infrastructure

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Environmental Management

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. GIS/Remote Sensing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 7

No.	Subject
1	Advanced Structural Analysis
2	Smart Infrastructure
3	Project Phase I
4	Elective
5	Internship
6	Seminar

1. Advanced Structural Analysis

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Smart Infrastructure

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Project Phase I

Unit 1 – Problem Definition: Problem statement; Objectives; Requirements

Unit 2 – Literature Review: Research papers; Existing systems; Gap analysis

Unit 3 – System Design: Architecture; Modules; Technology selection

Unit 4 – Initial Implementation: Prototype; Testing; Initial results

Unit 5 – Project Report: Documentation; Presentation; Future plan

4. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics

Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation

Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment

Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting

Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

6. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

Semester 8

No.	Subject
1	Major Project
2	Internship
3	Seminar
4	Elective
5	Report
6	Viva

1. Major Project

Unit 1 – Problem and Requirements: Problem statement; Objectives; Scope; Requirements
Unit 2 – Architecture and Design: System architecture; Modules; Data flow; Technology stack
Unit 3 – Implementation: Development; Integration; Testing; Deployment
Unit 4 – Results: Evaluation; Metrics; Comparison; Limitations
Unit 5 – Documentation and Viva: Final report; Presentation; Demonstration; Future work

2. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics
Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation
Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment
Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting
Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

3. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope
Unit 2 – Literature: Research papers; Comparison; Gap identification
Unit 3 – Technical Study: Methodology; Architecture; Results
Unit 4 – Presentation: Slides; Technical explanation; Questions
Unit 5 – Report: Citation; References; Summary; Future scope

4. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture
Unit 2 – Methods: Algorithms; Design methods; Implementation
Unit 3 – Applications: Use cases; Tools; Case studies
Unit 4 – Evaluation: Performance; Testing; Security/quality
Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Report

Unit 1 – Structure: Title; Abstract; Introduction
Unit 2 – Literature: Related work; References; Research gap
Unit 3 – Methodology: Design; Implementation; Evaluation
Unit 4 – Results: Results; Discussion; Limitations
Unit 5 – Conclusion: Conclusions; Future work; References

6. Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications
Unit 2 – Specialization: Advanced concepts; Problem solving
Unit 3 – Project: Architecture; Implementation; Testing; Results
Unit 4 – Professional Practice: Standards; Ethics; Industry tools
Unit 5 – Defense: Technical questions; Future scope; Communication

B.Tech – Mechatronics Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Engineering Mechanics
6	Programming Fundamentals

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Engineering Mechanics

Unit 1 – Statics: Forces and moments; Free-body diagrams; Equilibrium; Friction

Unit 2 – Structures: Trusses; Frames; Centroids; Moments of inertia

Unit 3 – Kinematics: Particle motion; Rigid body motion; Relative motion

Unit 4 – Dynamics: Newton's laws; Work-energy; Impulse-momentum

Unit 5 – Applications: Virtual work; Simple machines; Engineering applications

6. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Circuit Theory
3	Digital Electronics
4	Manufacturing
5	Communication Skills
6	Workshop

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Circuit Theory

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Digital Electronics

Unit 1 – Number Systems: Binary; Octal; Hexadecimal; Boolean algebra; Logic gates

Unit 2 – Combinational Logic: Adders; Subtractors; Multiplexers; Decoders; Encoders

Unit 3 – Sequential Logic: Latches; Flip-flops; Counters; Registers

Unit 4 – Memory: RAM; ROM; Cache concepts; Programmable logic

Unit 5 – Digital Design: State machines; Timing; A/D and D/A basics

4. Manufacturing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Workshop

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 3

No.	Subject
1	Sensors and Transducers
2	Microcontrollers
3	Control Systems
4	Mechanical Design
5	Fluid Power
6	Manufacturing

1. Sensors and Transducers

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Microcontrollers

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Control Systems

Unit 1: Modeling; Transfer functions; Block diagrams

Unit 2: Time response; Stability

Unit 3: Root locus; Frequency response

Unit 4: State-space methods

Unit 5: Controllers; Industrial applications

4. Mechanical Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Fluid Power

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Manufacturing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 4

No.	Subject
1	PLC and Automation
2	Robotics
3	Embedded Systems
4	CAD/CAM
5	Instrumentation
6	Machine Vision

1. PLC and Automation

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Robotics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Embedded Systems

Unit 1: Embedded architecture; Microcontrollers

Unit 2: Real-time concepts; RTOS

Unit 3: Interfaces; Peripherals

Unit 4: Embedded networking

Unit 5: Testing; Deployment

4. CAD/CAM

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Instrumentation

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Machine Vision

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 5

No.	Subject
1	Industrial Automation
2	Motion Control
3	IoT
4	Mechatronic System Design
5	Machine Learning Basics
6	Elective

1. Industrial Automation

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Motion Control

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. IoT

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Mechatronic System Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Machine Learning Basics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 6

No.	Subject
1	Advanced Robotics
2	Industrial IoT
3	Digital Twins
4	Smart Manufacturing
5	Computer Vision
6	Elective

1. Advanced Robotics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Industrial IoT

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Digital Twins

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Smart Manufacturing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Computer Vision

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 7

No.	Subject
1	Autonomous Systems
2	Advanced Control
3	Project Phase I
4	Elective
5	Internship
6	Seminar

1. Autonomous Systems

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Control

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Project Phase I

Unit 1 – Problem Definition: Problem statement; Objectives; Requirements

Unit 2 – Literature Review: Research papers; Existing systems; Gap analysis

Unit 3 – System Design: Architecture; Modules; Technology selection

Unit 4 – Initial Implementation: Prototype; Testing; Initial results

Unit 5 – Project Report: Documentation; Presentation; Future plan

4. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics

Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation

Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment

Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting

Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

6. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

Semester 8

No.	Subject
1	Major Project
2	Internship
3	Seminar
4	Elective
5	Report
6	Viva

1. Major Project

Unit 1 – Problem and Requirements: Problem statement; Objectives; Scope; Requirements
Unit 2 – Architecture and Design: System architecture; Modules; Data flow; Technology stack
Unit 3 – Implementation: Development; Integration; Testing; Deployment
Unit 4 – Results: Evaluation; Metrics; Comparison; Limitations
Unit 5 – Documentation and Viva: Final report; Presentation; Demonstration; Future work

2. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics
Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation
Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment
Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting
Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

3. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope
Unit 2 – Literature: Research papers; Comparison; Gap identification
Unit 3 – Technical Study: Methodology; Architecture; Results
Unit 4 – Presentation: Slides; Technical explanation; Questions
Unit 5 – Report: Citation; References; Summary; Future scope

4. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture
Unit 2 – Methods: Algorithms; Design methods; Implementation
Unit 3 – Applications: Use cases; Tools; Case studies
Unit 4 – Evaluation: Performance; Testing; Security/quality
Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Report

Unit 1 – Structure: Title; Abstract; Introduction
Unit 2 – Literature: Related work; References; Research gap
Unit 3 – Methodology: Design; Implementation; Evaluation
Unit 4 – Results: Results; Discussion; Limitations
Unit 5 – Conclusion: Conclusions; Future work; References

6. Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications
Unit 2 – Specialization: Advanced concepts; Problem solving
Unit 3 – Project: Architecture; Implementation; Testing; Results
Unit 4 – Professional Practice: Standards; Ethics; Industry tools
Unit 5 – Defense: Technical questions; Future scope; Communication

B.Tech – Automobile Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Engineering Mechanics
6	Programming Fundamentals

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Engineering Mechanics

Unit 1 – Statics: Forces and moments; Free-body diagrams; Equilibrium; Friction

Unit 2 – Structures: Trusses; Frames; Centroids; Moments of inertia

Unit 3 – Kinematics: Particle motion; Rigid body motion; Relative motion

Unit 4 – Dynamics: Newton's laws; Work-energy; Impulse-momentum

Unit 5 – Applications: Virtual work; Simple machines; Engineering applications

6. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Thermodynamics
3	Strength of Materials
4	Basic Electrical
5	Communication Skills
6	Workshop

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Thermodynamics

Unit 1 – Fundamentals: Systems and properties; Processes; Zeroth law; First law

Unit 2 – Second Law: Entropy; Reversibility; Carnot cycle; Clausius inequality

Unit 3 – Properties: Pure substances; Steam tables; Ideal gases; Real gases

Unit 4 – Cycles: Otto; Diesel; Dual; Rankine cycles

Unit 5 – Applications: Nozzles; Compressors; Turbines; Heat engines

3. Strength of Materials

Unit 1 – Stress and Strain: Normal stress; Shear stress; Strain; Hooke's law; Elastic constants

Unit 2 – Bending: Shear force; Bending moment; Bending stress; Beam deflection

Unit 3 – Torsion: Torsion equation; Shafts; Power transmission

Unit 4 – Columns: Buckling; Euler theory; Rankine theory

Unit 5 – Failure: Principal stresses; Mohr's circle; Failure theories

4. Basic Electrical

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Workshop

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 3

No.	Subject
1	Automotive Engines
2	Fluid Mechanics
3	Manufacturing Processes
4	Vehicle Chassis
5	Automotive Materials
6	Automotive Lab

1. Automotive Engines

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Fluid Mechanics

Unit 1: Fluid properties; Pressure

Unit 2: Fluid kinematics

Unit 3: Bernoulli equation

Unit 4: Pipe flow

Unit 5: Boundary layers; Turbomachinery basics

3. Manufacturing Processes

Unit 1: Casting

Unit 2: Forming

Unit 3: Machining

Unit 4: Joining

Unit 5: Advanced manufacturing

4. Vehicle Chassis

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Automotive Materials

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Automotive Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Automotive Transmission
2	Vehicle Dynamics
3	Automotive Electrical
4	Machine Design
5	Automotive Electronics
6	Laboratory

1. Automotive Transmission

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Vehicle Dynamics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Automotive Electrical

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Machine Design

Unit 1: Design principles; Failure theories

Unit 2: Shafts; Keys; Couplings

Unit 3: Gears

Unit 4: Bearings; Springs

Unit 5: Design optimization; CAD

5. Automotive Electronics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Emission Control
2	Vehicle Diagnostics
3	Hybrid Vehicles
4	EV Technology
5	Automotive Manufacturing
6	Elective

1. Emission Control

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Vehicle Diagnostics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Hybrid Vehicles

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. EV Technology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Automotive Manufacturing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 6

No.	Subject
1	Advanced EV Systems
2	Connected Vehicles
3	Automotive Safety
4	CAD/CAE
5	Industrial Management
6	Elective

1. Advanced EV Systems

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Connected Vehicles

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Automotive Safety

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. CAD/CAE

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Industrial Management

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 7

No.	Subject
1	Advanced Vehicle Systems
2	Project Phase I
3	Elective
4	Internship
5	Seminar
6	Industry Project

1. Advanced Vehicle Systems

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Project Phase I

Unit 1 – Problem Definition: Problem statement; Objectives; Requirements

Unit 2 – Literature Review: Research papers; Existing systems; Gap analysis

Unit 3 – System Design: Architecture; Modules; Technology selection

Unit 4 – Initial Implementation: Prototype; Testing; Initial results

Unit 5 – Project Report: Documentation; Presentation; Future plan

3. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics

Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation

Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment

Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting

Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

5. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

6. Industry Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

Semester 8

No.	Subject
1	Major Project
2	Internship
3	Seminar
4	Elective
5	Report
6	Viva

1. Major Project

Unit 1 – Problem and Requirements: Problem statement; Objectives; Scope; Requirements
Unit 2 – Architecture and Design: System architecture; Modules; Data flow; Technology stack
Unit 3 – Implementation: Development; Integration; Testing; Deployment
Unit 4 – Results: Evaluation; Metrics; Comparison; Limitations
Unit 5 – Documentation and Viva: Final report; Presentation; Demonstration; Future work

2. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics
Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation
Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment
Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting
Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

3. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope
Unit 2 – Literature: Research papers; Comparison; Gap identification
Unit 3 – Technical Study: Methodology; Architecture; Results
Unit 4 – Presentation: Slides; Technical explanation; Questions
Unit 5 – Report: Citation; References; Summary; Future scope

4. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture
Unit 2 – Methods: Algorithms; Design methods; Implementation
Unit 3 – Applications: Use cases; Tools; Case studies
Unit 4 – Evaluation: Performance; Testing; Security/quality
Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Report

Unit 1 – Structure: Title; Abstract; Introduction
Unit 2 – Literature: Related work; References; Research gap
Unit 3 – Methodology: Design; Implementation; Evaluation
Unit 4 – Results: Results; Discussion; Limitations
Unit 5 – Conclusion: Conclusions; Future work; References

6. Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications
Unit 2 – Specialization: Advanced concepts; Problem solving
Unit 3 – Project: Architecture; Implementation; Testing; Results
Unit 4 – Professional Practice: Standards; Ethics; Industry tools
Unit 5 – Defense: Technical questions; Future scope; Communication

B.Tech – Aerospace Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Engineering Mechanics
6	Programming Fundamentals

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Engineering Mechanics

Unit 1 – Statics: Forces and moments; Free-body diagrams; Equilibrium; Friction

Unit 2 – Structures: Trusses; Frames; Centroids; Moments of inertia

Unit 3 – Kinematics: Particle motion; Rigid body motion; Relative motion

Unit 4 – Dynamics: Newton's laws; Work-energy; Impulse-momentum

Unit 5 – Applications: Virtual work; Simple machines; Engineering applications

6. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Thermodynamics
3	Strength of Materials
4	Fluid Mechanics
5	Communication Skills
6	Workshop

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Thermodynamics

Unit 1 – Fundamentals: Systems and properties; Processes; Zeroth law; First law

Unit 2 – Second Law: Entropy; Reversibility; Carnot cycle; Clausius inequality

Unit 3 – Properties: Pure substances; Steam tables; Ideal gases; Real gases

Unit 4 – Cycles: Otto; Diesel; Dual; Rankine cycles

Unit 5 – Applications: Nozzles; Compressors; Turbines; Heat engines

3. Strength of Materials

Unit 1 – Stress and Strain: Normal stress; Shear stress; Strain; Hooke's law; Elastic constants

Unit 2 – Bending: Shear force; Bending moment; Bending stress; Beam deflection

Unit 3 – Torsion: Torsion equation; Shafts; Power transmission

Unit 4 – Columns: Buckling; Euler theory; Rankine theory

Unit 5 – Failure: Principal stresses; Mohr's circle; Failure theories

4. Fluid Mechanics

Unit 1: Fluid properties; Pressure

Unit 2: Fluid kinematics

Unit 3: Bernoulli equation

Unit 4: Pipe flow

Unit 5: Boundary layers; Turbomachinery basics

5. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Workshop

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 3

No.	Subject
1	Aerodynamics I
2	Aircraft Structures I
3	Propulsion I
4	Flight Mechanics
5	Materials
6	Laboratory

1. Aerodynamics I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Aircraft Structures I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Propulsion I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Flight Mechanics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Materials

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Aerodynamics II
2	Aircraft Structures II
3	Propulsion II
4	Avionics Basics
5	Control Systems
6	Laboratory

1. Aerodynamics II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Aircraft Structures II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Propulsion II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Avionics Basics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Control Systems

Unit 1: Modeling; Transfer functions; Block diagrams

Unit 2: Time response; Stability

Unit 3: Root locus; Frequency response

Unit 4: State-space methods

Unit 5: Controllers; Industrial applications

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Aerospace Manufacturing
2	Space Mechanics
3	CFD
4	Aircraft Design
5	Composite Materials
6	Elective

1. Aerospace Manufacturing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Space Mechanics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. CFD

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Aircraft Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Composite Materials

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 6

No.	Subject
1	Advanced Aerodynamics
2	UAV Systems
3	Rocket Propulsion
4	Aerospace Control
5	Systems Engineering
6	Elective

1. Advanced Aerodynamics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. UAV Systems

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Rocket Propulsion

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Aerospace Control

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Systems Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 7

No.	Subject
1	Advanced Aerospace Elective
2	Project Phase I
3	Internship
4	Seminar
5	Elective
6	Industry Project

1. Advanced Aerospace Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Project Phase I

Unit 1 – Problem Definition: Problem statement; Objectives; Requirements

Unit 2 – Literature Review: Research papers; Existing systems; Gap analysis

Unit 3 – System Design: Architecture; Modules; Technology selection

Unit 4 – Initial Implementation: Prototype; Testing; Initial results

Unit 5 – Project Report: Documentation; Presentation; Future plan

3. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics

Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation

Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment

Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting

Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

4. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

5. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Industry Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

Semester 8

No.	Subject
1	Major Project
2	Internship
3	Seminar
4	Elective
5	Report
6	Viva

1. Major Project

Unit 1 – Problem and Requirements: Problem statement; Objectives; Scope; Requirements
Unit 2 – Architecture and Design: System architecture; Modules; Data flow; Technology stack
Unit 3 – Implementation: Development; Integration; Testing; Deployment
Unit 4 – Results: Evaluation; Metrics; Comparison; Limitations
Unit 5 – Documentation and Viva: Final report; Presentation; Demonstration; Future work

2. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics
Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation
Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment
Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting
Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

3. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope
Unit 2 – Literature: Research papers; Comparison; Gap identification
Unit 3 – Technical Study: Methodology; Architecture; Results
Unit 4 – Presentation: Slides; Technical explanation; Questions
Unit 5 – Report: Citation; References; Summary; Future scope

4. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture
Unit 2 – Methods: Algorithms; Design methods; Implementation
Unit 3 – Applications: Use cases; Tools; Case studies
Unit 4 – Evaluation: Performance; Testing; Security/quality
Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Report

Unit 1 – Structure: Title; Abstract; Introduction
Unit 2 – Literature: Related work; References; Research gap
Unit 3 – Methodology: Design; Implementation; Evaluation
Unit 4 – Results: Results; Discussion; Limitations
Unit 5 – Conclusion: Conclusions; Future work; References

6. Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications
Unit 2 – Specialization: Advanced concepts; Problem solving
Unit 3 – Project: Architecture; Implementation; Testing; Results
Unit 4 – Professional Practice: Standards; Ethics; Industry tools
Unit 5 – Defense: Technical questions; Future scope; Communication

B.Tech – Biotechnology

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Biology for Engineers
5	Programming Fundamentals
6	Engineering Graphics

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Biology for Engineers

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

6. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Biochemistry
3	Cell Biology
4	Genetics
5	Communication Skills
6	Biotechnology Lab

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Biochemistry

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Cell Biology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Genetics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Biotechnology Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 3

No.	Subject
1	Microbiology
2	Molecular Biology
3	Bioprocess Principles
4	Bioinformatics
5	Biostatistics
6	Laboratory

1. Microbiology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Molecular Biology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Bioprocess Principles

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Bioinformatics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Biostatistics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Genetic Engineering
2	Immunology
3	Fermentation Technology
4	Downstream Processing
5	Instrumentation
6	Laboratory

1. Genetic Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Immunology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Fermentation Technology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Downstream Processing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Instrumentation

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Plant Biotechnology
2	Animal Biotechnology
3	Industrial Biotechnology
4	Food Biotechnology
5	Environmental Biotechnology
6	Elective

1. Plant Biotechnology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Animal Biotechnology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Industrial Biotechnology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Food Biotechnology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Environmental Biotechnology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 6

No.	Subject
1	Computational Biology
2	Bioprocess Design
3	Pharmaceutical Biotechnology
4	Biosensors
5	Entrepreneurship
6	Elective

1. Computational Biology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Bioprocess Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Pharmaceutical Biotechnology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Biosensors

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Entrepreneurship

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 7

No.	Subject
1	Advanced Biotechnology
2	Project Phase I
3	Internship
4	Seminar
5	Elective
6	Research Methods

1. Advanced Biotechnology

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Project Phase I

Unit 1 – Problem Definition: Problem statement; Objectives; Requirements

Unit 2 – Literature Review: Research papers; Existing systems; Gap analysis

Unit 3 – System Design: Architecture; Modules; Technology selection

Unit 4 – Initial Implementation: Prototype; Testing; Initial results

Unit 5 – Project Report: Documentation; Presentation; Future plan

3. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics

Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation

Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment

Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting

Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

4. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

5. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Research Methods

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 8

No.	Subject
1	Major Project
2	Internship
3	Seminar
4	Elective
5	Report
6	Viva

1. Major Project

Unit 1 – Problem and Requirements: Problem statement; Objectives; Scope; Requirements
Unit 2 – Architecture and Design: System architecture; Modules; Data flow; Technology stack
Unit 3 – Implementation: Development; Integration; Testing; Deployment
Unit 4 – Results: Evaluation; Metrics; Comparison; Limitations
Unit 5 – Documentation and Viva: Final report; Presentation; Demonstration; Future work

2. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics
Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation
Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment
Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting
Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

3. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope
Unit 2 – Literature: Research papers; Comparison; Gap identification
Unit 3 – Technical Study: Methodology; Architecture; Results
Unit 4 – Presentation: Slides; Technical explanation; Questions
Unit 5 – Report: Citation; References; Summary; Future scope

4. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture
Unit 2 – Methods: Algorithms; Design methods; Implementation
Unit 3 – Applications: Use cases; Tools; Case studies
Unit 4 – Evaluation: Performance; Testing; Security/quality
Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Report

Unit 1 – Structure: Title; Abstract; Introduction
Unit 2 – Literature: Related work; References; Research gap
Unit 3 – Methodology: Design; Implementation; Evaluation
Unit 4 – Results: Results; Discussion; Limitations
Unit 5 – Conclusion: Conclusions; Future work; References

6. Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications
Unit 2 – Specialization: Advanced concepts; Problem solving
Unit 3 – Project: Architecture; Implementation; Testing; Results
Unit 4 – Professional Practice: Standards; Ethics; Industry tools
Unit 5 – Defense: Technical questions; Future scope; Communication

B.Tech – Chemical Engineering

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Engineering Physics
3	Engineering Chemistry
4	Engineering Graphics
5	Programming Fundamentals
6	Workshop

1. Engineering Mathematics I

Unit 1 – Differential Calculus: Limits and continuity; Differentiation rules; Taylor and Maclaurin series; Partial derivatives; Maxima and minima

Unit 2 – Integral Calculus: Indefinite integrals; Definite integrals; Applications of integration; Improper integrals

Unit 3 – Differential Equations: First-order equations; Higher-order linear equations; Applications

Unit 4 – Matrices: Matrix operations; Determinants; Rank; Eigenvalues and eigenvectors; Linear systems

Unit 5 – Vector Calculus: Vectors; Gradient; Divergence; Curl; Line and surface integrals

2. Engineering Physics

Unit 1 – Mechanics: Kinematics; Dynamics; Work and energy; Momentum; Rotational motion

Unit 2 – Waves and Optics: Wave motion; Interference; Diffraction; Polarization; Lasers

Unit 3 – Electricity and Magnetism: Electrostatics; Current electricity; Magnetism; Electromagnetic induction

Unit 4 – Modern Physics: Quantum concepts; Photoelectric effect; Atomic models; Nuclear physics

Unit 5 – Materials: Crystal structures; Semiconductors; Magnetic materials; Nanomaterials

3. Engineering Chemistry

Unit 1 – Water Chemistry: Hardness; Softening; Boiler problems; Water treatment

Unit 2 – Electrochemistry: Electrolytes; Galvanic cells; Electrolysis; Corrosion

Unit 3 – Materials: Polymers; Ceramics; Composites; Nanomaterials

Unit 4 – Fuels and Energy: Solid fuels; Liquid fuels; Gaseous fuels; Calorific value; Batteries

Unit 5 – Green Chemistry: Pollution; Waste treatment; Renewable chemistry; Sustainable processes

4. Engineering Graphics

Unit 1 – Drawing Basics: Instruments; Geometric constructions; Scales; Dimensioning

Unit 2 – Projection: Points; Lines; Planes; Solids

Unit 3 – Sections: Sectional views; True shapes; Intersections

Unit 4 – Development: Development of surfaces; Patterns; Transitions

Unit 5 – CAD: 2D drafting; Layers; Constraints; Basic 3D modeling

5. Programming Fundamentals

Unit 1 – Programming Basics: Algorithms and flowcharts; Variables and data types; Operators; Input/output

Unit 2 – Control Structures: Conditionals; Loops; Nested control; Functions

Unit 3 – Structured Data: Arrays; Strings; Structures/records; Pointers/references

Unit 4 – Modular Programming: Functions; Recursion; Libraries/modules; File handling

Unit 5 – Problem Solving: Searching; Sorting basics; Debugging; Testing; Complexity awareness

6. Workshop

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Material and Energy Balance
3	Thermodynamics I
4	Fluid Mechanics
5	Communication Skills
6	Laboratory

1. Engineering Mathematics II

Unit 1 – Multivariable Calculus: Multiple integrals; Jacobian; Change of variables; Applications

Unit 2 – Differential Equations: Partial differential equations; Fourier series; Boundary-value problems

Unit 3 – Complex Variables: Complex functions; Analytic functions; Cauchy-Riemann equations; Contour integration

Unit 4 – Probability: Random variables; Distributions; Expectation; Variance

Unit 5 – Statistics: Sampling; Estimation; Hypothesis testing; Correlation and regression

2. Material and Energy Balance

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Thermodynamics I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Fluid Mechanics

Unit 1: Fluid properties; Pressure

Unit 2: Fluid kinematics

Unit 3: Bernoulli equation

Unit 4: Pipe flow

Unit 5: Boundary layers; Turbomachinery basics

5. Communication Skills

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 3

No.	Subject
1	Heat Transfer
2	Mass Transfer I
3	Chemical Reaction Engineering I
4	Process Calculations
5	Mechanical Operations
6	Laboratory

1. Heat Transfer

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Mass Transfer I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Chemical Reaction Engineering I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Process Calculations

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Mechanical Operations

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 4

No.	Subject
1	Mass Transfer II
2	Reaction Engineering II
3	Process Control
4	Process Instrumentation
5	Thermodynamics II
6	Laboratory

1. Mass Transfer II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Reaction Engineering II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Process Control

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Process Instrumentation

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Thermodynamics II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Laboratory

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 5

No.	Subject
1	Process Design
2	Plant Design
3	Process Dynamics
4	Petroleum/Petrochemicals
5	Environmental Engineering
6	Elective

1. Process Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Plant Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Process Dynamics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Petroleum/Petrochemicals

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Environmental Engineering

Unit 1: Water quality

Unit 2: Water treatment

Unit 3: Wastewater treatment

Unit 4: Air pollution

Unit 5: Solid waste management

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 6

No.	Subject
1	Process Simulation
2	Advanced Separation
3	Safety Engineering
4	Biochemical Engineering
5	Energy Systems
6	Elective

1. Process Simulation

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Separation

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Safety Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Biochemical Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Energy Systems

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 7

No.	Subject
1	Advanced Chemical Elective
2	Project Phase I
3	Internship
4	Seminar
5	Plant Design Project
6	Research Methods

1. Advanced Chemical Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Project Phase I

Unit 1 – Problem Definition: Problem statement; Objectives; Requirements

Unit 2 – Literature Review: Research papers; Existing systems; Gap analysis

Unit 3 – System Design: Architecture; Modules; Technology selection

Unit 4 – Initial Implementation: Prototype; Testing; Initial results

Unit 5 – Project Report: Documentation; Presentation; Future plan

3. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics

Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation

Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment

Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting

Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

4. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

5. Plant Design Project

Unit 1 – Problem Definition: Problem statement; Objectives; Scope; Requirements

Unit 2 – Design: Architecture; Technology selection; Methodology; Planning

Unit 3 – Implementation: Development; Integration; Testing; Iteration

Unit 4 – Evaluation: Results; Metrics; Validation; Limitations

Unit 5 – Documentation: Report; Presentation; Future work; Viva

6. Research Methods

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

Semester 8

No.	Subject
1	Major Project
2	Internship
3	Seminar
4	Elective
5	Report
6	Viva

1. Major Project

Unit 1 – Problem and Requirements: Problem statement; Objectives; Scope; Requirements
Unit 2 – Architecture and Design: System architecture; Modules; Data flow; Technology stack
Unit 3 – Implementation: Development; Integration; Testing; Deployment
Unit 4 – Results: Evaluation; Metrics; Comparison; Limitations
Unit 5 – Documentation and Viva: Final report; Presentation; Demonstration; Future work

2. Internship

Unit 1 – Induction: Organization; Role; Safety; Professional ethics
Unit 2 – Domain Learning: Tools; Processes; Standards; Documentation
Unit 3 – Project Work: Requirements; Implementation; Testing; Deployment
Unit 4 – Professional Skills: Teamwork; Communication; Version control; Reporting
Unit 5 – Assessment: Deliverables; Presentation; Report; Viva

3. Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope
Unit 2 – Literature: Research papers; Comparison; Gap identification
Unit 3 – Technical Study: Methodology; Architecture; Results
Unit 4 – Presentation: Slides; Technical explanation; Questions
Unit 5 – Report: Citation; References; Summary; Future scope

4. Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture
Unit 2 – Methods: Algorithms; Design methods; Implementation
Unit 3 – Applications: Use cases; Tools; Case studies
Unit 4 – Evaluation: Performance; Testing; Security/quality
Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Report

Unit 1 – Structure: Title; Abstract; Introduction
Unit 2 – Literature: Related work; References; Research gap
Unit 3 – Methodology: Design; Implementation; Evaluation
Unit 4 – Results: Results; Discussion; Limitations
Unit 5 – Conclusion: Conclusions; Future work; References

6. Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications
Unit 2 – Specialization: Advanced concepts; Problem solving
Unit 3 – Project: Architecture; Implementation; Testing; Results
Unit 4 – Professional Practice: Standards; Ethics; Industry tools
Unit 5 – Defense: Technical questions; Future scope; Communication

PART III — M.TECH / M.E.

The PG section is specialization-oriented. AICTE PG model curricula include program cores, electives, research methodology/IPR, laboratories, mini projects and dissertation work; exact specializations and semester structures vary by institution.

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M.Tech – Computer Science & Engineering

Semester 1

No.	Subject
1	Mathematical Foundations of Computer Science
2	Advanced Data Structures
3	Advanced Algorithms
4	Research Methodology and IPR
5	Program Elective I
6	Advanced Data Structures Lab

1. Mathematical Foundations of Computer Science

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Data Structures

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Advanced Algorithms

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Research Methodology and IPR

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Program Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Advanced Data Structures Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 2

No.	Subject
1	Advanced Operating Systems
2	Advanced Computer Networks
3	Machine Learning
4	Program Elective II
5	Advanced Systems Lab
6	Mini Project with Seminar

1. Advanced Operating Systems

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Computer Networks

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Machine Learning

Unit 1 – ML Foundations: Learning paradigms; Data preprocessing; Feature engineering; Train/test split

Unit 2 – Regression: Linear regression; Polynomial regression; Regularization; Metrics

Unit 3 – Classification: Logistic regression; KNN; Naive Bayes; SVM

Unit 4 – Trees and Ensembles: Decision trees; Random forests; Boosting; Bagging

Unit 5 – Unsupervised Learning: K-means; Hierarchical clustering; PCA; Anomaly detection

4. Program Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Advanced Systems Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

6. Mini Project with Seminar

Unit 1 – Problem: Problem statement; Objectives; Scope

Unit 2 – Literature: Existing work; Research gap; References

Unit 3 – Prototype: Design; Implementation; Testing

Unit 4 – Results: Evaluation; Discussion

Unit 5 – Seminar: Presentation; Report; Viva

Semester 3

No.	Subject
1	Advanced Elective I
2	Advanced Elective II
3	Research Seminar
4	Dissertation Phase I
5	Publication/Research Work

1. Advanced Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Research Seminar

Unit 1 – Research Topic: Problem; Objectives; Scope

Unit 2 – Literature: Search strategy; Paper analysis; Research gap

Unit 3 – Method: Methodology; Experiments; Evaluation

Unit 4 – Findings: Results; Discussion; Limitations

Unit 5 – Presentation: Technical presentation; Citation; Questions

4. Dissertation Phase I

Unit 1 – Proposal: Problem statement; Objectives; Literature review

Unit 2 – Methodology: Research design; Tools; Experimental plan

Unit 3 – Implementation: Prototype; Data collection; Experiments

Unit 4 – Preliminary Results: Evaluation; Analysis; Discussion

Unit 5 – Report: Progress report; References; Future plan

5. Publication/Research Work

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

Semester 4

No.	Subject
1	Dissertation Phase II
2	Thesis
3	Final Seminar
4	Viva Voce

1. Dissertation Phase II

Unit 1 – Final Implementation: Complete system; Integration; Validation

Unit 2 – Results: Experiments; Metrics; Statistical analysis

Unit 3 – Discussion: Findings; Comparison; Limitations

Unit 4 – Thesis: Chapters; Figures; Tables; References

Unit 5 – Defense: Presentation; Questions; Viva; Archiving

2. Thesis

Unit 1 – Research Problem: Problem; Objectives; Scope

Unit 2 – Literature and Method: Literature review; Methodology; Experimental setup

Unit 3 – Results: Results; Analysis; Comparison

Unit 4 – Conclusions: Conclusions; Contributions; Limitations

Unit 5 – Future Work: Extensions; Applications; Publication plan

3. Final Seminar

Unit 1 – Topic Selection: Problem identification; Literature search; Scope

Unit 2 – Literature: Research papers; Comparison; Gap identification

Unit 3 – Technical Study: Methodology; Architecture; Results

Unit 4 – Presentation: Slides; Technical explanation; Questions

Unit 5 – Report: Citation; References; Summary; Future scope

4. Viva Voce

Unit 1 – Foundations: Core concepts; Definitions; Principles

Unit 2 – Specialization: Advanced concepts; Applications

Unit 3 – Project: Problem; Method; Implementation; Results

Unit 4 – Research: Literature; Contribution; Limitations

Unit 5 – Defense: Presentation; Technical questions; Future work

M.Tech – Artificial Intelligence & Machine Learning

Semester 1

No.	Subject
1	Advanced Mathematics for AI
2	Statistical Learning
3	Advanced Machine Learning
4	Research Methodology and IPR
5	AI Elective I
6	ML Lab

1. Advanced Mathematics for AI

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Statistical Learning

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Advanced Machine Learning

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Research Methodology and IPR

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. AI Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. ML Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 2

No.	Subject
1	Deep Learning
2	Computer Vision
3	Natural Language Processing
4	AI Elective II
5	Deep Learning Lab
6	Mini Project

1. Deep Learning

Unit 1 – Neural Networks: Perceptron; Activation functions; Loss functions; Backpropagation

Unit 2 – Optimization: Gradient descent; Learning rates; Regularization; Batch normalization

Unit 3 – CNN: Convolution; Pooling; Architectures; Transfer learning

Unit 4 – Sequence Models: RNN; LSTM; GRU; Sequence prediction

Unit 5 – Generative Models: Autoencoders; GANs; Transformers overview; Generative applications

2. Computer Vision

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Natural Language Processing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. AI Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Deep Learning Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

6. Mini Project

Unit 1 – Problem Definition: Problem; Objectives; Scope

Unit 2 – Design: Architecture; Components; Method

Unit 3 – Development: Implementation; Testing; Debugging

Unit 4 – Results: Evaluation; Discussion; Limitations

Unit 5 – Report: Documentation; Presentation; Future work

Semester 3

No.	Subject
1	Reinforcement Learning
2	Generative AI
3	Advanced AI Elective
4	Dissertation Phase I
5	Research Seminar

1. Reinforcement Learning

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Generative AI

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Advanced AI Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Dissertation Phase I

Unit 1 – Proposal: Problem statement; Objectives; Literature review

Unit 2 – Methodology: Research design; Tools; Experimental plan

Unit 3 – Implementation: Prototype; Data collection; Experiments

Unit 4 – Preliminary Results: Evaluation; Analysis; Discussion

Unit 5 – Report: Progress report; References; Future plan

5. Research Seminar

Unit 1 – Research Topic: Problem; Objectives; Scope

Unit 2 – Literature: Search strategy; Paper analysis; Research gap

Unit 3 – Method: Methodology; Experiments; Evaluation

Unit 4 – Findings: Results; Discussion; Limitations

Unit 5 – Presentation: Technical presentation; Citation; Questions

Semester 4

No.	Subject
1	Dissertation Phase II
2	Thesis
3	Publication
4	Final Viva

1. Dissertation Phase II

Unit 1 – Final Implementation: Complete system; Integration; Validation

Unit 2 – Results: Experiments; Metrics; Statistical analysis

Unit 3 – Discussion: Findings; Comparison; Limitations

Unit 4 – Thesis: Chapters; Figures; Tables; References

Unit 5 – Defense: Presentation; Questions; Viva; Archiving

2. Thesis

Unit 1 – Research Problem: Problem; Objectives; Scope

Unit 2 – Literature and Method: Literature review; Methodology; Experimental setup

Unit 3 – Results: Results; Analysis; Comparison

Unit 4 – Conclusions: Conclusions; Contributions; Limitations

Unit 5 – Future Work: Extensions; Applications; Publication plan

3. Publication

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

4. Final Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications

Unit 2 – Specialization: Advanced concepts; Problem solving

Unit 3 – Project: Architecture; Implementation; Testing; Results

Unit 4 – Professional Practice: Standards; Ethics; Industry tools

Unit 5 – Defense: Technical questions; Future scope; Communication

M.Tech – Data Science

Semester 1

No.	Subject
1	Advanced Probability and Statistics
2	Data Mining
3	Advanced Algorithms
4	Research Methodology and IPR
5	Data Science Elective I
6	Data Analytics Lab

1. Advanced Probability and Statistics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Data Mining

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Advanced Algorithms

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Research Methodology and IPR

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Data Science Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Data Analytics Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 2

No.	Subject
1	Big Data Analytics
2	Distributed Data Processing
3	Machine Learning
4	Data Science Elective II
5	Big Data Lab
6	Mini Project

1. Big Data Analytics

Unit 1 – Big Data: Characteristics; Data sources; Distributed storage; Batch vs stream

Unit 2 – Distributed Processing: MapReduce; Spark concepts; Data partitioning

Unit 3 – NoSQL: Key-value; Document; Column; Graph databases

Unit 4 – Analytics: Data preparation; Descriptive analytics; Predictive analytics; Visualization

Unit 5 – Platforms: Data lakes; Pipelines; Governance; Security

2. Distributed Data Processing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Machine Learning

Unit 1 – ML Foundations: Learning paradigms; Data preprocessing; Feature engineering; Train/test split

Unit 2 – Regression: Linear regression; Polynomial regression; Regularization; Metrics

Unit 3 – Classification: Logistic regression; KNN; Naive Bayes; SVM

Unit 4 – Trees and Ensembles: Decision trees; Random forests; Boosting; Bagging

Unit 5 – Unsupervised Learning: K-means; Hierarchical clustering; PCA; Anomaly detection

4. Data Science Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Big Data Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

6. Mini Project

Unit 1 – Problem Definition: Problem; Objectives; Scope

Unit 2 – Design: Architecture; Components; Method

Unit 3 – Development: Implementation; Testing; Debugging

Unit 4 – Results: Evaluation; Discussion; Limitations

Unit 5 – Report: Documentation; Presentation; Future work

Semester 3

No.	Subject
1	Advanced Data Engineering
2	Advanced Analytics
3	Dissertation Phase I
4	Research Seminar
5	Publication

1. Advanced Data Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Analytics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Dissertation Phase I

Unit 1 – Proposal: Problem statement; Objectives; Literature review

Unit 2 – Methodology: Research design; Tools; Experimental plan

Unit 3 – Implementation: Prototype; Data collection; Experiments

Unit 4 – Preliminary Results: Evaluation; Analysis; Discussion

Unit 5 – Report: Progress report; References; Future plan

4. Research Seminar

Unit 1 – Research Topic: Problem; Objectives; Scope

Unit 2 – Literature: Search strategy; Paper analysis; Research gap

Unit 3 – Method: Methodology; Experiments; Evaluation

Unit 4 – Findings: Results; Discussion; Limitations

Unit 5 – Presentation: Technical presentation; Citation; Questions

5. Publication

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

Semester 4

No.	Subject
1	Dissertation Phase II
2	Thesis
3	Publication
4	Final Viva

1. Dissertation Phase II

Unit 1 – Final Implementation: Complete system; Integration; Validation

Unit 2 – Results: Experiments; Metrics; Statistical analysis

Unit 3 – Discussion: Findings; Comparison; Limitations

Unit 4 – Thesis: Chapters; Figures; Tables; References

Unit 5 – Defense: Presentation; Questions; Viva; Archiving

2. Thesis

Unit 1 – Research Problem: Problem; Objectives; Scope

Unit 2 – Literature and Method: Literature review; Methodology; Experimental setup

Unit 3 – Results: Results; Analysis; Comparison

Unit 4 – Conclusions: Conclusions; Contributions; Limitations

Unit 5 – Future Work: Extensions; Applications; Publication plan

3. Publication

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

4. Final Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications

Unit 2 – Specialization: Advanced concepts; Problem solving

Unit 3 – Project: Architecture; Implementation; Testing; Results

Unit 4 – Professional Practice: Standards; Ethics; Industry tools

Unit 5 – Defense: Technical questions; Future scope; Communication

M.Tech – VLSI Design

Semester 1

No.	Subject
1	Advanced VLSI Design
2	Digital System Design
3	Semiconductor Device Modeling
4	Research Methodology and IPR
5	VLSI Elective I
6	VLSI Lab

1. Advanced VLSI Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Digital System Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Semiconductor Device Modeling

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Research Methodology and IPR

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. VLSI Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. VLSI Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 2

No.	Subject
1	ASIC Design
2	FPGA Design
3	Low Power VLSI
4	VLSI Elective II
5	EDA Lab
6	Mini Project

1. ASIC Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. FPGA Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Low Power VLSI

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. VLSI Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. EDA Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

6. Mini Project

Unit 1 – Problem Definition: Problem; Objectives; Scope

Unit 2 – Design: Architecture; Components; Method

Unit 3 – Development: Implementation; Testing; Debugging

Unit 4 – Results: Evaluation; Discussion; Limitations

Unit 5 – Report: Documentation; Presentation; Future work

Semester 3

No.	Subject
1	Advanced VLSI Elective
2	Research Seminar
3	Dissertation Phase I
4	Publication

1. Advanced VLSI Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Research Seminar

Unit 1 – Research Topic: Problem; Objectives; Scope

Unit 2 – Literature: Search strategy; Paper analysis; Research gap

Unit 3 – Method: Methodology; Experiments; Evaluation

Unit 4 – Findings: Results; Discussion; Limitations

Unit 5 – Presentation: Technical presentation; Citation; Questions

3. Dissertation Phase I

Unit 1 – Proposal: Problem statement; Objectives; Literature review

Unit 2 – Methodology: Research design; Tools; Experimental plan

Unit 3 – Implementation: Prototype; Data collection; Experiments

Unit 4 – Preliminary Results: Evaluation; Analysis; Discussion

Unit 5 – Report: Progress report; References; Future plan

4. Publication

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

Semester 4

No.	Subject
1	Dissertation Phase II
2	Thesis
3	Publication
4	Final Viva

1. Dissertation Phase II

Unit 1 – Final Implementation: Complete system; Integration; Validation

Unit 2 – Results: Experiments; Metrics; Statistical analysis

Unit 3 – Discussion: Findings; Comparison; Limitations

Unit 4 – Thesis: Chapters; Figures; Tables; References

Unit 5 – Defense: Presentation; Questions; Viva; Archiving

2. Thesis

Unit 1 – Research Problem: Problem; Objectives; Scope

Unit 2 – Literature and Method: Literature review; Methodology; Experimental setup

Unit 3 – Results: Results; Analysis; Comparison

Unit 4 – Conclusions: Conclusions; Contributions; Limitations

Unit 5 – Future Work: Extensions; Applications; Publication plan

3. Publication

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

4. Final Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications

Unit 2 – Specialization: Advanced concepts; Problem solving

Unit 3 – Project: Architecture; Implementation; Testing; Results

Unit 4 – Professional Practice: Standards; Ethics; Industry tools

Unit 5 – Defense: Technical questions; Future scope; Communication

M.Tech – Embedded Systems

Semester 1

No.	Subject
1	Advanced Microcontrollers
2	Embedded System Architecture
3	Real-Time Systems
4	Research Methodology and IPR
5	Embedded Elective I
6	Embedded Lab

1. Advanced Microcontrollers

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Embedded System Architecture

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Real-Time Systems

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Research Methodology and IPR

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Embedded Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Embedded Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 2

No.	Subject
1	Embedded Linux
2	IoT Systems
3	Embedded Networking
4	Embedded Elective II
5	System Design Lab
6	Mini Project

1. Embedded Linux

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. IoT Systems

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Embedded Networking

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Embedded Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. System Design Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

6. Mini Project

Unit 1 – Problem Definition: Problem; Objectives; Scope

Unit 2 – Design: Architecture; Components; Method

Unit 3 – Development: Implementation; Testing; Debugging

Unit 4 – Results: Evaluation; Discussion; Limitations

Unit 5 – Report: Documentation; Presentation; Future work

Semester 3

No.	Subject
1	Edge AI
2	Advanced Embedded Design
3	Dissertation Phase I
4	Research Seminar
5	Publication

1. Edge AI

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Embedded Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Dissertation Phase I

Unit 1 – Proposal: Problem statement; Objectives; Literature review

Unit 2 – Methodology: Research design; Tools; Experimental plan

Unit 3 – Implementation: Prototype; Data collection; Experiments

Unit 4 – Preliminary Results: Evaluation; Analysis; Discussion

Unit 5 – Report: Progress report; References; Future plan

4. Research Seminar

Unit 1 – Research Topic: Problem; Objectives; Scope

Unit 2 – Literature: Search strategy; Paper analysis; Research gap

Unit 3 – Method: Methodology; Experiments; Evaluation

Unit 4 – Findings: Results; Discussion; Limitations

Unit 5 – Presentation: Technical presentation; Citation; Questions

5. Publication

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

Semester 4

No.	Subject
1	Dissertation Phase II
2	Thesis
3	Publication
4	Final Viva

1. Dissertation Phase II

Unit 1 – Final Implementation: Complete system; Integration; Validation

Unit 2 – Results: Experiments; Metrics; Statistical analysis

Unit 3 – Discussion: Findings; Comparison; Limitations

Unit 4 – Thesis: Chapters; Figures; Tables; References

Unit 5 – Defense: Presentation; Questions; Viva; Archiving

2. Thesis

Unit 1 – Research Problem: Problem; Objectives; Scope

Unit 2 – Literature and Method: Literature review; Methodology; Experimental setup

Unit 3 – Results: Results; Analysis; Comparison

Unit 4 – Conclusions: Conclusions; Contributions; Limitations

Unit 5 – Future Work: Extensions; Applications; Publication plan

3. Publication

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

4. Final Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications

Unit 2 – Specialization: Advanced concepts; Problem solving

Unit 3 – Project: Architecture; Implementation; Testing; Results

Unit 4 – Professional Practice: Standards; Ethics; Industry tools

Unit 5 – Defense: Technical questions; Future scope; Communication

M.Tech – Electronics & Communication Engineering

Semester 1

No.	Subject
1	Advanced Communication Systems
2	Advanced Signal Processing
3	Wireless Networks
4	Research Methodology and IPR
5	ECE Elective I
6	DSP/Communication Lab

1. Advanced Communication Systems

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Signal Processing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Wireless Networks

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Research Methodology and IPR

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. ECE Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. DSP/Communication Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 2

No.	Subject
1	Optical Communication
2	RF and Microwave Engineering
3	Advanced Digital Signal Processing
4	ECE Elective II
5	Advanced Lab
6	Mini Project

1. Optical Communication

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. RF and Microwave Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Advanced Digital Signal Processing

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. ECE Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Advanced Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

6. Mini Project

Unit 1 – Problem Definition: Problem; Objectives; Scope

Unit 2 – Design: Architecture; Components; Method

Unit 3 – Development: Implementation; Testing; Debugging

Unit 4 – Results: Evaluation; Discussion; Limitations

Unit 5 – Report: Documentation; Presentation; Future work

Semester 3

No.	Subject
1	5G/6G Networks
2	Advanced ECE Elective
3	Dissertation Phase I
4	Research Seminar
5	Publication

1. 5G/6G Networks

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced ECE Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Dissertation Phase I

Unit 1 – Proposal: Problem statement; Objectives; Literature review

Unit 2 – Methodology: Research design; Tools; Experimental plan

Unit 3 – Implementation: Prototype; Data collection; Experiments

Unit 4 – Preliminary Results: Evaluation; Analysis; Discussion

Unit 5 – Report: Progress report; References; Future plan

4. Research Seminar

Unit 1 – Research Topic: Problem; Objectives; Scope

Unit 2 – Literature: Search strategy; Paper analysis; Research gap

Unit 3 – Method: Methodology; Experiments; Evaluation

Unit 4 – Findings: Results; Discussion; Limitations

Unit 5 – Presentation: Technical presentation; Citation; Questions

5. Publication

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

Semester 4

No.	Subject
1	Dissertation Phase II
2	Thesis
3	Publication
4	Final Viva

1. Dissertation Phase II

Unit 1 – Final Implementation: Complete system; Integration; Validation

Unit 2 – Results: Experiments; Metrics; Statistical analysis

Unit 3 – Discussion: Findings; Comparison; Limitations

Unit 4 – Thesis: Chapters; Figures; Tables; References

Unit 5 – Defense: Presentation; Questions; Viva; Archiving

2. Thesis

Unit 1 – Research Problem: Problem; Objectives; Scope

Unit 2 – Literature and Method: Literature review; Methodology; Experimental setup

Unit 3 – Results: Results; Analysis; Comparison

Unit 4 – Conclusions: Conclusions; Contributions; Limitations

Unit 5 – Future Work: Extensions; Applications; Publication plan

3. Publication

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

4. Final Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications

Unit 2 – Specialization: Advanced concepts; Problem solving

Unit 3 – Project: Architecture; Implementation; Testing; Results

Unit 4 – Professional Practice: Standards; Ethics; Industry tools

Unit 5 – Defense: Technical questions; Future scope; Communication

M.Tech – Power Systems

Semester 1

No.	Subject
1	Advanced Power System Analysis
2	Power System Dynamics
3	Power System Protection
4	Research Methodology and IPR
5	Power Elective I
6	Power Systems Lab

1. Advanced Power System Analysis

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Power System Dynamics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Power System Protection

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Research Methodology and IPR

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Power Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Power Systems Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 2

No.	Subject
1	Smart Grid
2	Power Quality
3	Renewable Energy Integration
4	Power Elective II
5	Simulation Lab
6	Mini Project

1. Smart Grid

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Power Quality

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Renewable Energy Integration

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Power Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Simulation Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

6. Mini Project

Unit 1 – Problem Definition: Problem; Objectives; Scope

Unit 2 – Design: Architecture; Components; Method

Unit 3 – Development: Implementation; Testing; Debugging

Unit 4 – Results: Evaluation; Discussion; Limitations

Unit 5 – Report: Documentation; Presentation; Future work

Semester 3

No.	Subject
1	Advanced Grid Technologies
2	Energy Systems Research
3	Dissertation Phase I
4	Research Seminar
5	Publication

1. Advanced Grid Technologies

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Energy Systems Research

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Dissertation Phase I

Unit 1 – Proposal: Problem statement; Objectives; Literature review

Unit 2 – Methodology: Research design; Tools; Experimental plan

Unit 3 – Implementation: Prototype; Data collection; Experiments

Unit 4 – Preliminary Results: Evaluation; Analysis; Discussion

Unit 5 – Report: Progress report; References; Future plan

4. Research Seminar

Unit 1 – Research Topic: Problem; Objectives; Scope

Unit 2 – Literature: Search strategy; Paper analysis; Research gap

Unit 3 – Method: Methodology; Experiments; Evaluation

Unit 4 – Findings: Results; Discussion; Limitations

Unit 5 – Presentation: Technical presentation; Citation; Questions

5. Publication

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

Semester 4

No.	Subject
1	Dissertation Phase II
2	Thesis
3	Publication
4	Final Viva

1. Dissertation Phase II

Unit 1 – Final Implementation: Complete system; Integration; Validation

Unit 2 – Results: Experiments; Metrics; Statistical analysis

Unit 3 – Discussion: Findings; Comparison; Limitations

Unit 4 – Thesis: Chapters; Figures; Tables; References

Unit 5 – Defense: Presentation; Questions; Viva; Archiving

2. Thesis

Unit 1 – Research Problem: Problem; Objectives; Scope

Unit 2 – Literature and Method: Literature review; Methodology; Experimental setup

Unit 3 – Results: Results; Analysis; Comparison

Unit 4 – Conclusions: Conclusions; Contributions; Limitations

Unit 5 – Future Work: Extensions; Applications; Publication plan

3. Publication

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

4. Final Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications

Unit 2 – Specialization: Advanced concepts; Problem solving

Unit 3 – Project: Architecture; Implementation; Testing; Results

Unit 4 – Professional Practice: Standards; Ethics; Industry tools

Unit 5 – Defense: Technical questions; Future scope; Communication

M.Tech – Mechanical Design

Semester 1

No.	Subject
1	Advanced Stress Analysis
2	Advanced Vibrations and Acoustics
3	Advanced Machine Design
4	Research Methodology and IPR
5	Design Elective I
6	Design Lab

1. Advanced Stress Analysis

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Vibrations and Acoustics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Advanced Machine Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Research Methodology and IPR

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Design Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Design Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 2

No.	Subject
1	Finite Element Method
2	Computer Aided Design
3	Advanced Materials
4	Design Elective II
5	CAD/CAE Lab
6	Mini Project

1. Finite Element Method

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Computer Aided Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Advanced Materials

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Design Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. CAD/CAE Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

6. Mini Project

Unit 1 – Problem Definition: Problem; Objectives; Scope

Unit 2 – Design: Architecture; Components; Method

Unit 3 – Development: Implementation; Testing; Debugging

Unit 4 – Results: Evaluation; Discussion; Limitations

Unit 5 – Report: Documentation; Presentation; Future work

Semester 3

No.	Subject
1	Advanced Design Elective
2	Research Seminar
3	Dissertation Phase I
4	Publication

1. Advanced Design Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Research Seminar

Unit 1 – Research Topic: Problem; Objectives; Scope

Unit 2 – Literature: Search strategy; Paper analysis; Research gap

Unit 3 – Method: Methodology; Experiments; Evaluation

Unit 4 – Findings: Results; Discussion; Limitations

Unit 5 – Presentation: Technical presentation; Citation; Questions

3. Dissertation Phase I

Unit 1 – Proposal: Problem statement; Objectives; Literature review

Unit 2 – Methodology: Research design; Tools; Experimental plan

Unit 3 – Implementation: Prototype; Data collection; Experiments

Unit 4 – Preliminary Results: Evaluation; Analysis; Discussion

Unit 5 – Report: Progress report; References; Future plan

4. Publication

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

Semester 4

No.	Subject
1	Dissertation Phase II
2	Thesis
3	Publication
4	Final Viva

1. Dissertation Phase II

Unit 1 – Final Implementation: Complete system; Integration; Validation

Unit 2 – Results: Experiments; Metrics; Statistical analysis

Unit 3 – Discussion: Findings; Comparison; Limitations

Unit 4 – Thesis: Chapters; Figures; Tables; References

Unit 5 – Defense: Presentation; Questions; Viva; Archiving

2. Thesis

Unit 1 – Research Problem: Problem; Objectives; Scope

Unit 2 – Literature and Method: Literature review; Methodology; Experimental setup

Unit 3 – Results: Results; Analysis; Comparison

Unit 4 – Conclusions: Conclusions; Contributions; Limitations

Unit 5 – Future Work: Extensions; Applications; Publication plan

3. Publication

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

4. Final Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications

Unit 2 – Specialization: Advanced concepts; Problem solving

Unit 3 – Project: Architecture; Implementation; Testing; Results

Unit 4 – Professional Practice: Standards; Ethics; Industry tools

Unit 5 – Defense: Technical questions; Future scope; Communication

M.Tech – Structural Engineering

Semester 1

No.	Subject
1	Advanced Structural Analysis
2	Advanced Solid Mechanics
3	Theory of Plates and Shells
4	Research Methodology and IPR
5	Structural Elective I
6	Structural Design Lab

1. Advanced Structural Analysis

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Solid Mechanics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Theory of Plates and Shells

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Research Methodology and IPR

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Structural Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Structural Design Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 2

No.	Subject
1	Finite Element Method
2	Structural Dynamics
3	Advanced Steel/Concrete Design
4	Structural Elective II
5	Numerical Analysis Lab
6	Mini Project

1. Finite Element Method

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Structural Dynamics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Advanced Steel/Concrete Design

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Structural Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Numerical Analysis Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

6. Mini Project

Unit 1 – Problem Definition: Problem; Objectives; Scope

Unit 2 – Design: Architecture; Components; Method

Unit 3 – Development: Implementation; Testing; Debugging

Unit 4 – Results: Evaluation; Discussion; Limitations

Unit 5 – Report: Documentation; Presentation; Future work

Semester 3

No.	Subject
1	Structural Health Monitoring
2	Advanced Structural Elective
3	Dissertation Phase I
4	Research Seminar
5	Publication

1. Structural Health Monitoring

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Structural Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Dissertation Phase I

Unit 1 – Proposal: Problem statement; Objectives; Literature review

Unit 2 – Methodology: Research design; Tools; Experimental plan

Unit 3 – Implementation: Prototype; Data collection; Experiments

Unit 4 – Preliminary Results: Evaluation; Analysis; Discussion

Unit 5 – Report: Progress report; References; Future plan

4. Research Seminar

Unit 1 – Research Topic: Problem; Objectives; Scope

Unit 2 – Literature: Search strategy; Paper analysis; Research gap

Unit 3 – Method: Methodology; Experiments; Evaluation

Unit 4 – Findings: Results; Discussion; Limitations

Unit 5 – Presentation: Technical presentation; Citation; Questions

5. Publication

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

Semester 4

No.	Subject
1	Dissertation Phase II
2	Thesis
3	Publication
4	Final Viva

1. Dissertation Phase II

Unit 1 – Final Implementation: Complete system; Integration; Validation

Unit 2 – Results: Experiments; Metrics; Statistical analysis

Unit 3 – Discussion: Findings; Comparison; Limitations

Unit 4 – Thesis: Chapters; Figures; Tables; References

Unit 5 – Defense: Presentation; Questions; Viva; Archiving

2. Thesis

Unit 1 – Research Problem: Problem; Objectives; Scope

Unit 2 – Literature and Method: Literature review; Methodology; Experimental setup

Unit 3 – Results: Results; Analysis; Comparison

Unit 4 – Conclusions: Conclusions; Contributions; Limitations

Unit 5 – Future Work: Extensions; Applications; Publication plan

3. Publication

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

4. Final Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications

Unit 2 – Specialization: Advanced concepts; Problem solving

Unit 3 – Project: Architecture; Implementation; Testing; Results

Unit 4 – Professional Practice: Standards; Ethics; Industry tools

Unit 5 – Defense: Technical questions; Future scope; Communication

M.Tech – Geotechnical Engineering

Semester 1

No.	Subject
1	Advanced Soil Mechanics
2	Advanced Foundation Engineering
3	Soil Dynamics
4	Research Methodology and IPR
5	Geotechnical Elective I
6	Geotechnical Lab

1. Advanced Soil Mechanics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Foundation Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Soil Dynamics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Research Methodology and IPR

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Geotechnical Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Geotechnical Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 2

No.	Subject
1	Earth Retaining Structures
2	Ground Improvement
3	Geotechnical Earthquake Engineering
4	Geotechnical Elective II
5	Numerical Lab
6	Mini Project

1. Earth Retaining Structures

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Ground Improvement

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Geotechnical Earthquake Engineering

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Geotechnical Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Numerical Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

6. Mini Project

Unit 1 – Problem Definition: Problem; Objectives; Scope

Unit 2 – Design: Architecture; Components; Method

Unit 3 – Development: Implementation; Testing; Debugging

Unit 4 – Results: Evaluation; Discussion; Limitations

Unit 5 – Report: Documentation; Presentation; Future work

Semester 3

No.	Subject
1	Advanced Geotechnical Elective
2	Research Seminar
3	Dissertation Phase I
4	Publication

1. Advanced Geotechnical Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Research Seminar

Unit 1 – Research Topic: Problem; Objectives; Scope

Unit 2 – Literature: Search strategy; Paper analysis; Research gap

Unit 3 – Method: Methodology; Experiments; Evaluation

Unit 4 – Findings: Results; Discussion; Limitations

Unit 5 – Presentation: Technical presentation; Citation; Questions

3. Dissertation Phase I

Unit 1 – Proposal: Problem statement; Objectives; Literature review

Unit 2 – Methodology: Research design; Tools; Experimental plan

Unit 3 – Implementation: Prototype; Data collection; Experiments

Unit 4 – Preliminary Results: Evaluation; Analysis; Discussion

Unit 5 – Report: Progress report; References; Future plan

4. Publication

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

Semester 4

No.	Subject
1	Dissertation Phase II
2	Thesis
3	Publication
4	Final Viva

1. Dissertation Phase II

Unit 1 – Final Implementation: Complete system; Integration; Validation

Unit 2 – Results: Experiments; Metrics; Statistical analysis

Unit 3 – Discussion: Findings; Comparison; Limitations

Unit 4 – Thesis: Chapters; Figures; Tables; References

Unit 5 – Defense: Presentation; Questions; Viva; Archiving

2. Thesis

Unit 1 – Research Problem: Problem; Objectives; Scope

Unit 2 – Literature and Method: Literature review; Methodology; Experimental setup

Unit 3 – Results: Results; Analysis; Comparison

Unit 4 – Conclusions: Conclusions; Contributions; Limitations

Unit 5 – Future Work: Extensions; Applications; Publication plan

3. Publication

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

4. Final Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications

Unit 2 – Specialization: Advanced concepts; Problem solving

Unit 3 – Project: Architecture; Implementation; Testing; Results

Unit 4 – Professional Practice: Standards; Ethics; Industry tools

Unit 5 – Defense: Technical questions; Future scope; Communication

M.Tech – Thermal Engineering

Semester 1

No.	Subject
1	Advanced Thermodynamics
2	Advanced Heat Transfer
3	Computational Fluid Dynamics
4	Research Methodology and IPR
5	Thermal Elective I
6	Thermal Lab

1. Advanced Thermodynamics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Advanced Heat Transfer

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Computational Fluid Dynamics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Research Methodology and IPR

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. Thermal Elective I

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

6. Thermal Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

Semester 2

No.	Subject
1	Advanced IC Engines
2	Refrigeration and Cryogenics
3	Energy Systems
4	Thermal Elective II
5	CFD Lab
6	Mini Project

1. Advanced IC Engines

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Refrigeration and Cryogenics

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

3. Energy Systems

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

4. Thermal Elective II

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

5. CFD Lab

Unit 1 – Safety: Lab safety; Equipment handling; Measurement practice

Unit 2 – Fundamentals: Core experiments; Observation; Data collection

Unit 3 – Analysis: Calculations; Graphs; Error analysis

Unit 4 – Implementation: Mini experiments; Troubleshooting; Documentation

Unit 5 – Viva: Concept review; Experiment interpretation; Reporting

6. Mini Project

Unit 1 – Problem Definition: Problem; Objectives; Scope

Unit 2 – Design: Architecture; Components; Method

Unit 3 – Development: Implementation; Testing; Debugging

Unit 4 – Results: Evaluation; Discussion; Limitations

Unit 5 – Report: Documentation; Presentation; Future work

Semester 3

No.	Subject
1	Advanced Thermal Elective
2	Research Seminar
3	Dissertation Phase I
4	Publication

1. Advanced Thermal Elective

Unit 1 – Foundations: Core concepts; Terminology; Architecture

Unit 2 – Methods: Algorithms; Design methods; Implementation

Unit 3 – Applications: Use cases; Tools; Case studies

Unit 4 – Evaluation: Performance; Testing; Security/quality

Unit 5 – Advanced Topics: Current trends; Research directions; Industry practices

2. Research Seminar

Unit 1 – Research Topic: Problem; Objectives; Scope

Unit 2 – Literature: Search strategy; Paper analysis; Research gap

Unit 3 – Method: Methodology; Experiments; Evaluation

Unit 4 – Findings: Results; Discussion; Limitations

Unit 5 – Presentation: Technical presentation; Citation; Questions

3. Dissertation Phase I

Unit 1 – Proposal: Problem statement; Objectives; Literature review

Unit 2 – Methodology: Research design; Tools; Experimental plan

Unit 3 – Implementation: Prototype; Data collection; Experiments

Unit 4 – Preliminary Results: Evaluation; Analysis; Discussion

Unit 5 – Report: Progress report; References; Future plan

4. Publication

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

Semester 4

No.	Subject
1	Dissertation Phase II
2	Thesis
3	Publication
4	Final Viva

1. Dissertation Phase II

Unit 1 – Final Implementation: Complete system; Integration; Validation

Unit 2 – Results: Experiments; Metrics; Statistical analysis

Unit 3 – Discussion: Findings; Comparison; Limitations

Unit 4 – Thesis: Chapters; Figures; Tables; References

Unit 5 – Defense: Presentation; Questions; Viva; Archiving

2. Thesis

Unit 1 – Research Problem: Problem; Objectives; Scope

Unit 2 – Literature and Method: Literature review; Methodology; Experimental setup

Unit 3 – Results: Results; Analysis; Comparison

Unit 4 – Conclusions: Conclusions; Contributions; Limitations

Unit 5 – Future Work: Extensions; Applications; Publication plan

3. Publication

Unit 1 – Paper Preparation: Research question; Contribution; Literature

Unit 2 – Methodology: Experimental design; Data; Evaluation

Unit 3 – Results: Analysis; Figures; Tables

Unit 4 – Writing: Abstract; Introduction; Method; Results; Discussion

Unit 5 – Submission: Citation; Formatting; Peer review; Revision

4. Final Viva

Unit 1 – Core Concepts: Core theory; Key definitions; Applications

Unit 2 – Specialization: Advanced concepts; Problem solving

Unit 3 – Project: Architecture; Implementation; Testing; Results

Unit 4 – Professional Practice: Standards; Ethics; Industry tools

Unit 5 – Defense: Technical questions; Future scope; Communication

Reference and Scope

- AICTE Model Curriculum for First Year Undergraduate Degree Courses in Engineering & Technology.
- AICTE Model Curriculum for Undergraduate Engineering branches, including branch-specific curricula.
- AICTE Model Curriculum for Postgraduate Degree Courses in Engineering & Technology.
- MeitY C2S model-curriculum information for semiconductor-focused programs.

Important: This PDF is a standardized master learning map. It is intentionally broader than a single college syllabus and should not be treated as the official examination syllabus of a specific university. For an exact syllabus, the university/board and regulation year must be specified.

PART IV - CREATIVE, DIGITAL MEDIA, COMPUTING & DESIGN PROGRAMS

Expanded program family covering computer graphics, animation, VFX, design, multimedia, UI/UX, games, digital media, photography, film, web production and related computing programs. These are standardized curriculum maps; official university/board syllabi can differ.

A. POLYTECHNIC / DIPLOMA - CREATIVE & DIGITAL PROGRAMS

Polytechnic - Animation & VFX

Semester 1

No.	Subject
1	Design Foundations
2	Drawing and Visual Development
3	Digital Imaging Basics
4	Computer Fundamentals
5	Communication Skills
6	Lab / Studio Practice

1. Design Foundations

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Drawing and Visual Development

Unit 1 - Drawing Basics: Line, Shape, Form, Proportion, Perspective

Unit 2 - Observation: Still life, Environment studies, Figure studies, Gesture

Unit 3 - Perspective: One-point, Two-point, Three-point, Camera angles

Unit 4 - Visual Development: Silhouette, Shape language, Thumbnailing, Value studies

Unit 5 - Digital Practice: Drawing tablets, Brushes, Layers, Digital painting workflow

3. Digital Imaging Basics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Computer Fundamentals

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Communication Skills

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Lab / Studio Practice

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 2

No.	Subject
1	Graphic Design
2	Digital Illustration
3	2D Animation
4	Video Editing
5	Web Design and Front-End
6	Lab / Studio Practice

1. Graphic Design

Unit 1 - Graphic Design: Composition, Layout, Branding, Visual identity
Unit 2 - Typography: Type selection, Hierarchy, Pairing, Editorial typography
Unit 3 - Branding: Logo systems, Brand guidelines, Color systems, Stationery
Unit 4 - Publication: Posters, Brochures, Editorial layouts, Packaging
Unit 5 - Digital Design: Social graphics, Digital ads, Web graphics, Export formats

2. Digital Illustration

Unit 1 - Illustration: Visual storytelling, Styles, Composition, Concept sketches
Unit 2 - Digital Tools: Brushes, Layers, Masks, Blending, Coloring
Unit 3 - Character Illustration: Anatomy, Expressions, Costume, Posing
Unit 4 - Environment Illustration: Architecture, Lighting, Atmosphere, Perspective
Unit 5 - Portfolio: Series creation, Art direction, Presentation, Client briefs

3. 2D Animation

Unit 1 - 2D Basics: Frame rates, Keyframes, In-betweens, Exposure
Unit 2 - Character Animation: Walk cycles, Run cycles, Dialogue, Acting
Unit 3 - Backgrounds: Layout, Perspective, Color keys, Painted backgrounds
Unit 4 - Compositing: Layers, Masks, Camera, Effects
Unit 5 - Production: Pipeline, Shot tracking, Rendering, Delivery

4. Video Editing

Unit 1 - Editing Fundamentals: Cuts, Continuity, Pacing, Story structure
Unit 2 - Narrative Editing: Scene construction, Montage, Transitions, Rhythm
Unit 3 - Audio: Dialogue, Music, Sound effects, Mixing basics
Unit 4 - Color: Color correction, Color grading, Scopes, Look creation
Unit 5 - Delivery: Codecs, Resolutions, Aspect ratios, Export

5. Web Design and Front-End

Unit 1 - Web Foundations: Internet basics, HTTP, HTML, Semantic markup
Unit 2 - Styling: CSS, Layouts, Responsive design, Accessibility
Unit 3 - JavaScript: DOM, Events, Fetch, Modules
Unit 4 - UI Implementation: Components, Forms, State concepts, API integration
Unit 5 - Deployment: Version control, Builds, Hosting, Performance

6. Lab / Studio Practice

Unit 1 - Safety and Setup: Equipment, File management, Workspace
Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration
Unit 3 - Applied Exercises: Brief-based work, Testing, Revision
Unit 4 - Production Practice: Workflow, Quality control, Deadlines
Unit 5 - Submission: Documentation, Presentation, Viva

Semester 3

No.	Subject
1	3D Modeling
2	Texturing and Materials
3	Animation Principles
4	Compositing Fundamentals
5	Photography and Digital Imaging
6	Lab / Studio Practice

1. 3D Modeling

Unit 1 - 3D Foundations: 3D coordinate systems, Primitives, Topology, Transformations

Unit 2 - Polygon Modeling: Extrusion, Inset, Bevel, Edge flow, Subdivision

Unit 3 - Hard Surface: Mechanical forms, Booleans, Detailing, Optimization

Unit 4 - Organic Modeling: Sculpting, Base meshes, Anatomical forms, Retopology

Unit 5 - Production: UVs, Asset organization, Export, Game-ready models

2. Texturing and Materials

Unit 1 - Texturing Basics: UV mapping, Texture coordinates, Resolution, Seam management

Unit 2 - PBR: Albedo, Roughness, Metallic, Normal maps, Ambient occlusion

Unit 3 - Procedural Materials: Nodes, Masks, Generators, Material variations

Unit 4 - Surface Detail: Microdetail, Weathering, Imperfections, Decals

Unit 5 - Look Development: Material libraries, Consistency, Optimization, Export

3. Animation Principles

Unit 1 - Foundations: Timing, Spacing, Squash and stretch, Anticipation

Unit 2 - Motion: Arcs, Overlap, Follow-through, Secondary action

Unit 3 - Acting: Poses, Facial expressions, Body language, Staging

Unit 4 - Workflow: Storyboards, Animatics, Blocking, Polish

Unit 5 - Production: Shot management, Review, Rendering, Delivery

4. Compositing Fundamentals

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Photography and Digital Imaging

Unit 1 - Camera Basics: Exposure triangle, Lenses, Focus, Metering

Unit 2 - Composition: Framing, Perspective, Depth, Visual balance

Unit 3 - Lighting: Natural light, Studio light, Modifiers, Light ratios

Unit 4 - Image Processing: RAW, Color correction, Retouching, Compositing

Unit 5 - Publishing: Portfolio, Print, Web, Rights and metadata

6. Lab / Studio Practice

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 4

No.	Subject
1	3D Animation
2	Visual Effects (VFX)
3	Motion Graphics
4	Film and Video Production
5	Audio for Media
6	Lab / Studio Practice

1. 3D Animation

Unit 1 - Character Setup: Rigging concepts, Controllers, Skinning, Constraints

Unit 2 - Body Mechanics: Walk, Run, Weight, Balance

Unit 3 - Character Acting: Poses, Dialogue, Facial animation, Lip sync

Unit 4 - Advanced Animation: Motion blending, Crowds, Procedural motion

Unit 5 - Production: Shot workflow, Caching, Review, Final render

2. Visual Effects (VFX)

Unit 1 - VFX Foundations: VFX pipeline, Shot planning, Reference gathering, Continuity

Unit 2 - Compositing: Keying, Rotoscopy, Tracking, Color matching

Unit 3 - Effects: Particles, Smoke, Fire, Destruction, Fluid effects

Unit 4 - 3D Integration: Matchmoving, Camera tracking, CG integration, Lighting

Unit 5 - Finishing: Color correction, Grain, Cleanup, Delivery

3. Motion Graphics

Unit 1 - Motion Design: Graphic composition, Timing, Transitions, Kinetic type

Unit 2 - Animation Systems: Keyframes, Expressions, Procedural motion, Graph editors

Unit 3 - 3D Motion: 3D layers, Cameras, Lighting, Materials

Unit 4 - Broadcast Design: Titles, Lower thirds, Intros, Promos

Unit 5 - Delivery: Formats, Compression, Broadcast specs, Social media output

4. Film and Video Production

Unit 1 - Pre-production: Script, Concept, Research, Budget, Scheduling

Unit 2 - Cinematography: Cameras, Lenses, Composition, Exposure, Lighting

Unit 3 - Production: Directing, Sound recording, Continuity, Set workflow

Unit 4 - Post-production: Editing, Color, Sound, VFX

Unit 5 - Distribution: Mastering, Subtitles, Platforms, Promotion

5. Audio for Media

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Lab / Studio Practice

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 5

No.	Subject
1	Advanced VFX
2	Character Animation
3	Game Art and 3D
4	Color Grading
5	Portfolio and Professional Practice
6	Project

1. Advanced VFX

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Character Animation

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Game Art and 3D

Unit 1 - Concept Art: Ideation, Silhouette, Style guides, Reference boards

Unit 2 - 3D Assets: Modeling, UVs, Texturing, Optimization

Unit 3 - Characters: Sculpting, Retopology, Rigging, Skinning

Unit 4 - Environments: Modular kits, Materials, Lighting, World building

Unit 5 - Real-time Art: Engine import, LODs, Shaders, Performance

4. Color Grading

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Portfolio and Professional Practice

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

6. Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

Semester 6

No.	Subject
1	Advanced 3D/VFX
2	Studio Production
3	Real-time Graphics
4	Industry Workflow
5	Portfolio and Professional Practice
6	Major Project

1. Advanced 3D/VFX

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Studio Production

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Real-time Graphics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Industry Workflow

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Portfolio and Professional Practice

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

6. Major Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

Polytechnic - Graphic Design & Multimedia

Semester 1

No.	Subject
1	Design Foundations
2	Drawing and Visual Development
3	Computer Fundamentals
4	Communication Skills
5	Digital Imaging Basics
6	Studio Practice

1. Design Foundations

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Drawing and Visual Development

Unit 1 - Drawing Basics: Line, Shape, Form, Proportion, Perspective

Unit 2 - Observation: Still life, Environment studies, Figure studies, Gesture

Unit 3 - Perspective: One-point, Two-point, Three-point, Camera angles

Unit 4 - Visual Development: Silhouette, Shape language, Thumbnailing, Value studies

Unit 5 - Digital Practice: Drawing tablets, Brushes, Layers, Digital painting workflow

3. Computer Fundamentals

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Communication Skills

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Digital Imaging Basics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 2

No.	Subject
1	Graphic Design
2	Typography
3	Digital Illustration
4	Photography and Digital Imaging
5	Web Design and Front-End
6	Studio Practice

1. Graphic Design

Unit 1 - Graphic Design: Composition, Layout, Branding, Visual identity

Unit 2 - Typography: Type selection, Hierarchy, Pairing, Editorial typography

Unit 3 - Branding: Logo systems, Brand guidelines, Color systems, Stationery

Unit 4 - Publication: Posters, Brochures, Editorial layouts, Packaging

Unit 5 - Digital Design: Social graphics, Digital ads, Web graphics, Export formats

2. Typography

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Digital Illustration

Unit 1 - Illustration: Visual storytelling, Styles, Composition, Concept sketches

Unit 2 - Digital Tools: Brushes, Layers, Masks, Blending, Coloring

Unit 3 - Character Illustration: Anatomy, Expressions, Costume, Posing

Unit 4 - Environment Illustration: Architecture, Lighting, Atmosphere, Perspective

Unit 5 - Portfolio: Series creation, Art direction, Presentation, Client briefs

4. Photography and Digital Imaging

Unit 1 - Camera Basics: Exposure triangle, Lenses, Focus, Metering

Unit 2 - Composition: Framing, Perspective, Depth, Visual balance

Unit 3 - Lighting: Natural light, Studio light, Modifiers, Light ratios

Unit 4 - Image Processing: RAW, Color correction, Retouching, Compositing

Unit 5 - Publishing: Portfolio, Print, Web, Rights and metadata

5. Web Design and Front-End

Unit 1 - Web Foundations: Internet basics, HTTP, HTML, Semantic markup

Unit 2 - Styling: CSS, Layouts, Responsive design, Accessibility

Unit 3 - JavaScript: DOM, Events, Fetch, Modules

Unit 4 - UI Implementation: Components, Forms, State concepts, API integration

Unit 5 - Deployment: Version control, Builds, Hosting, Performance

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 3

No.	Subject
1	Branding and Identity
2	Publication Design
3	Motion Graphics
4	Video Editing
5	UI/UX Design
6	Studio Practice

1. Branding and Identity

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Publication Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Motion Graphics

Unit 1 - Motion Design: Graphic composition, Timing, Transitions, Kinetic type

Unit 2 - Animation Systems: Keyframes, Expressions, Procedural motion, Graph editors

Unit 3 - 3D Motion: 3D layers, Cameras, Lighting, Materials

Unit 4 - Broadcast Design: Titles, Lower thirds, Intros, Promos

Unit 5 - Delivery: Formats, Compression, Broadcast specs, Social media output

4. Video Editing

Unit 1 - Editing Fundamentals: Cuts, Continuity, Pacing, Story structure

Unit 2 - Narrative Editing: Scene construction, Montage, Transitions, Rhythm

Unit 3 - Audio: Dialogue, Music, Sound effects, Mixing basics

Unit 4 - Color: Color correction, Color grading, Scopes, Look creation

Unit 5 - Delivery: Codecs, Resolutions, Aspect ratios, Export

5. UI/UX Design

Unit 1 - UX Foundations: User-centered design, Personas, Journey maps, Problem framing

Unit 2 - Research: Interviews, Surveys, Observation, Usability testing

Unit 3 - Information Architecture: User flows, Sitemaps, Navigation, Content structure

Unit 4 - UI Design: Wireframes, Design systems, Components, Responsive layouts

Unit 5 - Prototyping: Interactive prototypes, Testing, Accessibility, Handoff

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 4

No.	Subject
1	Advanced Graphic Design
2	Packaging Design
3	Digital Advertising
4	3D Foundations
5	Creative Coding
6	Studio Practice

1. Advanced Graphic Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Packaging Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Digital Advertising

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. 3D Foundations

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Creative Coding

Unit 1 - Programming for Creatives: Variables, Functions, Loops, Events

Unit 2 - Graphics Programming: 2D primitives, Transforms, Color, Animation

Unit 3 - Interactive Media: Input, Audio, Interaction, State

Unit 4 - Generative Art: Procedural systems, Noise, Pattern generation, Simulation

Unit 5 - Creative Projects: Concept, Prototype, Iteration, Presentation

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 5

No.	Subject
1	Art Direction
2	UX/UI Production
3	Motion Design
4	Portfolio and Professional Practice
5	Freelancing
6	Project

1. Art Direction

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. UX/UI Production

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Motion Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Portfolio and Professional Practice

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

5. Freelancing

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

Semester 6

No.	Subject
1	Advanced Multimedia Production
2	Digital Campaigns
3	Studio Workflow
4	Portfolio
5	Major Project
6	Internship

1. Advanced Multimedia Production

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Digital Campaigns

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Studio Workflow

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Portfolio

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

5. Major Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

6. Internship

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Polytechnic - Game Design & Development

Semester 1

No.	Subject
1	Design Foundations
2	Programming Fundamentals
3	Drawing and Visual Development
4	Computer Fundamentals
5	Mathematics
6	Lab Practice

1. Design Foundations

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Programming Fundamentals

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Drawing and Visual Development

Unit 1 - Drawing Basics: Line, Shape, Form, Proportion, Perspective

Unit 2 - Observation: Still life, Environment studies, Figure studies, Gesture

Unit 3 - Perspective: One-point, Two-point, Three-point, Camera angles

Unit 4 - Visual Development: Silhouette, Shape language, Thumbnailing, Value studies

Unit 5 - Digital Practice: Drawing tablets, Brushes, Layers, Digital painting workflow

4. Computer Fundamentals

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Mathematics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Lab Practice

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 2

No.	Subject
1	Game Design and Development
2	Game Programming
3	Game Art and 3D
4	Web Design and Front-End
5	Digital Illustration
6	Lab Practice

1. Game Design and Development

Unit 1 - Game Design: Game mechanics, Rules, Feedback, Player experience

Unit 2 - Game Programming: Scripting, Input, Physics, Game loops

Unit 3 - Game Art: 2D assets, 3D assets, Materials, Animation

Unit 4 - Game Systems: UI, Audio, Saving, AI for games

Unit 5 - Production: Optimization, Testing, Publishing, Monetization basics

2. Game Programming

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Game Art and 3D

Unit 1 - Concept Art: Ideation, Silhouette, Style guides, Reference boards

Unit 2 - 3D Assets: Modeling, UVs, Texturing, Optimization

Unit 3 - Characters: Sculpting, Retopology, Rigging, Skinning

Unit 4 - Environments: Modular kits, Materials, Lighting, World building

Unit 5 - Real-time Art: Engine import, LODs, Shaders, Performance

4. Web Design and Front-End

Unit 1 - Web Foundations: Internet basics, HTTP, HTML, Semantic markup

Unit 2 - Styling: CSS, Layouts, Responsive design, Accessibility

Unit 3 - JavaScript: DOM, Events, Fetch, Modules

Unit 4 - UI Implementation: Components, Forms, State concepts, API integration

Unit 5 - Deployment: Version control, Builds, Hosting, Performance

5. Digital Illustration

Unit 1 - Illustration: Visual storytelling, Styles, Composition, Concept sketches

Unit 2 - Digital Tools: Brushes, Layers, Masks, Blending, Coloring

Unit 3 - Character Illustration: Anatomy, Expressions, Costume, Posing

Unit 4 - Environment Illustration: Architecture, Lighting, Atmosphere, Perspective

Unit 5 - Portfolio: Series creation, Art direction, Presentation, Client briefs

6. Lab Practice

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 3

No.	Subject
1	3D Modeling
2	Game Art and 3D
3	Game Programming II
4	Animation Principles
5	Game Audio
6	Game Engine Lab

1. 3D Modeling

Unit 1 - 3D Foundations: 3D coordinate systems, Primitives, Topology, Transformations

Unit 2 - Polygon Modeling: Extrusion, Inset, Bevel, Edge flow, Subdivision

Unit 3 - Hard Surface: Mechanical forms, Booleans, Detailing, Optimization

Unit 4 - Organic Modeling: Sculpting, Base meshes, Anatomical forms, Retopology

Unit 5 - Production: UVs, Asset organization, Export, Game-ready models

2. Game Art and 3D

Unit 1 - Concept Art: Ideation, Silhouette, Style guides, Reference boards

Unit 2 - 3D Assets: Modeling, UVs, Texturing, Optimization

Unit 3 - Characters: Sculpting, Retopology, Rigging, Skinning

Unit 4 - Environments: Modular kits, Materials, Lighting, World building

Unit 5 - Real-time Art: Engine import, LODs, Shaders, Performance

3. Game Programming II

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Animation Principles

Unit 1 - Foundations: Timing, Spacing, Squash and stretch, Anticipation

Unit 2 - Motion: Arcs, Overlap, Follow-through, Secondary action

Unit 3 - Acting: Poses, Facial expressions, Body language, Staging

Unit 4 - Workflow: Storyboards, Animatics, Blocking, Polish

Unit 5 - Production: Shot management, Review, Rendering, Delivery

5. Game Audio

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Game Engine Lab

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 4

No.	Subject
1	Game Systems
2	3D Animation
3	Level Design
4	UI/UX Design
5	Creative Coding
6	Game Engine Lab

1. Game Systems

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. 3D Animation

Unit 1 - Character Setup: Rigging concepts, Controllers, Skinning, Constraints

Unit 2 - Body Mechanics: Walk, Run, Weight, Balance

Unit 3 - Character Acting: Poses, Dialogue, Facial animation, Lip sync

Unit 4 - Advanced Animation: Motion blending, Crowds, Procedural motion

Unit 5 - Production: Shot workflow, Caching, Review, Final render

3. Level Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. UI/UX Design

Unit 1 - UX Foundations: User-centered design, Personas, Journey maps, Problem framing

Unit 2 - Research: Interviews, Surveys, Observation, Usability testing

Unit 3 - Information Architecture: User flows, Sitemaps, Navigation, Content structure

Unit 4 - UI Design: Wireframes, Design systems, Components, Responsive layouts

Unit 5 - Prototyping: Interactive prototypes, Testing, Accessibility, Handoff

5. Creative Coding

Unit 1 - Programming for Creatives: Variables, Functions, Loops, Events

Unit 2 - Graphics Programming: 2D primitives, Transforms, Color, Animation

Unit 3 - Interactive Media: Input, Audio, Interaction, State

Unit 4 - Generative Art: Procedural systems, Noise, Pattern generation, Simulation

Unit 5 - Creative Projects: Concept, Prototype, Iteration, Presentation

6. Game Engine Lab

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 5

No.	Subject
1	Advanced Game Development
2	Game AI Basics
3	Real-time Graphics
4	Optimization
5	Portfolio and Professional Practice
6	Project

1. Advanced Game Development

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Game AI Basics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Real-time Graphics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Optimization

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Portfolio and Professional Practice

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

6. Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

Semester 6

No.	Subject
1	Game Production
2	Publishing
3	Technical Art
4	Studio Workflow
5	Major Project
6	Internship

1. Game Production

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Publishing

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Technical Art

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Studio Workflow

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Major Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

6. Internship

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Polytechnic - UI/UX & Digital Product Design

Semester 1

No.	Subject
1	Design Foundations
2	Drawing and Visual Development
3	Computer Fundamentals
4	Communication Skills
5	Digital Imaging Basics
6	Studio Practice

1. Design Foundations

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Drawing and Visual Development

Unit 1 - Drawing Basics: Line, Shape, Form, Proportion, Perspective

Unit 2 - Observation: Still life, Environment studies, Figure studies, Gesture

Unit 3 - Perspective: One-point, Two-point, Three-point, Camera angles

Unit 4 - Visual Development: Silhouette, Shape language, Thumbnailing, Value studies

Unit 5 - Digital Practice: Drawing tablets, Brushes, Layers, Digital painting workflow

3. Computer Fundamentals

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Communication Skills

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Digital Imaging Basics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 2

No.	Subject
1	UI/UX Design
2	Typography
3	Graphic Design
4	Web Design and Front-End
5	Research Methods
6	Studio Practice

1. UI/UX Design

Unit 1 - UX Foundations: User-centered design, Personas, Journey maps, Problem framing

Unit 2 - Research: Interviews, Surveys, Observation, Usability testing

Unit 3 - Information Architecture: User flows, Sitemaps, Navigation, Content structure

Unit 4 - UI Design: Wireframes, Design systems, Components, Responsive layouts

Unit 5 - Prototyping: Interactive prototypes, Testing, Accessibility, Handoff

2. Typography

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Graphic Design

Unit 1 - Graphic Design: Composition, Layout, Branding, Visual identity

Unit 2 - Typography: Type selection, Hierarchy, Pairing, Editorial typography

Unit 3 - Branding: Logo systems, Brand guidelines, Color systems, Stationery

Unit 4 - Publication: Posters, Brochures, Editorial layouts, Packaging

Unit 5 - Digital Design: Social graphics, Digital ads, Web graphics, Export formats

4. Web Design and Front-End

Unit 1 - Web Foundations: Internet basics, HTTP, HTML, Semantic markup

Unit 2 - Styling: CSS, Layouts, Responsive design, Accessibility

Unit 3 - JavaScript: DOM, Events, Fetch, Modules

Unit 4 - UI Implementation: Components, Forms, State concepts, API integration

Unit 5 - Deployment: Version control, Builds, Hosting, Performance

5. Research Methods

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 3

No.	Subject
1	Interaction Design
2	Information Architecture
3	Prototyping
4	Design Systems
5	Creative Coding
6	Studio Practice

1. Interaction Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Information Architecture

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Prototyping

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Design Systems

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Creative Coding

Unit 1 - Programming for Creatives: Variables, Functions, Loops, Events

Unit 2 - Graphics Programming: 2D primitives, Transforms, Color, Animation

Unit 3 - Interactive Media: Input, Audio, Interaction, State

Unit 4 - Generative Art: Procedural systems, Noise, Pattern generation, Simulation

Unit 5 - Creative Projects: Concept, Prototype, Iteration, Presentation

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 4

No.	Subject
1	Usability Testing
2	Accessibility
3	Motion for Interfaces
4	Design Systems II
5	Front-End Production
6	Studio Practice

1. Usability Testing

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Accessibility

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Motion for Interfaces

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Design Systems II

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Front-End Production

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 5

No.	Subject
1	Product Design
2	Design Strategy
3	Service Design
4	Portfolio and Professional Practice
5	Freelancing
6	Project

1. Product Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Design Strategy

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Service Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Portfolio and Professional Practice

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

5. Freelancing

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

Semester 6

No.	Subject
1	Advanced Product Design
2	Industry Workflow
3	Portfolio
4	Major Project
5	Internship
6	Seminar

1. Advanced Product Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Industry Workflow

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Portfolio

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

4. Major Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

5. Internship

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Seminar

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

B. UNDERGRADUATE - CREATIVE, COMPUTING & DESIGN PROGRAMS

B.Sc - Animation & VFX

Semester 1

No.	Subject
1	Design Foundations
2	Drawing and Visual Development
3	Computer Graphics Basics
4	Digital Imaging Basics
5	Communication Skills
6	Studio Practice

1. Design Foundations

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Drawing and Visual Development

Unit 1 - Drawing Basics: Line, Shape, Form, Proportion, Perspective

Unit 2 - Observation: Still life, Environment studies, Figure studies, Gesture

Unit 3 - Perspective: One-point, Two-point, Three-point, Camera angles

Unit 4 - Visual Development: Silhouette, Shape language, Thumbnailing, Value studies

Unit 5 - Digital Practice: Drawing tablets, Brushes, Layers, Digital painting workflow

3. Computer Graphics Basics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Digital Imaging Basics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Communication Skills

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 2

No.	Subject
1	Graphic Design
2	2D Animation
3	Digital Illustration
4	Video Editing
5	Photography and Digital Imaging
6	Studio Practice

1. Graphic Design

Unit 1 - Graphic Design: Composition, Layout, Branding, Visual identity
Unit 2 - Typography: Type selection, Hierarchy, Pairing, Editorial typography
Unit 3 - Branding: Logo systems, Brand guidelines, Color systems, Stationery
Unit 4 - Publication: Posters, Brochures, Editorial layouts, Packaging
Unit 5 - Digital Design: Social graphics, Digital ads, Web graphics, Export formats

2. 2D Animation

Unit 1 - 2D Basics: Frame rates, Keyframes, In-betweens, Exposure
Unit 2 - Character Animation: Walk cycles, Run cycles, Dialogue, Acting
Unit 3 - Backgrounds: Layout, Perspective, Color keys, Painted backgrounds
Unit 4 - Compositing: Layers, Masks, Camera, Effects
Unit 5 - Production: Pipeline, Shot tracking, Rendering, Delivery

3. Digital Illustration

Unit 1 - Illustration: Visual storytelling, Styles, Composition, Concept sketches
Unit 2 - Digital Tools: Brushes, Layers, Masks, Blending, Coloring
Unit 3 - Character Illustration: Anatomy, Expressions, Costume, Posing
Unit 4 - Environment Illustration: Architecture, Lighting, Atmosphere, Perspective
Unit 5 - Portfolio: Series creation, Art direction, Presentation, Client briefs

4. Video Editing

Unit 1 - Editing Fundamentals: Cuts, Continuity, Pacing, Story structure
Unit 2 - Narrative Editing: Scene construction, Montage, Transitions, Rhythm
Unit 3 - Audio: Dialogue, Music, Sound effects, Mixing basics
Unit 4 - Color: Color correction, Color grading, Scopes, Look creation
Unit 5 - Delivery: Codecs, Resolutions, Aspect ratios, Export

5. Photography and Digital Imaging

Unit 1 - Camera Basics: Exposure triangle, Lenses, Focus, Metering
Unit 2 - Composition: Framing, Perspective, Depth, Visual balance
Unit 3 - Lighting: Natural light, Studio light, Modifiers, Light ratios
Unit 4 - Image Processing: RAW, Color correction, Retouching, Compositing
Unit 5 - Publishing: Portfolio, Print, Web, Rights and metadata

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy
Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color
Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability
Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection
Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 3

No.	Subject
1	3D Modeling
2	Texturing and Materials
3	3D Animation
4	Compositing Fundamentals
5	Motion Graphics
6	Studio Practice

1. 3D Modeling

Unit 1 - 3D Foundations: 3D coordinate systems, Primitives, Topology, Transformations

Unit 2 - Polygon Modeling: Extrusion, Inset, Bevel, Edge flow, Subdivision

Unit 3 - Hard Surface: Mechanical forms, Booleans, Detailing, Optimization

Unit 4 - Organic Modeling: Sculpting, Base meshes, Anatomical forms, Retopology

Unit 5 - Production: UVs, Asset organization, Export, Game-ready models

2. Texturing and Materials

Unit 1 - Texturing Basics: UV mapping, Texture coordinates, Resolution, Seam management

Unit 2 - PBR: Albedo, Roughness, Metallic, Normal maps, Ambient occlusion

Unit 3 - Procedural Materials: Nodes, Masks, Generators, Material variations

Unit 4 - Surface Detail: Microdetail, Weathering, Imperfections, Decals

Unit 5 - Look Development: Material libraries, Consistency, Optimization, Export

3. 3D Animation

Unit 1 - Character Setup: Rigging concepts, Controllers, Skinning, Constraints

Unit 2 - Body Mechanics: Walk, Run, Weight, Balance

Unit 3 - Character Acting: Poses, Dialogue, Facial animation, Lip sync

Unit 4 - Advanced Animation: Motion blending, Crowds, Procedural motion

Unit 5 - Production: Shot workflow, Caching, Review, Final render

4. Compositing Fundamentals

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Motion Graphics

Unit 1 - Motion Design: Graphic composition, Timing, Transitions, Kinetic type

Unit 2 - Animation Systems: Keyframes, Expressions, Procedural motion, Graph editors

Unit 3 - 3D Motion: 3D layers, Cameras, Lighting, Materials

Unit 4 - Broadcast Design: Titles, Lower thirds, Intros, Promos

Unit 5 - Delivery: Formats, Compression, Broadcast specs, Social media output

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 4

No.	Subject
1	Visual Effects (VFX)
2	Advanced 3D Animation
3	Character Animation
4	Film and Video Production
5	Portfolio and Professional Practice
6	Major Project

1. Visual Effects (VFX)

Unit 1 - VFX Foundations: VFX pipeline, Shot planning, Reference gathering, Continuity

Unit 2 - Compositing: Keying, Rotoscopy, Tracking, Color matching

Unit 3 - Effects: Particles, Smoke, Fire, Destruction, Fluid effects

Unit 4 - 3D Integration: Matchmoving, Camera tracking, CG integration, Lighting

Unit 5 - Finishing: Color correction, Grain, Cleanup, Delivery

2. Advanced 3D Animation

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Character Animation

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Film and Video Production

Unit 1 - Pre-production: Script, Concept, Research, Budget, Scheduling

Unit 2 - Cinematography: Cameras, Lenses, Composition, Exposure, Lighting

Unit 3 - Production: Directing, Sound recording, Continuity, Set workflow

Unit 4 - Post-production: Editing, Color, Sound, VFX

Unit 5 - Distribution: Mastering, Subtitles, Platforms, Promotion

5. Portfolio and Professional Practice

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

6. Major Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

B.Sc - Multimedia & Digital Media

Semester 1

No.	Subject
1	Design Foundations
2	Drawing and Visual Development
3	Computer Fundamentals
4	Digital Imaging Basics
5	Communication Skills
6	Studio Practice

1. Design Foundations

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Drawing and Visual Development

Unit 1 - Drawing Basics: Line, Shape, Form, Proportion, Perspective

Unit 2 - Observation: Still life, Environment studies, Figure studies, Gesture

Unit 3 - Perspective: One-point, Two-point, Three-point, Camera angles

Unit 4 - Visual Development: Silhouette, Shape language, Thumbnailing, Value studies

Unit 5 - Digital Practice: Drawing tablets, Brushes, Layers, Digital painting workflow

3. Computer Fundamentals

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Digital Imaging Basics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Communication Skills

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 2

No.	Subject
1	Graphic Design
2	Web Design and Front-End
3	Digital Illustration
4	Photography and Digital Imaging
5	Video Editing
6	Studio Practice

1. Graphic Design

Unit 1 - Graphic Design: Composition, Layout, Branding, Visual identity
Unit 2 - Typography: Type selection, Hierarchy, Pairing, Editorial typography
Unit 3 - Branding: Logo systems, Brand guidelines, Color systems, Stationery
Unit 4 - Publication: Posters, Brochures, Editorial layouts, Packaging
Unit 5 - Digital Design: Social graphics, Digital ads, Web graphics, Export formats

2. Web Design and Front-End

Unit 1 - Web Foundations: Internet basics, HTTP, HTML, Semantic markup
Unit 2 - Styling: CSS, Layouts, Responsive design, Accessibility
Unit 3 - JavaScript: DOM, Events, Fetch, Modules
Unit 4 - UI Implementation: Components, Forms, State concepts, API integration
Unit 5 - Deployment: Version control, Builds, Hosting, Performance

3. Digital Illustration

Unit 1 - Illustration: Visual storytelling, Styles, Composition, Concept sketches
Unit 2 - Digital Tools: Brushes, Layers, Masks, Blending, Coloring
Unit 3 - Character Illustration: Anatomy, Expressions, Costume, Posing
Unit 4 - Environment Illustration: Architecture, Lighting, Atmosphere, Perspective
Unit 5 - Portfolio: Series creation, Art direction, Presentation, Client briefs

4. Photography and Digital Imaging

Unit 1 - Camera Basics: Exposure triangle, Lenses, Focus, Metering
Unit 2 - Composition: Framing, Perspective, Depth, Visual balance
Unit 3 - Lighting: Natural light, Studio light, Modifiers, Light ratios
Unit 4 - Image Processing: RAW, Color correction, Retouching, Compositing
Unit 5 - Publishing: Portfolio, Print, Web, Rights and metadata

5. Video Editing

Unit 1 - Editing Fundamentals: Cuts, Continuity, Pacing, Story structure
Unit 2 - Narrative Editing: Scene construction, Montage, Transitions, Rhythm
Unit 3 - Audio: Dialogue, Music, Sound effects, Mixing basics
Unit 4 - Color: Color correction, Color grading, Scopes, Look creation
Unit 5 - Delivery: Codecs, Resolutions, Aspect ratios, Export

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy
Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color
Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability
Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection
Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 3

No.	Subject
1	UI/UX Design
2	Motion Graphics
3	3D Modeling
4	Creative Coding
5	Game Design and Development
6	Studio Practice

1. UI/UX Design

Unit 1 - UX Foundations: User-centered design, Personas, Journey maps, Problem framing

Unit 2 - Research: Interviews, Surveys, Observation, Usability testing

Unit 3 - Information Architecture: User flows, Sitemaps, Navigation, Content structure

Unit 4 - UI Design: Wireframes, Design systems, Components, Responsive layouts

Unit 5 - Prototyping: Interactive prototypes, Testing, Accessibility, Handoff

2. Motion Graphics

Unit 1 - Motion Design: Graphic composition, Timing, Transitions, Kinetic type

Unit 2 - Animation Systems: Keyframes, Expressions, Procedural motion, Graph editors

Unit 3 - 3D Motion: 3D layers, Cameras, Lighting, Materials

Unit 4 - Broadcast Design: Titles, Lower thirds, Intros, Promos

Unit 5 - Delivery: Formats, Compression, Broadcast specs, Social media output

3. 3D Modeling

Unit 1 - 3D Foundations: 3D coordinate systems, Primitives, Topology, Transformations

Unit 2 - Polygon Modeling: Extrusion, Inset, Bevel, Edge flow, Subdivision

Unit 3 - Hard Surface: Mechanical forms, Booleans, Detailing, Optimization

Unit 4 - Organic Modeling: Sculpting, Base meshes, Anatomical forms, Retopology

Unit 5 - Production: UVs, Asset organization, Export, Game-ready models

4. Creative Coding

Unit 1 - Programming for Creatives: Variables, Functions, Loops, Events

Unit 2 - Graphics Programming: 2D primitives, Transforms, Color, Animation

Unit 3 - Interactive Media: Input, Audio, Interaction, State

Unit 4 - Generative Art: Procedural systems, Noise, Pattern generation, Simulation

Unit 5 - Creative Projects: Concept, Prototype, Iteration, Presentation

5. Game Design and Development

Unit 1 - Game Design: Game mechanics, Rules, Feedback, Player experience

Unit 2 - Game Programming: Scripting, Input, Physics, Game loops

Unit 3 - Game Art: 2D assets, 3D assets, Materials, Animation

Unit 4 - Game Systems: UI, Audio, Saving, AI for games

Unit 5 - Production: Optimization, Testing, Publishing, Monetization basics

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 4

No.	Subject
1	VFX
2	Interactive Media
3	Advanced UI/UX
4	Portfolio and Professional Practice
5	Major Project
6	Industry Internship

1. VFX

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Interactive Media

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Advanced UI/UX

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Portfolio and Professional Practice

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

5. Major Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

6. Industry Internship

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

B.Sc - Computer Science

Semester 1

No.	Subject
1	Engineering Mathematics I
2	Programming Fundamentals
3	Computer Fundamentals
4	Digital Logic
5	Communication Skills
6	Programming Lab

1. Engineering Mathematics I

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Programming Fundamentals

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Computer Fundamentals

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Digital Logic

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Communication Skills

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Programming Lab

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 2

No.	Subject
1	Engineering Mathematics II
2	Data Structures
3	Object-Oriented Programming
4	Computer Organization
5	Web Design and Front-End
6	Lab

1. Engineering Mathematics II

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Data Structures

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Object-Oriented Programming

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Computer Organization

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Web Design and Front-End

Unit 1 - Web Foundations: Internet basics, HTTP, HTML, Semantic markup

Unit 2 - Styling: CSS, Layouts, Responsive design, Accessibility

Unit 3 - JavaScript: DOM, Events, Fetch, Modules

Unit 4 - UI Implementation: Components, Forms, State concepts, API integration

Unit 5 - Deployment: Version control, Builds, Hosting, Performance

6. Lab

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 3

No.	Subject
1	Database Management Systems
2	Operating Systems
3	Computer Networks
4	Algorithms
5	Software Engineering
6	Lab

1. Database Management Systems

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Operating Systems

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Computer Networks

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Algorithms

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Software Engineering

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Lab

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 4

No.	Subject
1	Artificial Intelligence
2	Machine Learning
3	Cloud Computing
4	Cyber Security
5	Big Data Analytics
6	Major Project

1. Artificial Intelligence

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Machine Learning

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Cloud Computing

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Cyber Security

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Big Data Analytics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Major Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

BCA - Bachelor of Computer Applications

Semester 1

No.	Subject
1	Mathematics for Computing
2	Programming Fundamentals
3	Computer Fundamentals
4	Web Design and Front-End
5	Communication Skills
6	Programming Lab

1. Mathematics for Computing

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Programming Fundamentals

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Computer Fundamentals

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Web Design and Front-End

Unit 1 - Web Foundations: Internet basics, HTTP, HTML, Semantic markup

Unit 2 - Styling: CSS, Layouts, Responsive design, Accessibility

Unit 3 - JavaScript: DOM, Events, Fetch, Modules

Unit 4 - UI Implementation: Components, Forms, State concepts, API integration

Unit 5 - Deployment: Version control, Builds, Hosting, Performance

5. Communication Skills

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Programming Lab

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 2

No.	Subject
1	Data Structures
2	Object-Oriented Programming
3	Database Management Systems
4	Computer Organization
5	Operating Systems
6	Lab

1. Data Structures

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Object-Oriented Programming

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Database Management Systems

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Computer Organization

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Operating Systems

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Lab

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 3

No.	Subject
1	Computer Networks
2	Software Engineering
3	Web Application Development
4	Cloud Computing
5	Cyber Security
6	Project

1. Computer Networks

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Software Engineering

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Web Application Development

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Cloud Computing

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Cyber Security

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

Semester 4

No.	Subject
1	AI Fundamentals
2	Data Analytics
3	Mobile Application Development
4	DevOps Basics
5	Major Project
6	Internship

1. AI Fundamentals

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Data Analytics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Mobile Application Development

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. DevOps Basics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Major Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

6. Internship

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 5

No.	Subject
1	Machine Learning Fundamentals
2	Advanced Web Development
3	UI/UX Design
4	Cloud Applications
5	Professional Elective
6	Project

1. Machine Learning Fundamentals

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Advanced Web Development

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. UI/UX Design

Unit 1 - UX Foundations: User-centered design, Personas, Journey maps, Problem framing

Unit 2 - Research: Interviews, Surveys, Observation, Usability testing

Unit 3 - Information Architecture: User flows, Sitemaps, Navigation, Content structure

Unit 4 - UI Design: Wireframes, Design systems, Components, Responsive layouts

Unit 5 - Prototyping: Interactive prototypes, Testing, Accessibility, Handoff

4. Cloud Applications

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Professional Elective

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

6. Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

Semester 6

No.	Subject
1	Advanced Application Development
2	Data Engineering Basics
3	Cyber Security
4	Professional Elective
5	Major Project
6	Internship

1. Advanced Application Development

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Data Engineering Basics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Cyber Security

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Professional Elective

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

5. Major Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

6. Internship

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

B.Des - Communication Design

Semester 1

No.	Subject
1	Design Foundations
2	Drawing and Visual Development
3	Typography
4	History of Design
5	Photography and Digital Imaging
6	Studio Practice

1. Design Foundations

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Drawing and Visual Development

Unit 1 - Drawing Basics: Line, Shape, Form, Proportion, Perspective

Unit 2 - Observation: Still life, Environment studies, Figure studies, Gesture

Unit 3 - Perspective: One-point, Two-point, Three-point, Camera angles

Unit 4 - Visual Development: Silhouette, Shape language, Thumbnailing, Value studies

Unit 5 - Digital Practice: Drawing tablets, Brushes, Layers, Digital painting workflow

3. Typography

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. History of Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Photography and Digital Imaging

Unit 1 - Camera Basics: Exposure triangle, Lenses, Focus, Metering

Unit 2 - Composition: Framing, Perspective, Depth, Visual balance

Unit 3 - Lighting: Natural light, Studio light, Modifiers, Light ratios

Unit 4 - Image Processing: RAW, Color correction, Retouching, Compositing

Unit 5 - Publishing: Portfolio, Print, Web, Rights and metadata

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 2

No.	Subject
1	Graphic Design
2	Digital Illustration
3	Branding and Identity
4	Publication Design
5	UI/UX Design
6	Studio Practice

1. Graphic Design

Unit 1 - Graphic Design: Composition, Layout, Branding, Visual identity
Unit 2 - Typography: Type selection, Hierarchy, Pairing, Editorial typography
Unit 3 - Branding: Logo systems, Brand guidelines, Color systems, Stationery
Unit 4 - Publication: Posters, Brochures, Editorial layouts, Packaging
Unit 5 - Digital Design: Social graphics, Digital ads, Web graphics, Export formats

2. Digital Illustration

Unit 1 - Illustration: Visual storytelling, Styles, Composition, Concept sketches
Unit 2 - Digital Tools: Brushes, Layers, Masks, Blending, Coloring
Unit 3 - Character Illustration: Anatomy, Expressions, Costume, Posing
Unit 4 - Environment Illustration: Architecture, Lighting, Atmosphere, Perspective
Unit 5 - Portfolio: Series creation, Art direction, Presentation, Client briefs

3. Branding and Identity

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy
Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color
Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability
Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection
Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Publication Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy
Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color
Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability
Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection
Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. UI/UX Design

Unit 1 - UX Foundations: User-centered design, Personas, Journey maps, Problem framing
Unit 2 - Research: Interviews, Surveys, Observation, Usability testing
Unit 3 - Information Architecture: User flows, Sitemaps, Navigation, Content structure
Unit 4 - UI Design: Wireframes, Design systems, Components, Responsive layouts
Unit 5 - Prototyping: Interactive prototypes, Testing, Accessibility, Handoff

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy
Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color
Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability
Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection
Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 3

No.	Subject
1	Motion Graphics
2	Packaging Design
3	Information Design
4	Art Direction
5	Creative Coding
6	Studio Practice

1. Motion Graphics

Unit 1 - Motion Design: Graphic composition, Timing, Transitions, Kinetic type

Unit 2 - Animation Systems: Keyframes, Expressions, Procedural motion, Graph editors

Unit 3 - 3D Motion: 3D layers, Cameras, Lighting, Materials

Unit 4 - Broadcast Design: Titles, Lower thirds, Intros, Promos

Unit 5 - Delivery: Formats, Compression, Broadcast specs, Social media output

2. Packaging Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Information Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Art Direction

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Creative Coding

Unit 1 - Programming for Creatives: Variables, Functions, Loops, Events

Unit 2 - Graphics Programming: 2D primitives, Transforms, Color, Animation

Unit 3 - Interactive Media: Input, Audio, Interaction, State

Unit 4 - Generative Art: Procedural systems, Noise, Pattern generation, Simulation

Unit 5 - Creative Projects: Concept, Prototype, Iteration, Presentation

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 4

No.	Subject
1	Design Strategy
2	Service Design
3	Portfolio and Professional Practice
4	Major Project
5	Internship
6	Thesis/Exhibition

1. Design Strategy

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Service Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Portfolio and Professional Practice

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

4. Major Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

5. Internship

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Thesis/Exhibition

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

B.Des - UI/UX & Interaction Design

Semester 1

No.	Subject
1	Design Foundations
2	Drawing and Visual Development
3	Typography
4	Design History
5	Digital Imaging Basics
6	Studio Practice

1. Design Foundations

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Drawing and Visual Development

Unit 1 - Drawing Basics: Line, Shape, Form, Proportion, Perspective

Unit 2 - Observation: Still life, Environment studies, Figure studies, Gesture

Unit 3 - Perspective: One-point, Two-point, Three-point, Camera angles

Unit 4 - Visual Development: Silhouette, Shape language, Thumbnailing, Value studies

Unit 5 - Digital Practice: Drawing tablets, Brushes, Layers, Digital painting workflow

3. Typography

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Design History

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Digital Imaging Basics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 2

No.	Subject
1	UI/UX Design
2	User Research
3	Information Architecture
4	Interaction Design
5	Web Design and Front-End
6	Studio Practice

1. UI/UX Design

Unit 1 - UX Foundations: User-centered design, Personas, Journey maps, Problem framing

Unit 2 - Research: Interviews, Surveys, Observation, Usability testing

Unit 3 - Information Architecture: User flows, Sitemaps, Navigation, Content structure

Unit 4 - UI Design: Wireframes, Design systems, Components, Responsive layouts

Unit 5 - Prototyping: Interactive prototypes, Testing, Accessibility, Handoff

2. User Research

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Information Architecture

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Interaction Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Web Design and Front-End

Unit 1 - Web Foundations: Internet basics, HTTP, HTML, Semantic markup

Unit 2 - Styling: CSS, Layouts, Responsive design, Accessibility

Unit 3 - JavaScript: DOM, Events, Fetch, Modules

Unit 4 - UI Implementation: Components, Forms, State concepts, API integration

Unit 5 - Deployment: Version control, Builds, Hosting, Performance

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 3

No.	Subject
1	Design Systems
2	Prototyping
3	Usability Testing
4	Accessibility
5	Creative Coding
6	Studio Practice

1. Design Systems

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Prototyping

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Usability Testing

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Accessibility

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Creative Coding

Unit 1 - Programming for Creatives: Variables, Functions, Loops, Events

Unit 2 - Graphics Programming: 2D primitives, Transforms, Color, Animation

Unit 3 - Interactive Media: Input, Audio, Interaction, State

Unit 4 - Generative Art: Procedural systems, Noise, Pattern generation, Simulation

Unit 5 - Creative Projects: Concept, Prototype, Iteration, Presentation

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 4

No.	Subject
1	Product Design
2	Service Design
3	Design Strategy
4	Portfolio and Professional Practice
5	Major Project
6	Internship

1. Product Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Service Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Design Strategy

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Portfolio and Professional Practice

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

5. Major Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

6. Internship

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

B.Des - Game Design

Semester 1

No.	Subject
1	Design Foundations
2	Drawing and Visual Development
3	Game Design and Development
4	Computer Fundamentals
5	Communication Skills
6	Studio Practice

1. Design Foundations

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Drawing and Visual Development

Unit 1 - Drawing Basics: Line, Shape, Form, Proportion, Perspective

Unit 2 - Observation: Still life, Environment studies, Figure studies, Gesture

Unit 3 - Perspective: One-point, Two-point, Three-point, Camera angles

Unit 4 - Visual Development: Silhouette, Shape language, Thumbnailing, Value studies

Unit 5 - Digital Practice: Drawing tablets, Brushes, Layers, Digital painting workflow

3. Game Design and Development

Unit 1 - Game Design: Game mechanics, Rules, Feedback, Player experience

Unit 2 - Game Programming: Scripting, Input, Physics, Game loops

Unit 3 - Game Art: 2D assets, 3D assets, Materials, Animation

Unit 4 - Game Systems: UI, Audio, Saving, AI for games

Unit 5 - Production: Optimization, Testing, Publishing, Monetization basics

4. Computer Fundamentals

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Communication Skills

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 2

No.	Subject
1	Game Art and 3D
2	Game Programming
3	Level Design
4	UI/UX Design
5	Animation Principles
6	Studio Practice

1. Game Art and 3D

Unit 1 - Concept Art: Ideation, Silhouette, Style guides, Reference boards

Unit 2 - 3D Assets: Modeling, UVs, Texturing, Optimization

Unit 3 - Characters: Sculpting, Retopology, Rigging, Skinning

Unit 4 - Environments: Modular kits, Materials, Lighting, World building

Unit 5 - Real-time Art: Engine import, LODs, Shaders, Performance

2. Game Programming

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Level Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. UI/UX Design

Unit 1 - UX Foundations: User-centered design, Personas, Journey maps, Problem framing

Unit 2 - Research: Interviews, Surveys, Observation, Usability testing

Unit 3 - Information Architecture: User flows, Sitemaps, Navigation, Content structure

Unit 4 - UI Design: Wireframes, Design systems, Components, Responsive layouts

Unit 5 - Prototyping: Interactive prototypes, Testing, Accessibility, Handoff

5. Animation Principles

Unit 1 - Foundations: Timing, Spacing, Squash and stretch, Anticipation

Unit 2 - Motion: Arcs, Overlap, Follow-through, Secondary action

Unit 3 - Acting: Poses, Facial expressions, Body language, Staging

Unit 4 - Workflow: Storyboards, Animatics, Blocking, Polish

Unit 5 - Production: Shot management, Review, Rendering, Delivery

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 3

No.	Subject
1	3D Modeling
2	Character Design
3	Game AI Basics
4	Game Audio
5	Creative Coding
6	Studio Practice

1. 3D Modeling

Unit 1 - 3D Foundations: 3D coordinate systems, Primitives, Topology, Transformations

Unit 2 - Polygon Modeling: Extrusion, Inset, Bevel, Edge flow, Subdivision

Unit 3 - Hard Surface: Mechanical forms, Booleans, Detailing, Optimization

Unit 4 - Organic Modeling: Sculpting, Base meshes, Anatomical forms, Retopology

Unit 5 - Production: UVs, Asset organization, Export, Game-ready models

2. Character Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Game AI Basics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Game Audio

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Creative Coding

Unit 1 - Programming for Creatives: Variables, Functions, Loops, Events

Unit 2 - Graphics Programming: 2D primitives, Transforms, Color, Animation

Unit 3 - Interactive Media: Input, Audio, Interaction, State

Unit 4 - Generative Art: Procedural systems, Noise, Pattern generation, Simulation

Unit 5 - Creative Projects: Concept, Prototype, Iteration, Presentation

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 4

No.	Subject
1	Advanced Game Development
2	Technical Art
3	Game Production
4	Portfolio and Professional Practice
5	Major Project
6	Internship

1. Advanced Game Development

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Technical Art

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Game Production

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Portfolio and Professional Practice

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

5. Major Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

6. Internship

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

B.Des - Animation Film Design

Semester 1

No.	Subject
1	Design Foundations
2	Drawing and Visual Development
3	Animation Principles
4	Storytelling
5	Digital Imaging Basics
6	Studio Practice

1. Design Foundations

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Drawing and Visual Development

Unit 1 - Drawing Basics: Line, Shape, Form, Proportion, Perspective

Unit 2 - Observation: Still life, Environment studies, Figure studies, Gesture

Unit 3 - Perspective: One-point, Two-point, Three-point, Camera angles

Unit 4 - Visual Development: Silhouette, Shape language, Thumbnailing, Value studies

Unit 5 - Digital Practice: Drawing tablets, Brushes, Layers, Digital painting workflow

3. Animation Principles

Unit 1 - Foundations: Timing, Spacing, Squash and stretch, Anticipation

Unit 2 - Motion: Arcs, Overlap, Follow-through, Secondary action

Unit 3 - Acting: Poses, Facial expressions, Body language, Staging

Unit 4 - Workflow: Storyboards, Animatics, Blocking, Polish

Unit 5 - Production: Shot management, Review, Rendering, Delivery

4. Storytelling

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Digital Imaging Basics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 2

No.	Subject
1	2D Animation
2	Character Design
3	Background Design
4	Storyboarding
5	Video Editing
6	Studio Practice

1. 2D Animation

Unit 1 - 2D Basics: Frame rates, Keyframes, In-betweens, Exposure

Unit 2 - Character Animation: Walk cycles, Run cycles, Dialogue, Acting

Unit 3 - Backgrounds: Layout, Perspective, Color keys, Painted backgrounds

Unit 4 - Compositing: Layers, Masks, Camera, Effects

Unit 5 - Production: Pipeline, Shot tracking, Rendering, Delivery

2. Character Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Background Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Storyboarding

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Video Editing

Unit 1 - Editing Fundamentals: Cuts, Continuity, Pacing, Story structure

Unit 2 - Narrative Editing: Scene construction, Montage, Transitions, Rhythm

Unit 3 - Audio: Dialogue, Music, Sound effects, Mixing basics

Unit 4 - Color: Color correction, Color grading, Scopes, Look creation

Unit 5 - Delivery: Codecs, Resolutions, Aspect ratios, Export

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 3

No.	Subject
1	3D Modeling
2	3D Animation
3	Character Animation
4	Compositing Fundamentals
5	Motion Graphics
6	Studio Practice

1. 3D Modeling

Unit 1 - 3D Foundations: 3D coordinate systems, Primitives, Topology, Transformations

Unit 2 - Polygon Modeling: Extrusion, Inset, Bevel, Edge flow, Subdivision

Unit 3 - Hard Surface: Mechanical forms, Booleans, Detailing, Optimization

Unit 4 - Organic Modeling: Sculpting, Base meshes, Anatomical forms, Retopology

Unit 5 - Production: UVs, Asset organization, Export, Game-ready models

2. 3D Animation

Unit 1 - Character Setup: Rigging concepts, Controllers, Skinning, Constraints

Unit 2 - Body Mechanics: Walk, Run, Weight, Balance

Unit 3 - Character Acting: Poses, Dialogue, Facial animation, Lip sync

Unit 4 - Advanced Animation: Motion blending, Crowds, Procedural motion

Unit 5 - Production: Shot workflow, Caching, Review, Final render

3. Character Animation

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Compositing Fundamentals

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Motion Graphics

Unit 1 - Motion Design: Graphic composition, Timing, Transitions, Kinetic type

Unit 2 - Animation Systems: Keyframes, Expressions, Procedural motion, Graph editors

Unit 3 - 3D Motion: 3D layers, Cameras, Lighting, Materials

Unit 4 - Broadcast Design: Titles, Lower thirds, Intros, Promos

Unit 5 - Delivery: Formats, Compression, Broadcast specs, Social media output

6. Studio Practice

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 4

No.	Subject
1	Visual Effects (VFX)
2	Advanced Animation
3	Film and Video Production
4	Portfolio and Professional Practice
5	Major Project
6	Internship

1. Visual Effects (VFX)

Unit 1 - VFX Foundations: VFX pipeline, Shot planning, Reference gathering, Continuity

Unit 2 - Compositing: Keying, Rotoscopy, Tracking, Color matching

Unit 3 - Effects: Particles, Smoke, Fire, Destruction, Fluid effects

Unit 4 - 3D Integration: Matchmoving, Camera tracking, CG integration, Lighting

Unit 5 - Finishing: Color correction, Grain, Cleanup, Delivery

2. Advanced Animation

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Film and Video Production

Unit 1 - Pre-production: Script, Concept, Research, Budget, Scheduling

Unit 2 - Cinematography: Cameras, Lenses, Composition, Exposure, Lighting

Unit 3 - Production: Directing, Sound recording, Continuity, Set workflow

Unit 4 - Post-production: Editing, Color, Sound, VFX

Unit 5 - Distribution: Mastering, Subtitles, Platforms, Promotion

4. Portfolio and Professional Practice

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

5. Major Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

6. Internship

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

C. POSTGRADUATE - CREATIVE, COMPUTING & DESIGN PROGRAMS

M.Des - Communication / Graphic Design

Semester 1

No.	Subject
1	Advanced Design Foundations
2	Design Research
3	Advanced Graphic Design
4	Typography and Editorial Systems
5	Design Strategy
6	Studio Lab

1. Advanced Design Foundations

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Design Research

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Advanced Graphic Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Typography and Editorial Systems

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Design Strategy

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Studio Lab

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 2

No.	Subject
1	Branding Systems
2	Information Design
3	Motion Graphics
4	UI/UX Design
5	Creative Coding
6	Studio Lab

1. Branding Systems

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Information Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Motion Graphics

Unit 1 - Motion Design: Graphic composition, Timing, Transitions, Kinetic type

Unit 2 - Animation Systems: Keyframes, Expressions, Procedural motion, Graph editors

Unit 3 - 3D Motion: 3D layers, Cameras, Lighting, Materials

Unit 4 - Broadcast Design: Titles, Lower thirds, Intros, Promos

Unit 5 - Delivery: Formats, Compression, Broadcast specs, Social media output

4. UI/UX Design

Unit 1 - UX Foundations: User-centered design, Personas, Journey maps, Problem framing

Unit 2 - Research: Interviews, Surveys, Observation, Usability testing

Unit 3 - Information Architecture: User flows, Sitemaps, Navigation, Content structure

Unit 4 - UI Design: Wireframes, Design systems, Components, Responsive layouts

Unit 5 - Prototyping: Interactive prototypes, Testing, Accessibility, Handoff

5. Creative Coding

Unit 1 - Programming for Creatives: Variables, Functions, Loops, Events

Unit 2 - Graphics Programming: 2D primitives, Transforms, Color, Animation

Unit 3 - Interactive Media: Input, Audio, Interaction, State

Unit 4 - Generative Art: Procedural systems, Noise, Pattern generation, Simulation

Unit 5 - Creative Projects: Concept, Prototype, Iteration, Presentation

6. Studio Lab

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 3

No.	Subject
1	Design Management
2	Advanced Art Direction
3	Portfolio and Professional Practice
4	Research Seminar
5	Dissertation / Major Project

1. Design Management

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Advanced Art Direction

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Portfolio and Professional Practice

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

4. Research Seminar

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Dissertation / Major Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

Semester 4

No.	Subject
1	Dissertation / Major Project
2	Thesis
3	Exhibition / Portfolio
4	Final Viva

1. Dissertation / Major Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

2. Thesis

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Exhibition / Portfolio

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

4. Final Viva

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

M.Des - Interaction / UX Design

Semester 1

No.	Subject
1	Advanced UX Foundations
2	User Research
3	Interaction Design
4	Information Architecture
5	Design Systems
6	UX Lab

1. Advanced UX Foundations

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. User Research

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Interaction Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Information Architecture

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Design Systems

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. UX Lab

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 2

No.	Subject
1	Service Design
2	Usability Engineering
3	Accessible Design
4	Design Strategy
5	Creative Coding
6	UX Lab

1. Service Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Usability Engineering

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Accessible Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Design Strategy

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Creative Coding

Unit 1 - Programming for Creatives: Variables, Functions, Loops, Events

Unit 2 - Graphics Programming: 2D primitives, Transforms, Color, Animation

Unit 3 - Interactive Media: Input, Audio, Interaction, State

Unit 4 - Generative Art: Procedural systems, Noise, Pattern generation, Simulation

Unit 5 - Creative Projects: Concept, Prototype, Iteration, Presentation

6. UX Lab

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 3

No.	Subject
1	Advanced Product Design
2	Research Seminar
3	Portfolio and Professional Practice
4	Dissertation Phase I
5	Industry Project

1. Advanced Product Design

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Research Seminar

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Portfolio and Professional Practice

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

4. Dissertation Phase I

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Industry Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

Semester 4

No.	Subject
1	Dissertation Phase II
2	Thesis
3	Portfolio
4	Final Viva

1. Dissertation Phase II

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Thesis

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Portfolio

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

4. Final Viva

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

M.Sc - Animation & VFX

Semester 1

No.	Subject
1	Advanced Animation Principles
2	Advanced 3D Modeling
3	Texturing and Materials
4	VFX Foundations
5	Research Methodology
6	Studio Lab

1. Advanced Animation Principles

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Advanced 3D Modeling

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Texturing and Materials

Unit 1 - Texturing Basics: UV mapping, Texture coordinates, Resolution, Seam management

Unit 2 - PBR: Albedo, Roughness, Metallic, Normal maps, Ambient occlusion

Unit 3 - Procedural Materials: Nodes, Masks, Generators, Material variations

Unit 4 - Surface Detail: Microdetail, Weathering, Imperfections, Decals

Unit 5 - Look Development: Material libraries, Consistency, Optimization, Export

4. VFX Foundations

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Research Methodology

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Studio Lab

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 2

No.	Subject
1	Advanced 3D Animation
2	Character Animation
3	Compositing
4	Motion Graphics
5	VFX Simulation
6	Studio Lab

1. Advanced 3D Animation

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Character Animation

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Compositing

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Motion Graphics

Unit 1 - Motion Design: Graphic composition, Timing, Transitions, Kinetic type

Unit 2 - Animation Systems: Keyframes, Expressions, Procedural motion, Graph editors

Unit 3 - 3D Motion: 3D layers, Cameras, Lighting, Materials

Unit 4 - Broadcast Design: Titles, Lower thirds, Intros, Promos

Unit 5 - Delivery: Formats, Compression, Broadcast specs, Social media output

5. VFX Simulation

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Studio Lab

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 3

No.	Subject
1	Advanced VFX
2	Technical Art
3	Production Management
4	Research Seminar
5	Dissertation Phase I
6	Industry Project

1. Advanced VFX

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Technical Art

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Production Management

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Research Seminar

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Dissertation Phase I

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Industry Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

Semester 4

No.	Subject
1	Dissertation Phase II
2	Thesis
3	Showreel / Portfolio
4	Final Viva

1. Dissertation Phase II

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Thesis

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Showreel / Portfolio

Unit 1 - Portfolio Development: Project selection, Curation, Presentation

Unit 2 - Personal Brand: Identity, Resume, Website, Social presence

Unit 3 - Freelancing: Briefs, Pricing, Contracts, Client communication

Unit 4 - Studio Practice: Pipeline, File management, Versioning, Review

Unit 5 - Career Readiness: Interviews, Showreel, Networking, Continuous learning

4. Final Viva

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

M.Sc - Computer Graphics

Semester 1

No.	Subject
1	Advanced Computer Graphics
2	Geometric Modeling
3	Rendering Algorithms
4	Image Processing
5	Mathematics for Graphics
6	Graphics Lab

1. Advanced Computer Graphics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Geometric Modeling

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Rendering Algorithms

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Image Processing

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Mathematics for Graphics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Graphics Lab

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 2

No.	Subject
1	Real-time Rendering
2	Computer Vision
3	Animation Systems
4	Shader Programming
5	Graphics Programming
6	Advanced Lab

1. Real-time Rendering

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Computer Vision

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Animation Systems

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Shader Programming

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Graphics Programming

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Advanced Lab

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 3

No.	Subject
1	Advanced Rendering
2	Simulation
3	Research Seminar
4	Dissertation Phase I
5	Industry Project

1. Advanced Rendering

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Simulation

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Research Seminar

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Dissertation Phase I

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Industry Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

Semester 4

No.	Subject
1	Dissertation Phase II
2	Thesis
3	Publication
4	Final Viva

1. Dissertation Phase II

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Thesis

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Publication

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Final Viva

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

MCA - Master of Computer Applications

Semester 1

No.	Subject
1	Advanced Programming
2	Advanced Data Structures
3	Advanced Database Systems
4	Computer Networks
5	Research Methodology
6	Programming Lab

1. Advanced Programming

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Advanced Data Structures

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Advanced Database Systems

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Computer Networks

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Research Methodology

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Programming Lab

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 2

No.	Subject
1	Software Engineering
2	Cloud Computing
3	Cyber Security
4	Machine Learning
5	Advanced Web Applications
6	Cloud/ML Lab

1. Software Engineering

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Cloud Computing

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Cyber Security

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Machine Learning

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Advanced Web Applications

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Cloud/ML Lab

Unit 1 - Safety and Setup: Equipment, File management, Workspace

Unit 2 - Core Exercises: Fundamental practice, Observation, Iteration

Unit 3 - Applied Exercises: Brief-based work, Testing, Revision

Unit 4 - Production Practice: Workflow, Quality control, Deadlines

Unit 5 - Submission: Documentation, Presentation, Viva

Semester 3

No.	Subject
1	Advanced AI / Data Analytics
2	DevOps
3	Elective I
4	Elective II
5	Research Seminar
6	Dissertation Phase I

1. Advanced AI / Data Analytics

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. DevOps

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

3. Elective I

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Elective II

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

5. Research Seminar

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

6. Dissertation Phase I

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Semester 4

No.	Subject
1	Dissertation Phase II
2	Major Project
3	Thesis
4	Internship / Industry Project
5	Final Viva

1. Dissertation Phase II

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

2. Major Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

3. Thesis

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

4. Internship / Industry Project

Unit 1 - Concept and Research: Problem/theme, References, Moodboard, Objectives

Unit 2 - Pre-production: Script/storyboard, Design bible, Schedule, Assets plan

Unit 3 - Production: Modeling/animation/design, Editing, Integration, Iteration

Unit 4 - Finishing: Compositing, Color, Audio, Quality control

Unit 5 - Delivery: Final master, Portfolio, Presentation, Viva

5. Final Viva

Unit 1 - Visual Language: Elements of design, Principles of design, Composition, Balance, Contrast, Hierarchy

Unit 2 - Color: Color wheel, Color harmony, Psychology of color, Color systems, Digital color

Unit 3 - Typography: Type anatomy, Classification, Pairing, Hierarchy, Readability

Unit 4 - Image and Layout: Cropping, Grid systems, Spacing, Visual flow, Image selection

Unit 5 - Design Practice: Briefs, Moodboards, Concept development, Critique, Portfolio

Added Program Families

- Animation & VFX
- Graphic Design & Multimedia
- Game Design & Development
- UI/UX & Digital Product Design
- 3D Modeling / 3D Art
- Motion Graphics
- Film & Video Production
- Photography & Digital Imaging
- Computer Graphics
- Web & Front-End
- Creative Coding
- Digital Media
- Communication Design
- Game Art
- Interaction Design
- MCA / Computer Applications
- Computer Science